

Supported Zero, Heat Collisions, and Holonomy

Split-Zero research programme

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This reader calculates supported-zero inversion in the full semilattice, the escaping endpoint's exact heat coordinates, and the original arithmetic heat and Weil receivers. It retains the pole and nilpotent defects at time zero. Complete derivations, source citations and exact auxiliary checks accompany the calculations. No Riemann-hypothesis or Navier–Stokes disproof is claimed.

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We calculate three different inverse operations on the original split-zero object, retaining its complete distributive support lattice. We then construct formal heat with its exact coefficient maps, its time-zero fibre, and its support adjunctions. Two nontrivial defect objects result. First, inverting the time coordinate before taking its zero fibre leaves the full support lattice as a boundary. Second, when support is permitted independently at every coefficient, forward and reverse heat leave an explicitly calculated idempotent support closure. Neither defect is replaced by a scalar Boolean quotient. Their maps to the original amplitude heat equation are proved below.

Original sources actually used. The first source is *A Note on the Globalization Semiring* $G(\mathbb{Z})$, March 2026, original `globalization_note.tex`, whose author field is empty. The user's attribution of the originating construction to Gemini 2.5 Pro is retained separately from that byline. Its definitions, projection, and localization statements are at source lines 53–139 and 432–563; the original localization source is a prior foundation, not a new result of this paper. The semilattice and adjunction source is The Clankers, *A Secondary Note on Split-Zero Globalization: Semimodule Classification, Multiplicative Reconstruction, and All-Orders Zeta Deformation*, `split_zero_secondary_note.tex`, source lines 459–819, especially `thm:fiber-modules`, `thm:equivalence-semimodules-diagrams`, and `cor:adjunctions`. The general idempotent universal property is in The Clankers, *Split-Zero Projective Geometry and Leibniz Monads*, `split_zero_projective_monads_surcomplex.tex`, lines 358–428, `thm:idempotent-localization`. The full lattice definition and its original localization theorem are in The Clankers, full-support Version 11, Definition `def:lattice-split`, and Theorem `thm:lattice-localization`. These exact original TeX passages were read. The formal heat and coefficient-support calculations below are new derivations from those objects; they are not attributed to the source authors as prior claims.

1 The whole support lattice and the three inverses

Let R be a nonzero commutative unital ring and L a nontrivial bounded distributive lattice. Equip L with semiring addition $\vee$, multiplication $\wedge$, zero 0_L and unit 1_L . The original carrier and every one of its coefficients are

$$S_R = G_L(R) = \{(0, \lambda) : \lambda \in L\} \cup \{(a, 1_L) : a \in R\} \subset R \times L. \quad (\text{SH1})$$

Write $z_\lambda = (0, \lambda)$, $\hat{a} = (a, 1_L)$, $\tau = z_{0_L}$, and $e = z_{1_L} = \hat{0}$. Operations are componentwise:

$$(a, \lambda) + (b, \mu) = (a + b, \lambda \vee \mu), \quad (a, \lambda)(b, \mu) = (ab, \lambda \wedge \mu). \quad (\text{SH2})$$

Closure follows because an element with nonzero amplitude has support 1_L . Distributivity is precisely ring distributivity and distributivity of meet over join. The semiring zero is τ and the multiplicative unit is $\hat{1}$; both differ from e . The two projections are unital semiring homomorphisms

$$p : S_R \rightarrow R, \quad (a, \lambda) \mapsto a, \quad \chi : S_R \rightarrow L, \quad (a, \lambda) \mapsto \lambda. \quad (\text{SH3})$$

The map (p, χ) is injective with the exact image in (SH1). Thus their simultaneous receiver preserves all amplitudes and all support labels.

Proposition 1.1 (Local additive inverse). *Multiplication by e is the projector*

$$\nu : S_R \rightarrow S_R, \quad \nu(x) = ex = z_{\chi(x)}. \quad (\text{SH4})$$

The additive monoid is a join-semilattice of abelian groups: the fibre of ν at z_λ is the singleton $\{z_\lambda\}$ for $\lambda < 1_L$, and is the additive group $\{\widehat{a} : a \in R\}$ at $\lambda = 1_L$, with zero e . Its local inverse map is

$$\iota(\widehat{a}) = \widehat{-a}, \quad \iota(z_\lambda) = z_\lambda, \quad x + \iota(x) = \nu(x). \quad (\text{SH5})$$

In particular the additive inverse of e in its own fibre is e . There is no y with $e + y = \tau$.

Proof. Multiplication in (SH2) proves the projector formula. Each stated fibre is closed under addition; its indicated point is the fibre zero. Formula (SH5) proves the inverse law within every fibre. It also proves $x + \iota(x) + x = x$ and $\iota(x) + x + \iota(x) = \iota(x)$. Uniqueness of such an inverse can be checked in the two coordinates: the two equalities force the support of a candidate inverse to equal the support of x , and its amplitude must be the negative amplitude of x . Finally $\chi(e + y) = 1_L$ for every y , whereas $\chi(\tau) = 0_L$. More generally, the only invertible element of the additive monoid with respect to its global identity τ is τ itself. $\square$

Proposition 1.2 (Local multiplicative inverse). *The local multiplicative monoid at the idempotent z_λ is*

$$z_\lambda S_R z_\lambda = \{z_\mu : \mu \leq \lambda\} \cong (\downarrow \lambda, \wedge), \quad 1_{\text{local}} = z_\lambda. \quad (\text{SH6})$$

Its only unit is z_λ , with inverse z_λ . In particular the inverse of e in the local monoid $eS_R e$ is e , while the global multiplicative unit group of S_R is $\{\widehat{a} : a \in R^\times\}$.

Proof. For a supported amplitude, $z_\lambda \widehat{a} = z_\lambda$; for a zero label, $z_\lambda z_\mu = z_{\lambda \wedge \mu}$. These formulas prove the local monoid identification. If $\mu, \eta \leq \lambda$ and $\mu \wedge \eta = \lambda$, then $\mu = \eta = \lambda$, proving the unit assertion. Globally a product equal to $\widehat{1}$ must have both amplitudes nonzero and multiply to 1_R , so both factors are the claimed ring units. The equation $ey = e$ alone has all supported amplitudes as solutions; it does not say that those elements are inverses in $eS_R e$, because they do not belong to that local monoid unless they equal e . If a reflexive generalized inverse is required to satisfy $eye = e$ and $yey = y$, the first identity forces y to be supported and the second then forces $y = e$. $\square$

Theorem 1.3 (The genuine inverse-monoid structure over a field). *Let K be a field. The entire multiplicative monoid of $G_L(K)$ is an inverse monoid. Its unique generalized inverse is*

$$(\widehat{a})^\dagger = \widehat{a^{-1}} \quad (a \neq 0), \quad z_\lambda^\dagger = z_\lambda \quad (\lambda \in L). \quad (\text{SH6a})$$

It obeys $xx^\dagger x = x$ and $x^\dagger x x^\dagger = x^\dagger$ for every element, including the genuine identity $e^\dagger = e$. The corresponding multiplicative local identity is

$$\eta(x) = xx^\dagger = \begin{cases} \widehat{1}, & p(x) \neq 0, \\ z_\lambda, & x = z_\lambda. \end{cases} \quad (\text{SH6b})$$

All multiplicative idempotents form the meet-semilattice $\mathcal{E} = \{\widehat{1}\} \sqcup \{z_\lambda : \lambda \in L\}$, in which $\widehat{1}$ is above every z_λ and the original lattice L remains unchanged beneath it.

Proof. The two generalized-inverse equations follow by direct multiplication. To prove uniqueness, if $x = \widehat{a}$ with $a \neq 0$, a lower-label candidate y makes xyx have zero amplitude, and cannot give x . For $y = \widehat{b}$ the first equation gives $a^2 b = a$, hence $b = a^{-1}$ in the field. If $x = z_\lambda$ and a candidate y has nonzero amplitude, the product xyx has zero amplitude and cannot equal y . Thus $y = z_\mu$; the two equations are $\lambda \wedge \mu = \lambda$ and $\lambda \wedge \mu = \mu$, which force $\lambda = \mu$. This proves uniqueness for every element. A multiplicative idempotent with nonzero amplitude has amplitude satisfying $a^2 = a$ in a field, hence is $\widehat{1}$; all other idempotents are exactly the z_λ . The monoid is commutative, so the idempotents commute, and their multiplication is the asserted meet. This also proves the inverse-monoid assertion directly. $\square$

The map $\eta : G_L(K) \rightarrow \mathcal{E}$ is a multiplicative monoid homomorphism: a product of nonzero field amplitudes is nonzero, and every remaining product meets its zero labels. The full original support map factors through it by $\widehat{1} \mapsto 1_L$, $z_\lambda \mapsto \lambda$. This last map identifies the two different idempotents $\widehat{1}$ and e ; η itself retains their difference. At an actual supported scalar difference $d = \widehat{a - b}$ one has the exact two cases

$$dd^\dagger = \begin{cases} \widehat{1}, & a \neq b, \\ e, & a = b. \end{cases} \quad (\text{SH6c})$$

Thus inverse-monoid division by the supported zero is well-defined, with its specified local identity e . For any $f \in G_L(K)$ the exact division equation is $d(d^\dagger f) = \eta(d)f$. On the distinct-value locus it is f ; at $a = b$ it is $ef = \nu(f)$. This supplies the precise idempotent defect in a rational identity whose ordinary derivation canceled d . It does not assume that the unit $\widehat{1}$ remains its local identity at a collision.

Theorem 1.4 (Universal unital inversion). *For every $\lambda \in L$, the universal unital localization is*

$$S_R[z_\lambda^{-1}] \cong \downarrow \lambda, \quad q_\lambda(x) = \lambda \wedge \chi(x). \quad (\text{SH7})$$

The target has unit λ . In particular

$$S_R[e^{-1}] \cong L, \quad q_1 = \chi, \quad S_R[\tau^{-1}] \cong \{0\}. \quad (\text{SH8})$$

The localization at e keeps every label of L and identifies all supported amplitudes with 1_L .

Proof. For any commutative semiring S and idempotent a , the subset aS is a semiring with unit a and the same zero as S . The map $x \mapsto ax$ is unital into this new target because $\widehat{1}$ maps to its unit a . If $f : S \rightarrow T$ is unital and $f(a)$ is invertible, $f(a)^2 = f(a)$ forces $f(a) = 1_T$. It follows that $ax = ay$ implies $f(x) = f(y)$, so $\bar{f}(ax) = f(x)$ is a well-defined unique factorization. This proves the universal property of aS . Apply it to $a = z_\lambda$ and use (SH6). The two endpoint cases are $\lambda = 1_L, 0_L$. $\square$

The kernel *congruence* of χ is exactly

$$\mathcal{R}_\chi = \{(\widehat{a}, \widehat{b}) : a, b \in R\} \cup \{(z_\lambda, z_\lambda) : \lambda < 1_L\}. \quad (\text{SH9})$$

Its zero ideal $\chi^{-1}(0_L) = \{\tau\}$ does not detect the identified amplitudes. The kernel-pair object $S_R \times_L S_R$, its two projections to S_R , and the amplitude difference

$$d : S_R \times_L S_R \rightarrow (R, +), \quad d(x, y) = p(x) - p(y)$$

give an exact description of that identification: d is additive and surjective, and $d^{-1}(0)$ is the diagonal copy of S_R . Surjectivity uses $(\widehat{a}, e)$; the kernel assertion follows from injectivity of (p, χ) . This supplies an explicit receiver for the amplitudes identified by localization.

The ring reflection p is universal among unital maps to rings. Indeed $e + z_\lambda = e$ forces every z_λ to map to zero in a ring by additive cancellation; the remaining restriction is then a ring homomorphism. Consequently the pushout of $S_R \rightarrow L$ and $S_R \rightarrow R$ in commutative unital semirings is the one-element ring: in the pushout the same element e is both 1 and 0. This proves the exact relation between the support localization and the ring reflection.

2 The original semilattice adjunctions with the full lattice

Let M be a unital S_R -semimodule, meaning a commutative monoid under addition with a distributive S_R action. Its global zero is 0_M . Put

$$\epsilon_M(m) = em, \quad P_M = eM, \quad M_\ell = \{m : em = \ell\} \quad (\ell \in P_M). \quad (\text{SH10})$$

The module support semilattice P_M is distinguished from the base support lattice L .

Theorem 2.1 (Full-lattice semilattice description). *The set P_M is a join-semilattice with zero 0_M under the addition in M , and each M_ℓ is an R -module with zero ℓ . For $\ell \leq \ell'$ the transition*

$$T_{\ell, \ell'} : M_\ell \rightarrow M_{\ell'}, \quad m \mapsto m + \ell' \quad (\text{SH11})$$

is R -linear and these transitions compose. The remaining full lattice action is precisely an L -semimodule action on P_M :

$$\lambda \cdot \ell = z_\lambda \ell, \quad z_\lambda m = z_\lambda(em) = \lambda \cdot \ell \in P_M \quad (m \in M_\ell). \quad (\text{SH12})$$

Conversely, a join-semilattice P with zero carrying an L -semimodule action, together with a diagram of R -modules M_ℓ and R -linear transitions over P , reconstructs an S_R -semimodule by the following explicit operations:

$$\begin{aligned} M &= \bigsqcup_{\ell \in P} M_\ell, & m + n &= T_{\ell, \ell \vee \eta} m + T_{\eta, \ell \vee \eta} n \quad (m \in M_\ell, n \in M_\eta), \\ \widehat{a} m &= am \in M_\ell, & z_\lambda m &= 0_{M_{\lambda \cdot \ell}}. \end{aligned} \quad (\text{SH13})$$

The overlap $z_{1_L} = \widehat{0}$ has identical actions in these two formulas.

Proof. The identities $e^2 = e$, $e + e = e$, and $\widehat{1} + e = \widehat{1}$ give $\epsilon_M^2 = \epsilon_M$, $em + em = em$, and $m + em = m$. Thus P_M is the asserted semilattice. In a fibre M_ℓ , the last identity makes ℓ its zero, and $m + \widehat{1}m = em = \ell$ proves the additive inverse. Supported scalars preserve the fibre because $e\widehat{a} = e$, giving its R -module structure. The formula for transitions preserves fibres, zero, addition and scalar multiplication; composing transitions uses $\ell + \ell' = \ell'$ for $\ell \leq \ell'$. Addition between fibres is their sum transported to the join fibre because $em + en$ is that join. These are the original semilattice constructions.

For the full L action, $z_\lambda e = z_\lambda$ proves (SH12). All semimodule laws on P_M follow from the operations on the zero fibre of S_R . In particular $\lambda \cdot \ell \leq \ell$, since $\lambda \vee 1_L = 1_L$. Conversely the usual diagram sum in (SH13) is associative: transport three summands to their common join, where ordinary module associativity applies. Zero, scalar multiplication by supported amplitudes, and fibre inverses follow from their module counterparts. Scalar identities within L are the L -semimodule identities on P . For mixed scalars, $(\widehat{a} + z_\lambda)m = \widehat{a}m$ holds because the zero in fibre $\lambda \cdot \ell \leq \ell$ transports to the zero in M_ℓ . The identity $\widehat{a}z_\lambda = z_\lambda$ holds because an R scalar sends every fibre zero to itself. The zero scalar z_{0_L} acts as the zero of M_{0_P} . Thus every semimodule law holds. Extraction and reconstruction return the same elements and all the same operations. $\square$

The adjunctions retain all these support fibres:

$$\begin{aligned} \text{Hom}_{S_R}(M, \chi^* N) &\cong \text{Hom}_L(eM, N), \\ K_M &= \{m : em = 0_M\} = M_{0_M}, & \text{Hom}_{S_R}(p^* A, M) &\cong \text{Hom}_R(A, K_M). \end{aligned} \quad (\text{SH14})$$

Here N is an L -semimodule and A an R -module. For the first identity, $f(m) = ef(m) = f(em)$ forces unique factorization through $m \mapsto em$; conversely an L -linear map on eM composed with this projector is S_R -linear. For the second, $eg(a) = g(ea) = g(0) = 0_M$ forces the image into K_M ; on that fibre every z_λ acts by zero since $z_\lambda m = z_\lambda em$, so precisely the R -linear maps occur. These arguments prove both bijections and their naturality.

For the regular module $M = S_R$, one has $P_M = L$, top fibre R , and singleton fibres below the top. In particular K_{S_R} is the singleton $\{\tau\}$, whereas the ring reflection is R . The right adjoint K_M is not being substituted for the amplitude projection p . The first adjoint is precisely universal inversion of the idempotent action: if an S_R -linear map receives e as an invertible operator, that idempotent operator must be the identity, so the map factors uniquely through eM .

3 The exact integral coefficient heat operator

Let $A = R[[t, z]]$, with t the formal heat parameter. An element F is a family of coefficients $f_{mn} \in R$: $F = \sum_{m,n \geq 0} f_{mn} t^m z^n$. The variable z is the unchanged spatial coordinate. For $\sigma \in \{+1, -1\}$ define

$$[t^m z^n] \mathcal{H}_\sigma F = \sum_{j=0}^m \sigma^j \frac{(n+2j)!}{n! j!} f_{m-j, n+2j}. \quad (\text{SH15})$$

Each displayed integer is integral: $(n+2j)!/(n!j!) = \binom{n+2j}{2j} (2j)!/j!$. Thus the definition makes sense over every R , with no division operation in R . Every coefficient is a finite sum. The notation $\mathcal{H}_\sigma = \exp(\sigma t \partial_z^2)$ records this definition, not an analytic convergence assertion. The Newman heat sign $e^{tu^2} \cos(zu)$ corresponds to $\sigma = -1$, because $\partial_z^2 \cos(zu) = -u^2 \cos(zu)$.

Theorem 3.1 (Formal inverse and time-zero value). *The maps $\mathcal{H}_\sigma$ are $R[[t]]$ -linear continuous automorphisms for the t -adic topology, and*

$$\mathcal{H}_+ \mathcal{H}_- = \mathcal{H}_- \mathcal{H}_+ = I, \quad \text{ev}_0 \mathcal{H}_\sigma = \text{ev}_0, \quad \text{ev}_0(F) = F(0, z). \quad (\text{SH16})$$

They commute with ∂_z and with every coefficient-ring homomorphism. For initial data $f(z) \in R[[z]]$, $F = \mathcal{H}_\sigma f$ satisfies

$$\partial_t F = \sigma \partial_z^2 F, \quad F(0, z) = f(z). \quad (\text{SH17})$$

When R is a $\mathbb{Q}$ -algebra these conditions uniquely determine F . All assertions extend to coefficients in any R -module.

Proof. The formula uses only integer scalar maps, sums and shifts of indices, proving linearity, coefficient naturality and commutation with ∂_z . It preserves the ideal $t^k A$ for every k ; and multiplication by t commutes with it, proving continuity and $R[[t]]$ -linearity. To calculate the inverse, collect terms whose two heat indices sum to k . Over $\mathbb{Q}$ their coefficient in the composite is

$$\frac{(n+2k)!}{n!} \sum_{j=0}^k \frac{(-1)^j}{j!(k-j)!} = \frac{(n+2k)!}{n!k!} (1-1)^k.$$

Every summand before collection is an integer, being a product of the integer coefficients in (SH15); therefore this integer identity holds over every R . It is 1 at $k = 0$ and zero otherwise, proving both inverse formulas. At $m = 0$ only $j = 0$ occurs, proving evaluation at zero.

For initial data the coefficients are $[t^m z^n] F = \sigma^m (n+2m)!/(n!m!) f_{n+2m}$. Multiplying the coefficient with index $m+1$ by $m+1$ gives exactly $\sigma(n+2)(n+1)$ times the coefficient with indices $m, n+2$. This proves the differential equation without division in R . If all positive integers are invertible in R , the recurrence $(m+1)F_{m+1}(z) = \sigma F_m''(z)$ uniquely constructs the solution from $F_0 = f$. Module coefficients obey the same identities, proving the final assertion. $\square$

The $\mathbb{Q}$ -algebra condition is needed for that PDE uniqueness claim, not for (SH15) or its inverse. For example over $R = \mathbf{F}_p$, the nonzero series t^p has zero time-zero value and satisfies $\partial_t t^p = \sigma \partial_z^2 t^p = 0$. The exact additional solution object in any characteristic is

$$\mathcal{N}_R = \{(F_m)_{m \geq 0} : F_0 = 0, (m+1)F_{m+1} = \sigma F_m''\} \subset tR[[t, z]]. \quad (\text{SH18})$$

The difference of any two solutions with the same initial value lies in $\mathcal{N}_R$, and adding an element of $\mathcal{N}_R$ to a solution gives another with that initial value. The $\mathbb{Q}$ -algebra calculation proves $\mathcal{N}_R = 0$ there.

The heat map is generally not a ring homomorphism. With the original coordinate z and sign σ ,

$$\mathcal{H}_\sigma z = z, \quad \mathcal{H}_\sigma(z^2) = z^2 + 2\sigma t, \quad \mathcal{H}_\sigma(z^2) - (\mathcal{H}_\sigma z)^2 = 2\sigma t. \quad (\text{SH19})$$

Its complete multiplication defect has an exact receiver:

$$\mathcal{H}_\sigma(FG) - (\mathcal{H}_\sigma F)(\mathcal{H}_\sigma G) = \sum_{k \geq 1} \frac{(2\sigma t)^k}{k!} \partial_z^k(\mathcal{H}_\sigma F) \partial_z^k(\mathcal{H}_\sigma G). \quad (\text{SH20})$$

This is coefficientwise integral as well: the product of the two derivative coefficients is divisible by $k!$ as an integer. To prove the identity, first work over $\mathbb{Q}$ with independent variables z_1, z_2 , so that differentiation of the product is $\partial_1 + \partial_2$. The three commuting differential operators in $(\partial_1 + \partial_2)^2 = \partial_1^2 + 2\partial_1\partial_2 + \partial_2^2$ exponentiate and give the formula, including its $k = 0$ term. Each fixed output coefficient involves finitely many input coefficients and has integral coefficients on both sides; the identity in their polynomial ring over $\mathbb{Q}$ is consequently an integral identity, so descends to every R .

An exact repaired multiplication, rather than an imposed multiplicativity claim, is

$$F \star_\sigma G = \sum_{k \geq 0} \frac{(2\sigma t)^k}{k!} (\partial_z^k F)(\partial_z^k G) = \mathcal{H}_\sigma((\mathcal{H}_{-\sigma} F)(\mathcal{H}_{-\sigma} G)). \quad (\text{SH21})$$

Transport of multiplication through the proved additive automorphism makes this an associative commutative $R[[t]]$ -algebra with unit 1, and makes $\mathcal{H}_\sigma : (A, \cdot) \rightarrow (A, \star_\sigma)$ a unital ring isomorphism. At $t = 0$ this multiplication is the original one. This construction retains the precise product defect and does not assume that the original heat map was multiplicative.

Multiplicative reciprocals of actual units have an equally exact map. A series $F \in A$ is a unit precisely when its coefficient f_{00} is a unit in R : necessity follows by constant-term evaluation, and sufficiency follows from the coefficientwise convergent geometric series after factoring out f_{00} . Heat preserves this constant coefficient, hence preserves the set of units. For every such F ,

$$\begin{aligned} (\mathcal{H}_\sigma F) \star_\sigma \mathcal{H}_\sigma(F^{-1}) &= 1, \\ (\mathcal{H}_\sigma F) \mathcal{H}_\sigma(F^{-1}) - 1 &= - \sum_{k \geq 1} \frac{(2\sigma t)^k}{k!} \partial_z^k(\mathcal{H}_\sigma F) \partial_z^k(\mathcal{H}_\sigma(F^{-1})). \end{aligned} \quad (\text{SH21a})$$

The first is the transported multiplication law; the second is (SH20) applied to $FF^{-1} = 1$. This proves the complete ordinary-inversion defect and its divisibility by t , so at time zero the ordinary reciprocal is retained. For example over $\mathbb{Q}$, $F = 1 + z$ has $\mathcal{H}_\sigma F = 1 + z$, whereas $\mathcal{H}_\sigma(F^{-1}) = (1 + z)^{-1} + 2\sigma t(1 + z)^{-3} + O(t^2)$. The defect is already nonzero at the displayed first order.

The field inverse-monoid theorem is applied in its stated coefficient category. The formal series ring $K[[t, z]]$ is not a field: t has no reflexive multiplicative inverse in it, since $tBt = t$ would imply $tB = 1$ in this integral domain, contradicting evaluation at $t = 0$. The exact extension receiving its inverse is the punctured chart in (SH28), whose full supported time-zero fibre is proved below.

4 Heat on the full semilattice and its adjunction maps

On the actual requested carrier $G_L(A)$ define

$$\widetilde{\mathcal{H}}_\sigma(\widehat{F}) = \widehat{\mathcal{H}_\sigma F}, \quad \widetilde{\mathcal{H}}_\sigma(z_\lambda) = z_\lambda. \quad (\text{SH22})$$

The two prescriptions agree at $e = \widehat{0} = z_{1_L}$. They give a $G_L(R[[t]])$ -linear automorphism of the underlying semimodule. Indeed additivity follows from heat additivity on the top fibre and from preservation of all lower zero fibres; supported scalars act by $R[[t]]$ multiplication, while z_λ sends every top-fibre element to z_λ and meets lower labels. Both operations commute with (SH22).

Its inverse is $\tilde{\mathcal{H}}_{-\sigma}$. It is generally not a homomorphism for the original semiring product, by (SH19); it is a unital semiring isomorphism to $G_L(A, \star_\sigma)$ by (SH21).

The exact receiving diagram is

$$\begin{array}{ccccc} A & \xleftarrow{p} & G_L(A) & \xrightarrow{\chi} & L \\ \mathcal{H}_\sigma \downarrow & & \tilde{\mathcal{H}}_\sigma \downarrow & & \downarrow I_L \\ A & \xleftarrow{p} & G_L(A) & \xrightarrow{\chi} & L. \end{array} \quad (\text{SH23})$$

Both squares commute directly from the coefficient formulas. Here the vertical maps are additive heat maps; the right square is also the exact induced map after universal unital inversion of e . On that localization the induced map is I_L . The restriction to the local monoid $eG_L(A)e = L$ is likewise I_L , and heat commutes with every local additive inverse in (SH5). For a general label the localized square is the same with L replaced by $\downarrow \lambda$ and χ replaced by q_λ . No Boolean branch has been chosen.

The construction also works on the original general semilattice diagram, without replacing it by the regular scalar fibre. For the data in Theorem 2.1, replace each module M_ℓ by $M_\ell[[t, z]]$ and each transition by its coefficientwise extension. The lattice P and its complete L action are unchanged. The heat maps of Theorem 3.1 commute with all these transitions because their coefficients are integers. They thus form a natural automorphism of the reconstructed semimodule. Under the two adjunctions (SH14), they induce the identity on the support object P and the ordinary coefficient heat on $M_{0P}[[t, z]]$, respectively. These claims follow by applying the explicit factorization maps in the proof of (SH14), not by replacing one adjoint with the other.

The derivative lifts also preserve every support label. Hence the differential equation is an equality in each module fibre: for example on the top fibre $\tilde{\partial}_t \hat{F} = \sigma \tilde{\partial}_z^2 \hat{F}$ means exactly $\partial_t F = \sigma \partial_z^2 F$. Bringing its two sides together gives the fibre zero e , not the external zero τ . On a lower singleton fibre both sides equal z_λ . This is the same fibrewise inverse operation as in (SH5).

5 The exact time-zero fibre and its punctured-time boundary

Set $A_0 = R[[z]]$. The time-zero map is the unital semiring map

$$\tilde{e}\tilde{v}_0 : G_L(A) \rightarrow G_L(A_0), \quad \hat{F} \mapsto \widehat{F(0, z)}, \quad z_\lambda \mapsto z_\lambda. \quad (\text{SH24})$$

It is onto, with constant-in- t section. Its kernel congruence has the following complete description:

$$\hat{F} \sim_0 \hat{G} \iff F - G \in tA, \quad z_\lambda \sim_0 z_\mu \iff \lambda = \mu; \quad (\text{SH25})$$

there is no additional identification between a lower label and a supported amplitude. In particular

$$\tilde{e}\tilde{v}_0^{-1}(\tau) = \{\tau\}, \quad \tilde{e}\tilde{v}_0^{-1}(e) = \{\widehat{tF} : F \in A\}. \quad (\text{SH26})$$

Thus the semiring-zero kernel ideal is trivial, while its congruence retains the full amplitude ideal tA . The ordinary exact sequence is $0 \rightarrow tA \rightarrow A \rightarrow A_0 \rightarrow 0$, and its full supported lift is precisely (SH25), not a quotient obtained by identifying supported zeros with τ .

The complete time-zero heat calculation is

$$\begin{array}{ccc} G_L(A) & \xrightarrow{\tilde{e}\tilde{v}_0} & G_L(A_0) \\ \tilde{e}\tilde{v}_0 \tilde{\mathcal{H}}_\sigma = \tilde{e}\tilde{v}_0, & \chi \downarrow & \downarrow \chi \\ L & \xrightarrow{I_L} & L. \end{array} \quad (\text{SH27})$$

Both claims follow from the $m = 0$ term in (SH15) and preservation of all labels. Thus localization at e and evaluation at $t = 0$ commute through the displayed full-lattice map.

Theorem 5.1 (Supported special fibre of punctured time). *Let $\hat{t} = (t, 1_L) \in G_L(A)$. There are canonical isomorphisms of unital semirings*

$$G_L(A)[\hat{t}^{-1}] \cong G_L(A[t^{-1}]), \quad G_L(A)[\hat{t}^{-1}] \amalg_{G_L(A)} G_L(A_0) \cong L. \quad (\text{SH28})$$

The second target is the actual pushout for time-zero base change of the punctured-time chart. For every $n \geq 0$,

$$G_L(A/(t^{n+1}))[\hat{t}^{-1}] \cong L. \quad (\text{SH29})$$

All transition maps between these localized finite thickenings are the identity on L .

Proof. Use fractions with denominators $\hat{t}^k$. The map $z_\lambda/\hat{t}^k \mapsto z_\lambda$ and $\hat{F}/\hat{t}^k \mapsto \widehat{F/\hat{t}^k}$ is surjective and respects both operations. Two fractions with lower labels are equal exactly when their labels agree, since multiplication by a supported denominator leaves such a label unchanged. A lower-label fraction cannot equal a top fraction. For two top fractions, their equality is exactly the usual ring localization relation. This proves the first isomorphism.

For any unital target T , compatible maps from the first pushout factors are exactly maps $G_L(A_0) \rightarrow T$ in which the image of $\hat{t}$ under time-zero evaluation is invertible. That image is e , so this universal property is the property of $G_L(A_0)[e^{-1}]$. Theorem 1.4 identifies it with L , proving the second isomorphism. In the n th thickening, $\hat{t}^{n+1} = e$, so inverting $\hat{t}$ forces e to be invertible. Conversely, after applying χ every supported amplitude, including $\hat{t}$, is already the unit 1_L . These two universal factorizations prove (SH29); they also force the stated identity transition maps. $\square$

The maps to the pushout in (SH28) are $\widehat{F/\hat{t}^k} \mapsto 1_L$ and $z_\lambda \mapsto \lambda$ from the punctured chart, and $\hat{f} \mapsto 1_L$, $z_\lambda \mapsto \lambda$ from its time-zero chart. Thus every amplitude is explicitly identified there, but no label is. On ordinary ring amplitudes the corresponding pushout is the zero ring, since an invertible t and $t = 0$ force $1 = 0$. This proves the exact map from the retained support boundary to the ordinary empty ring fibre. It is not a statement that the supported boundary is absent.

The spectral map retains its exact geometry. Every ideal of $G_L(A)$ either lies in the zero-label part, where it is a lattice ideal, or contains a supported amplitude and hence contains e and all z_λ ; its amplitudes then form a ring ideal. Testing products gives the two prime families $P_{\mathfrak{p}} = \{z_\lambda : \lambda \in \mathfrak{p}\}$ and $Q_{\mathfrak{q}} = \{z_\lambda : \lambda \in L\} \cup \{\hat{F} : F \in \mathfrak{q}\}$. Every $P_{\mathfrak{p}}$ omits e , and every $Q_{\mathfrak{q}}$ contains it. The basic-open formulas $D(z_\lambda) = D_L(\lambda)$ and $D(\hat{F}) = \text{Spec}_{\text{lat}} L \cup D_A(F)$ prove

$$\text{Spec } G_L(A) = \text{Spec}_{\text{lat}} L \blacktriangleright \text{Spec } A, \quad D(e) = \text{Spec}_{\text{lat}} L. \quad (\text{SH30})$$

The map on spectra from the pushout fibre L in (SH28) is exactly this full support stratum in the time-zero chart. It does not select one Boolean character.

The time-zero congruence map itself has the precise spectral image

$$\text{im}(\text{Spec } G_L(A_0) \rightarrow \text{Spec } G_L(A)) = \text{Spec}_{\text{lat}} L \cup V_A(t), \quad D(\hat{t}) = \text{Spec}_{\text{lat}} L \cup D_A(t). \quad (\text{SH30a})$$

Indeed pure-support primes pull back to the same lattice primes, and an arithmetic prime pulls back along $A \rightarrow A/(t)$ to a prime containing t . These are all such primes by the ordinary quotient correspondence. Their intersection with the punctured chart is exactly $\text{Spec}_{\text{lat}} L$. Thus the categorical fibre in (SH28) also has this explicit topological map. The time-zero quotient is the stated congruence quotient; it is not being identified with the semiring ideal quotient by its trivial zero ideal in (SH26).

Heat also extends to the punctured chart as an additive map: $\mathcal{H}_\sigma(F/t^k) = \mathcal{H}_\sigma(F)/t^k$. $R[[t]]$ -linearity makes the definition independent of the chosen denominator; its inverse is the opposite heat sign. Its support projection to the boundary L is the identity. All time thickenings before localization still keep their amplitudes: heat preserves $t^{n+1}A$ and induces inverse maps on $A/(t^{n+1})$. The inverse limit of their full split carriers is $G_L(A)$, because a compatible sequence has one fixed support label, and at top support is a compatible sequence of ring coefficients. This describes both the retained formal neighborhood and its localized support-only boundary.

6 Independent coefficient supports and the exact heat defect

The requested global carrier $G_L(R[[t, z]])$ has one support label for an entire series. The original semilattice theory also allows the larger coefficient-series semiring

$$\mathcal{T} = (G_L(R))[[t, z]].$$

An element X has at each bidegree a pair $X_{mn} = (f_{mn}, \lambda_{mn}) \in G_L(R)$. Addition is coefficientwise, and multiplication is the finite Cauchy sum. Its amplitude map goes to $R[[t, z]]$ and its coefficient support map goes to the full semiring $L[[t, z]]$. Neither collection of labels is collapsed.

Define coefficient heat using the same integers as before, lifted as supported scalars:

$$(\mathcal{H}_\sigma X)_{mn} = \sum_{j=0}^m \sigma^j \widehat{(n+2j)! / (n!j!)} X_{m-j, n+2j}. \quad (\text{SH31})$$

Even when that integer is zero in R , its displayed lift is e , not τ . Thus (SH31) has the exact amplitude and support formulas

$$p(\mathcal{H}_\sigma X) = \mathcal{H}_\sigma p(X), \quad \chi(\mathcal{H}_\sigma X)_{mn} = (C\lambda)_{mn} := \bigvee_{j=0}^m \lambda_{m-j, n+2j}. \quad (\text{SH32})$$

These are finite joins in L , so no completeness hypothesis on the lattice is needed.

Theorem 6.1 (The complete coefficient-support closure). *Define $\mathcal{C} : \mathcal{T} \rightarrow \mathcal{T}$ by keeping the amplitude of each coefficient and replacing its label by $C\lambda$ in (SH32). Then*

$$\mathcal{H}_+ \mathcal{H}_- = \mathcal{H}_- \mathcal{H}_+ = \mathcal{C}, \quad \mathcal{C}^2 = \mathcal{C}, \quad \mathcal{C} \mathcal{H}_\sigma = \mathcal{H}_\sigma \mathcal{C} = \mathcal{H}_\sigma. \quad (\text{SH33})$$

All these maps are additive $G_L(R)$ -linear maps. The image and fixed set of $\mathcal{C}$ coincide, and are specified exactly by

$$\lambda_{m-1, n+2} \leq \lambda_{mn} \quad (m \geq 1, n \geq 0). \quad (\text{SH34})$$

On that retained-support subspace the two heat signs are inverse automorphisms. On the entire coefficient space their failure to be inverse is exactly $\mathcal{C}$, with no amplitude defect.

Proof. The replacement gives a valid coefficient of $G_L(R)$ because the join includes its original label; any nonzero original amplitude already has label 1_L . The amplitude of either composite heat is the original one by (SH16). Its support is

$$\bigvee_{i=0}^m \bigvee_{j=0}^{m-i} \lambda_{m-i-j, n+2i+2j} = \bigvee_{k=0}^m \lambda_{m-k, n+2k},$$

because every $k = i + j$ from 0 to m occurs. This proves the first identity and the idempotence of the support closure. Applying this calculation to one heat image proves the two commutation identities. Distributivity of finite joins over joins proves additivity on support; amplitudes are additive. For a scalar with label η , distributivity $\eta \wedge \bigvee_j \lambda_j = \bigvee_j (\eta \wedge \lambda_j)$ proves scalar compatibility, including supported scalar zero.

The fixed condition implies (SH34) because the defining join includes $j = 1$. Conversely that inequality iterated j times gives $\lambda_{m-j, n+2j} \leq \lambda_{mn}$ for every j , so the join equals the term with $j = 0$. This proves the exact fixed set. Idempotence identifies it with the image; the first identity in (SH33) then proves mutual inverse heat maps on it. $\square$

The exact quotient object of this defect is also explicit. The two heat maps and the retraction $\mathcal{C}$ have the same kernel equivalence:

$$X \sim_{\mathcal{C}} Y \iff p(X) = p(Y) \text{ and } \mathcal{C}\chi(X) = \mathcal{C}\chi(Y). \quad (\text{SH34a})$$

For $\mathcal{C}$ this follows from its coordinate definition. For heat, equality of its images gives equality of input amplitudes by the inverse amplitude map, and equality of closed supports by (SH32); the converse is the same formula. Thus the quotient additive $G_L(R)$ -semimodule $\mathcal{T}/\sim_{\mathcal{C}}$ is isomorphic to $\text{im } \mathcal{C}$, with its two induced inverse heat maps. Inclusion and $\mathcal{C}$ give a specified split retraction onto that image. The preimage of the external zero under each heat map remains the single zero series, since every original label occurs in its closure. Thus its possible noninjectivity is encoded by (SH34a), not by its zero ideal.

For a literal nonzero defect, take the coefficient monomial $X = z^2$ in $\mathcal{T}$, meaning coefficient $\widehat{1}$ at $(m, n) = (0, 2)$ and τ at every other bidegree. Then

$$\mathcal{C}X : \quad X_{0,2} = \widehat{1}, \quad X_{1,0} = e, \quad X_{mn} = \tau \text{ at every other bidegree.} \quad (\text{SH35})$$

The amplitude is still z^2 , but the new coefficient e at t is a retained supported zero. This follows directly from (SH32): the only additional path from $(0, 2)$ is to $(1, 0)$. Over characteristic zero it is also the explicit cancellation of the $2t$ and $-2t$ amplitudes in the two heat steps. The same support computation remains valid in characteristic two, where each of those amplitudes is already zero but its coefficient is still supported.

There is also an exact infinite even-series example over $\mathbb{Q}$. Let X^{init} have coefficient $(-1)^n/(2n)!$ at $(0, 2n)$ for every $n \geq 0$, and τ elsewhere. Its amplitude is $\cos z$ and its initial time support is exactly the zeroth row. Then

$$p(\mathcal{H}_\sigma X^{\text{init}}) = e^{-\sigma t} \cos z, \quad [t^m z^{2n}] p(\mathcal{H}_\sigma X^{\text{init}}) = \frac{(-\sigma)^m (-1)^n}{m! (2n)!} \neq 0.$$

This follows either by direct substitution in (SH15), using $\partial_z^{2m} \cos z = (-1)^m \cos z$, or by the displayed coefficient formula itself. Its support is 1_L at every $(m, 2n)$ and 0_L at every odd spatial index. That support is $\mathcal{C}$ -fixed by (SH34). Reversing the heat therefore gives the initial amplitude $\cos z$ with supported zero e at every positive-time even coefficient. It is exactly $\mathcal{C}X^{\text{init}}$, not X^{init} . This explicitly realizes the even-coefficient support cone and its retained zero coefficients.

There is an exact comparison map from the global carrier:

$$\Gamma : G_L(R[[t, z]]) \hookrightarrow \mathcal{T}, \quad \Gamma(F, \lambda)_{mn} = (f_{mn}, \lambda). \quad (\text{SH36})$$

For $\lambda < 1_L$, necessarily every $f_{mn} = 0$. The map is injective, additive and multiplicative: at bidegree (m, n) every Cauchy summand has the same meet of the two global labels, and there is a nonempty finite set of summands. It need not send the unit to the standard unit of $\mathcal{T}$. Its image unit is the idempotent E with coefficient $\widehat{1}$ at $(0, 0)$ and e everywhere else. Thus Γ is a unital semiring map to its image, with that explicit unit, rather than a unital map to the whole standard power-series semiring. The image is the exact subsemiring of series with one constant label at every coefficient. It is fixed by $\mathcal{C}$, and

$$\mathcal{H}_\sigma \Gamma = \Gamma \widetilde{\mathcal{H}}_\sigma, \quad \mathcal{C} \Gamma = \Gamma. \quad (\text{SH37})$$

The amplitude identity is (SH15); the support identity is the join of finitely many identical labels. In particular the forward-reverse defect on arbitrary coefficient supports does not contradict the actual automorphism on the specified global-label carrier.

The coefficientwise support quotient itself is the universal inversion of the constant supported scalar e in $\mathcal{T}$:

$$\mathcal{T}[e^{-1}] \cong L[[t, z]], \quad X \mapsto \sum_{m,n} \lambda_{mn} t^m z^n. \quad (\text{SH38})$$

Indeed multiplication by that constant e sends every coefficient to its zero label and leaves its support unchanged; its image is exactly the displayed semiring, with the usual constant unit. The idempotent localization proof applies. On this quotient either heat sign induces the same map $\mathcal{C}$, not an inverse pair. This is the complete support receiver of the coefficient defect. Here the constant scalar being inverted has coefficient e at $(0,0)$ and τ at every other index; it differs from $\Gamma(e)$, whose every coefficient is e . Their connecting map is explicit: the support projection of Γ is $\Lambda : L \rightarrow L[[t, z]]$, with $\Lambda(\lambda)_{mn} = \lambda$ at every index. This map preserves both operations and has unit the all- 1_L series in its image, rather than the standard constant unit. It satisfies $\mathcal{C}\Lambda = \Lambda$ and $\chi_{\text{coeff}}\Gamma = \Lambda\chi$. Thus the two universal support receivers are connected without identifying their different coefficient units.

Finally time-zero evaluation in $\mathcal{S}$ keeps exactly the row $m = 0$. That row of (SH32) has only $j = 0$, so

$$\text{ev}_0 \mathcal{H}_\sigma = \text{ev}_0, \quad \text{ev}_0 \mathcal{C} = \text{ev}_0, \quad \text{ev}_0 \Gamma = \Gamma_0 \widetilde{\text{ev}}_0. \quad (\text{SH39})$$

Here $\Gamma_0 : G_L(R[[z]]) \rightarrow G_L(R)[[z]]$ is the same constant-label coefficient map in one variable. Thus even the enlarged support defect has the original time-zero coefficient fibre, with every original label retained.

7 The exact receiver of zero locations

Formal substitution $z = a$ into an arbitrary element of $R[[z]]$ is not assumed. For an analytic initial function $h(z)$ and a specified complex point a , instead use its actual local coordinate $w = z - a$ and Taylor series $h(a + w) \in \mathbb{C}[[w]]$. This yields a unital evaluation map on the formal germ:

$$G_L(\mathbb{C}[[t, w]]) \longrightarrow G_L(\mathbb{C}), \quad \widehat{F} \mapsto \widehat{F(0, 0)}, \quad z_\lambda \mapsto z_\lambda. \quad (\text{SH40})$$

For the supported initial germ and its formal heat evolution,

$$\mathcal{H}_\sigma(\widehat{h(a + w)}) \mapsto \widehat{h(a)}, \quad \widehat{h(a)} = e \iff h(a) = 0. \quad (\text{SH41})$$

This is exactly (SH27) followed by constant-term evaluation. The multiplicity is retained as well: the order of vanishing of $h(a + w)$ is the least n with nonzero amplitude coefficient, and every coefficient is unchanged on the time-zero row. None is replaced by a different number or a different local coordinate.

After localizing at e , every value in (SH41) maps to 1_L , regardless of whether $h(a)$ vanishes. The complete connecting receiver is the pair (p, χ) in (SH3); on the germ it recovers its actual complex amplitude and its full label. The failure of χ alone to distinguish zero values is precisely its kernel congruence (SH9). For a concrete test of that defect, the two initial polynomials $z^2 - 1$ and $z^2 + 1$ have different zero locations, while their supported values, formal heat families $z^2 - 1 + 2\sigma t$ and $z^2 + 1 + 2\sigma t$, and all their localized germs have the identical full-support projection 1_L . Their amplitude projections distinguish them exactly. This constructs the lost-information object and its recovery map rather than inferring any zero-location theorem from the support quotient.

The three inverse operations therefore have explicit, different receivers: local additive inverse stays in the fibre group; local multiplicative inverse changes the identity to its specified idempotent in the local monoid; universal unital inversion receives the full lattice via q_λ . Formal heat preserves their support maps and has the exact time-zero fibre (SH25). The supported boundary (SH28) and the coefficient closure (SH33) are additional calculated objects, with their amplitude maps proved. These results do not produce an off-real zero of the original initial function, nor a conclusion for or against the Riemann hypothesis.

8 The original escaping endpoint and the arithmetic heat family

EH1. Sources and exact scope. The two supplied versions of *The flip, the Fable fibre on the Riemann sphere, and Weil positivity* have identical SHA256 values; they are retained as `inputs/flip_a.txt` and `inputs/flip_b.txt`. Their supplied attribution is Claude (Anthropic), dated 23 September 2026. They are inputs to this calculation, not primary literature certificates. The original programme calculation of the escaping orbit is retained in `material_generator_endpoint.md` and `tex/inverse_fibre_heat.tex` in the 8 September fluid research edition. All identities used from that orbit are proved again below. Its time is written τ_f here to distinguish it from the adjoined semiring zero τ . This is an explicit renaming, not a change of its time scale. The original spatial third coordinate, called t in the polynomial, is written w . No coefficient is changed.

Rodgers–Tao’s original author source defines

$$\begin{aligned} H_t(Z) &= \int_0^\infty e^{tu^2} \Phi(u) \cos(Zu) du, & H_0(Z) &= \frac{1}{8} \xi \left(\frac{1}{2} + \frac{iZ}{2} \right), \\ \Phi(u) &= \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, & \xi(s) &= \frac{s(s-1)}{2} \pi^{-s/2} \Gamma(s/2) \zeta(s). \end{aligned} \quad (1)$$

These are equations `hoz`, `sas`, `phidef`, `htdef` in Rodgers–Tao, The de Bruijn–Newman constant is non-negative, original TeX lines 78–103. The factor $1/8$ in this definition is a nonzero scalar; the number $1/8$ in the escaping trajectory below is a time. Their different roles are resolved by an explicit map, not by matching the displayed numerals.

EH2. The original polynomial and its inverse-root chart. Let $F : \mathbb{C}_{x,y,w}^3 \rightarrow \mathbb{C}_{a,b,c}^3$ be

$$\begin{aligned} F_1 &= (1 + xy)^3 w + y^2(1 + xy)(4 + 3xy), \\ F_2 &= y + 3x(1 + xy)^2 w + 3xy^2(4 + 3xy), \\ F_3 &= 2x - 3x^2 y - x^3 w. \end{aligned}$$

On $x \neq 0$ set $v = 1 + xy$, $d = 2 - 3xy - x^2 w$, and $A = v^2 + v - v^3 d$, $B = 4v + 2 - 3v^2 d$. Then $F = (A/x^2, B/x, xd)$ and the coordinate Jacobian $\partial(x, v, d)/\partial(x, y, w)$ has determinant $-x^3$. The determinant of $\partial F/\partial(x, v, d)$ is x^{-3} times

$$\det \begin{pmatrix} -2A & 2v + 1 - 3v^2 d & -v^3 \\ -B & 4 - 6vd & -3v^2 \\ d & 0 & 1 \end{pmatrix} = 2.$$

Expansion reduces the last expression to $(4 - 6vd)(-2v^2 - 2v + 3v^3 d) + (2v + 1 - 3v^2 d)(4v + 2 - 6v^2 d) = 2$. Hence $\det DF = -2$ on a dense open set and therefore everywhere.

Put $P_{a,b,c}(r) = cr^3 - 2r^2 + br - 2a$. For $x \neq 0$, the exact inverse coordinates are

$$r = y + x^{-1}, \quad \alpha = x^{-1}, \quad (x, y, w) = (\alpha^{-1}, r - \alpha, 5\alpha^2 - 3r\alpha - c\alpha^3).$$

Substitution gives

$$F = (r^2 + r\alpha - cr^3, 4r + 2\alpha - 3cr^2, c).$$

Thus $F = (a, b, c)$ is equivalent to $P(r) = 0$ and $\alpha = P'(r)/2 \neq 0$. This proves both directions, including the excluded multiple-root locus. On $x = 0$ the formula is $F(0, y, w) = (w + 4y^2, y, 0)$, so the entire additional inverse point is $(0, b, a - 4b^2)$ when $c = 0$.

EH3. The escaping trajectory, with its original endpoint. For $\tau_f < 1/8$ put

$$z = \sqrt{1 - 8\tau_f} > 0, \quad \gamma(\tau_f) = (z^{-1}, -3z/2, 13z^2/2).$$

Direct substitution in the original polynomial gives

$$F(\gamma) = (-z^2/4, 0, 0) = (-1/4 + 2\tau_f, 0, 0), \quad r(\gamma) = -z/2, \quad \alpha(\gamma) = z.$$

Moreover $z' = -4/z$. Differentiating the displayed identity gives $DF(\gamma)\gamma' = (2, 0, 0)^\top$. For

$$U = 2(DF)^{-1}e_1 + 6F_3(DF)^{-1}e_2$$

this proves $\gamma' = U(\gamma)$, since $F_3(\gamma) = 0$ and DF is invertible. The trajectory has no finite affine limit because its first coordinate tends to infinity. Its target has the finite limit $(0, 0, 0)$. At that target the entire affine fibre is $(0, 0, 0)$, by the preceding inverse description: the double root $r = 0$ has $\alpha = 0$ and is excluded from the $x \neq 0$ chart. These are exact statements about this field; they require no claim concerning a separate Navier–Stokes solution.

EH4. The exact heat map and its time origin. Retain the incoming note's arithmetic coordinate $s = 1/2 + r$ and heat coordinate $Z = -2ir$. Define

$$h = 4\tau_f - \frac{1}{2} = 2a, \quad Q_h(Z) = Z^2 - 2h.$$

On $b = c = 0$ one has the polynomial identity

$$Q_h(Z) = 2P_{a,0,0}(r) \quad (Z = -2ir, \quad h = 2a).$$

Consequently $\partial_h Q_h = -2 = -\partial_Z^2 Q_h$, with exactly the backward heat sign of (1). Along the escaping branch $Z = iz$ and $h = -z^2/2$. The original times map as

$$\tau_f = 0 \mapsto h = -1/2, \quad \tau_f \uparrow 1/8 \mapsto h \uparrow 0.$$

There is no step identifying h with the absolute arithmetic time t of (1); the formula specifies the translated and scaled time of this particular polynomial solution completely. Its zeros are $\pm i\sqrt{-2h}$ for $h < 0$, a double zero at 0 for $h = 0$, and $\pm\sqrt{2h}$ for $h > 0$. Thus its all-real threshold is exactly $h = 0$, including the endpoint. The escape reaches that real double root. It does not leave a nonreal root at the endpoint.

The reciprocal of the escaping affine coordinate has an exact relation to the meromorphic heat-root velocity. On the same $b = c = 0$ chart, $P'(r) = -4r$, so the inverse formulas give

$$\alpha = -2r = -iZ, \quad x = \frac{i}{Z}, \quad \frac{dZ}{dh} = \frac{1}{Z}, \quad x = i\frac{dZ}{dh}. \quad (2)$$

Here the derivatives are along either root branch on $Z \neq 0$: differentiating $Z^2 = 2h$ proves $2Z dZ/dh = 2$. The original root coordinate and original flow time therefore satisfy, with no change of scale left implicit,

$$h = -2r^2, \quad \frac{dr}{dh} = -\frac{1}{4r}, \quad \frac{dr}{d\tau_f} = -\frac{1}{r}, \quad \frac{dh}{d\tau_f} = 4. \quad (3)$$

The ordered-pair rational energy of the two heat roots $Z, -Z$ is

$$\mathcal{E} = \frac{2}{(2Z)^2} = \frac{1}{2Z^2} = -\frac{1}{8r^2} = -\frac{x^2}{2}. \quad (4)$$

Each equality follows from $Z = -2ir$ and $x = i/Z$. In particular, on the original escaping branch this rational function is

$$\mathcal{E}(\gamma(\tau_f)) = -\frac{1}{2z^2} = -\frac{1}{2(1-8\tau_f)} \longrightarrow -\infty. \quad (5)$$

This is its analytic continuation along imaginary heat roots $Z = iz$. On the real-root branch $h > 0$ the same rational function equals $1/(4h) > 0$ instead. Thus the reciprocal bridge retains both the factor 4 in time and the negative sign on the incoming branch. These are calculations on Q_h ; the arithmetic time t in (1) has not been identified with h .

EH5. Retaining the collision algebra and its holonomy. The family

$$\mathcal{A} = \mathbb{C}[h, Z]/(Z^2 - 2h)$$

is free of rank two over $\mathbb{C}[h]$, with basis $(1, Z)$: division by the monic polynomial gives existence and uniqueness of this basis. Its zero fibre is $\mathbb{C}[\eta]/(\eta^2)$ by $Z \mapsto \eta$. For $h \neq 0$ its two roots are exchanged by one loop around $h = 0$, since $\sqrt{2h_0}e^{i\theta} = \sqrt{2h_0}e^{i\theta/2}$ for $0 \leq \theta \leq 2\pi$. On the two labelled roots this is the matrix

$$M_{\text{loop}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad M_{\text{loop}}^* M_{\text{loop}} = I, \quad M_{\text{loop}} \neq I.$$

This computes nontrivial *unitary* monodromy around a fibre whose only root is real. It supplies an explicit example against the implication “any nonidentity holonomy forces a nonreal zero”. The map from the family to this holonomy representation is the actual root covering over $\mathbb{C}^*$, not an asserted arithmetic spectral identification.

EH6. The form through the collision. For real h , conjugation on $\mathcal{A}_h$ fixes Z . The Hermitian trace form $q_h(f, g) = \text{Tr}_{\mathcal{A}_h/\mathbb{C}}(f\bar{g})$ has matrix $\text{diag}(2, 4h)$ in $(1, Z)$. Indeed multiplication by Z has matrix $\begin{pmatrix} 0 & 2h \\ 1 & 0 \end{pmatrix}$, so its trace is zero and the trace of Z^2 is $4h$. The signature is $(2, 0)$ for $h > 0$, $(1, 1)$ for $h < 0$, and a positive line plus the nilpotent radical at $h = 0$. In the original coordinate $r = iZ/2$ with involution $r \mapsto -r$, its matrix is $\text{diag}(2, h)$: multiplying the second basis vector by $i/2$ gives precisely that congruence. On point values the reflection $s \mapsto 1 - \bar{s}$ sends Z to $\bar{Z}$, because $s = 1/2 + iZ/2$. Its induced conjugate-linear coordinate-algebra involution, given by $f^\dagger(s) = \overline{f(1 - \bar{s})}$, fixes the formal generator $Z = -2i(s - 1/2)$:

$$Z^\dagger = 2i(1/2 - s) = -2i(s - 1/2) = Z, \quad r^\dagger = -r.$$

Every sign and the degeneracy at zero therefore agree in the two presentations. No strict positivity is asserted on the nilpotent radical.

EH7. The retained split boundary of the escaping chart. Let L be the original bounded distributive support lattice and let G_L denote the original split-support semiring. Let

$$B = G_L(\mathbb{C}[z]), \quad B' = G_L(\mathbb{C}[z, z^{-1}]), \quad C = G_L(\mathbb{C}),$$

where $B \rightarrow C$ is evaluation $z \mapsto e$ (supported amplitude zero). The chart map to the trajectory is the ring homomorphism

$$\mathbb{C}[x, y, w] \longrightarrow \mathbb{C}[z, z^{-1}], \quad (x, y, w) \longmapsto (z^{-1}, -3z/2, 13z^2/2),$$

and applying G_L gives its exact support-preserving semiring map. Since B' is the localization of B at the supported polynomial z , base change along $z \mapsto e$ proves

$$B' \amalg_B C \cong C[e^{-1}] \cong L. \tag{6}$$

For clarity, the second isomorphism follows directly: $e^2 = e$ makes the image of e equal to the unit when e is invertible. Then $ea = e$ for every supported amplitude a makes all those amplitudes equal to the top support, while each lower support label stays itself. The support character χ_L witnesses that no two distinct labels are identified. These equalities also prove the universal property. The construction has thus retained a whole support spectrum at the escaping endpoint. It has not created an affine complex value for z^{-1} at $z = 0$: any attempted nonzero

ring-valued base change would send the invertible supported e to ring zero and force $1 = 0$. The exact receiver is L , with all its labels, as in (6).

EH8. The actual arithmetic coefficients, without a finite-packet replacement. For every $u \geq 0$ each summand of Φ is strictly positive, since

$$2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u} = \pi n^2 e^{5u} (2\pi n^2 e^{4u} - 3) > 0.$$

For $a = \pi e^{4u} \geq \pi$, $\sum_{n \geq 1} n^4 e^{-an^2} \leq e^{-a/2} \sum_{n \geq 1} n^4 e^{-\pi n^2/2}$. Hence $0 < \Phi(u) \leq C e^{9u} e^{-\pi e^{4u}/2}$ with a finite constant C . This bound is integrable after multiplication by any $u^N e^{Tu^2 + Ru}$, for any fixed $N, T, R \geq 0$. Uniform domination on compact subsets of $(t, Z) \in \mathbb{C}^2$ therefore proves that (1) is entire jointly in (t, Z) , and permits all the following differentiations and series integrations.

Define the exact positive moments $M_j = \int_0^\infty u^{2j} \Phi(u) du > 0$ for $j \geq 0$. The full Taylor expansion, with every time and spatial coefficient retained, is

$$H_t(Z) = \sum_{m,n \geq 0} \frac{(-1)^n M_{m+n}}{m!(2n)!} t^m Z^{2n}. \quad (7)$$

Indeed the product of the absolute exponential and cosine Taylor series is bounded on $|t| \leq T, |Z| \leq R$ by $e^{Tu^2 + Ru}$, so Tonelli and dominated convergence prove the equality. All displayed coefficients are nonzero. All coefficients of odd powers of Z are zero. The identity $\partial_t H = -\partial_Z^2 H$ follows coefficient by coefficient, including the factorials, or directly under the integral.

The actual theta reciprocal can now be compared to (2), including its first discrepancy. Fix a real arithmetic time t and define $M_j(t) = \int_0^\infty u^{2j} e^{tu^2} \Phi(u) du$. The same domination proves that each moment is finite, and strict positivity of the integrand on $u > 0$ gives $M_j(t) > 0$. Taylor expansion at the spatial origin gives

$$H_t(Z) = M_0(t) - \frac{M_1(t)}{2} Z^2 + \frac{M_2(t)}{24} Z^4 + O(Z^6). \quad (8)$$

Consequently H_Z/Z is holomorphic and nonzero at $Z = 0$, with value $-M_1(t)$. It is a holomorphic unit in a neighbourhood of zero. Division in that punctured neighbourhood yields the exact identity and Laurent expansion

$$\frac{H_{ZZ}}{H_Z} - \frac{1}{Z} = \frac{ZH_{ZZ} - H_Z}{ZH_Z} = -\frac{M_2(t)}{3M_1(t)} Z + O(Z^3), \quad (9)$$

$$\frac{H_{ZZ}}{H_Z} = \frac{1}{Z} - \frac{M_2(t)}{3M_1(t)} Z + O(Z^3). \quad (10)$$

For completeness, differentiating (8) gives $H_Z = -M_1(t)Z + M_2(t)Z^3/6 + O(Z^5)$ and $H_{ZZ} = -M_1(t) + M_2(t)Z^2/2 + O(Z^4)$. Subtracting cancels the linear numerator terms and gives $ZH_{ZZ} - H_Z = M_2(t)Z^3/3 + O(Z^5)$; its denominator is $ZH_Z = -M_1(t)Z^2 + O(Z^4)$. This proves the coefficient and its negative sign in (9).

For the quadratic, $(Q_h)_{ZZ}/(Q_h)_Z = 1/Z$ identically wherever $Z \neq 0$. The theta correction just computed is nonzero because $M_2(t)/M_1(t) > 0$. Both quotients have a simple pole of residue one at the spatial origin, but $H_t(0) = M_0(t) > 0$ for every real t , including $t = 0$. Thus this pole is at a stationary point of the actual theta function, and is not a zero of that function. The quotient is a zero velocity only after restricting to an actual simple zero. The displayed identities compute the exact discrepancy between the two meromorphic quotients without identifying their zero spaces or their absolute time coordinates.

EH9. The exact coefficient-support defect for this particular source. Represent H_0 initially as a series independent of t , with top support on every even Z coefficient in time degree zero, and bottom support elsewhere. In the coefficient-labelled semiring $(G_L(\mathbb{C}))[[t, Z]]$, lift a nonzero scalar coefficient to its top-supported value and let the heat operators retain addition's

join of supports. The minus heat operator produces the coefficients in (7). Its support is top at every $(m, 2n)$ and bottom at every $(m, 2n + 1)$.

Applying the plus heat operator to this series cancels the amplitude of every positive time degree. At $(m, 2n)$ that amplitude is

$$\frac{(-1)^n M_{m+n}}{(2n)!} \sum_{j=0}^m \frac{(-1)^j}{j!(m-j)!},$$

which is zero for $m > 0$ by $(1 - 1)^m = 0$, and equals the original $(-1)^n M_n / (2n)!$ for $m = 0$. For $m > 0$ every summand has top support, so the cancelled coefficient is the supported zero e , not absence τ . All odd coefficients remain absent. This is a calculated support memory for the actual theta source. Evaluation at $t = 0$ retains exactly the original H_0 coefficients. The general closure and its inverse-on-fixed-support interpretation are proved in the accompanying semilattice chapter.

EH10. Why the support memory alone supplies neither sign of RH. The map evaluating a globally supported function at $(0, Z_0)$ is

$$G_L(\mathcal{O}(\mathbb{C}^2)) \longrightarrow G_L(\mathbb{C}), \quad \widehat{f} \longmapsto \widehat{f(0, Z_0)}.$$

It retains every lower support label. Applied to the original source, $\widehat{H}(0, Z_0) = e$ if and only if $H_0(Z_0) = 0$, equivalently $\xi(1/2 + iZ_0/2) = 0$. Applying χ_L afterwards maps both supported zero and every supported nonzero complex value to the same top label. Thus that additional map cannot test which Z_0 is a zero. This is proved by its displayed formula, without discarding the retained support boundary.

There is also an explicit test of the amount of information in (7)'s support. Take the positive measure δ_1 on $[0, \infty)$ in the same cosine-heat integral. It gives $K_t(Z) = e^t \cos Z$, all of whose even coefficients have exactly that same nonzero sign pattern for all m, n . Every zero is real for every real t , since $\cos Z = 0$ is equivalent to $e^{2iZ} = -1$ and hence $\Im Z = 0$. Alternatively $Q_t(Z) = Z^2 - 2t$ gives the preceding nontrivial collision and monodromy at a real-root time-zero fibre. These test families are connected to the actual family by the common linear integral operator $\mu \mapsto \int e^{tu^2} \cos(Zu) d\mu$; their measures and values are stated, not substituted for $\Phi(u)du$. They show that coefficient support, local invertibility at e , and collision monodromy, without the arithmetic moment values, do not force a failure of real-rootedness at zero.

EH11. The arithmetic threshold and the exact outstanding calculation. For the original H_t , the de Bruijn–Newman theorem gives real zeros exactly for $t \geq \Lambda$. Rodgers–Tao prove $\Lambda \geq 0$ (Theorem1.1; original TeX label `main`). Platt–Trudgian’s The Riemann hypothesis is true up to $3 \cdot 10^{12}$, section on the de Bruijn–Newman constant, original TeX lines135–149, proves the corollary $\Lambda \leq 0.2$ using D.H.J. Polymath, Table1. These cited results give $\text{RH} \iff \Lambda = 0$. The calculations above produce neither a nonreal zero of the actual H_0 nor a negative value of its whole Weil form. They identify the exact retained endpoint and support memory and the arithmetic evaluation map through which a proposed counterexample must pass. The assertion “RH is false because e is invertible” does not follow from these computations: the unital-localization receiver is L , and the precise local collision has a real double root. No verdict about RH is being inferred from the failure of that proposed implication.

9 Supported reciprocals in the actual zero-dynamics identities

RD1. The user’s proposed contradiction and the exact source. The claim being investigated is: supported zero is invertible, therefore the heat-flow monotonicity fails at time zero, therefore RH is false. The first step must use the inverse operation actually supplied by the

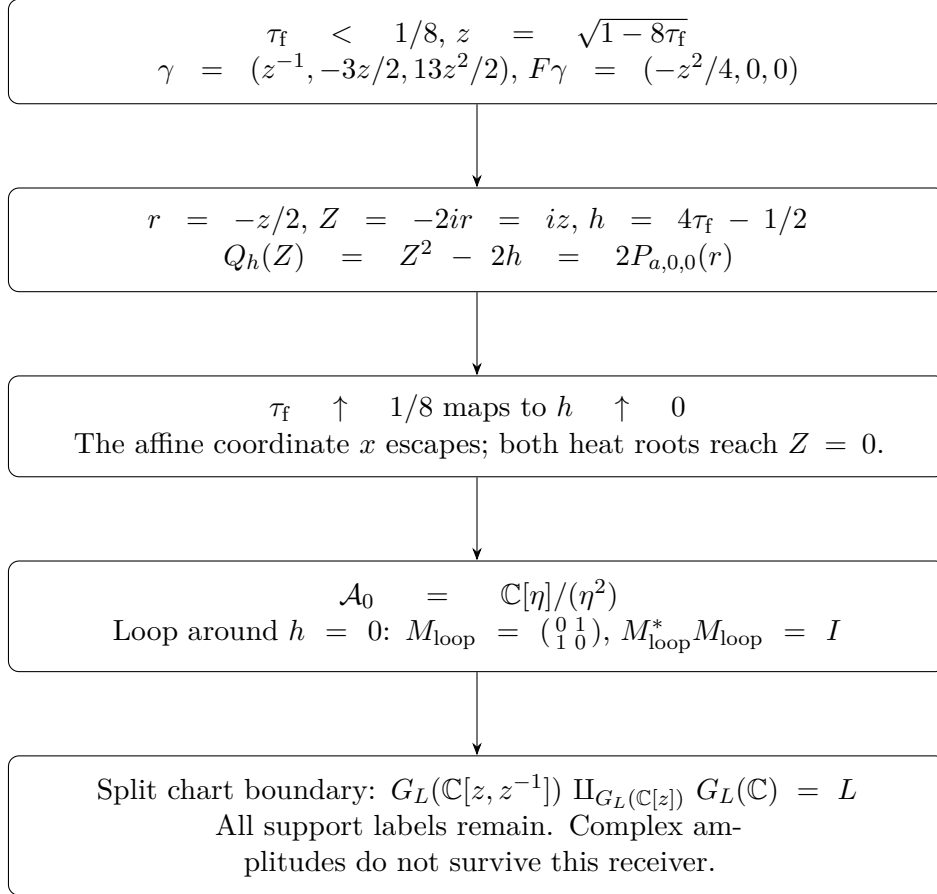


Figure 1: Exact endpoint maps of EH2–EH7. The final arrow means applying the split localization construction to the retained chart, not an identification of the dual-number algebra with L . The labels give the original coefficients and the complete time conversion. The heat coordinate convention is Rodgers–Tao’s; the polynomial orbit is the retained programme calculation rederived here.

semilattice. The second must refer to a specified monotonicity theorem. This section computes both, including a real failure of a totalized energy's monotonicity at a collision and its exact relation to heat.

The newly supplied `arXiv-1801.05914v5.tar.gz` is byte-identical to the retained Rodgers–Tao v5 original-author archive, SHA256 `cc3cd07ea418ec067bca6dce49ace0b4a2c67afe8d891fb82a1bdc5a1f63d6ed`. The actual passages read for this calculation are the definitions and proof setup, the full section *Dynamics of zeroes*, the gap propagation argument, and the full sections *Controlling the energy at time 0* and *Contradicting pair correlation*. Relevant original TeX labels are `ode`, `ident`, `tlar`, `tlar-iter`, and `picketfence`. The technical integrated energy argument is cited as part of their theorem; it is not claimed to have been reproduced here.

RD2. The original reciprocal, including supported zero. For $S = G_L(\mathbb{R})$ define

$$\widehat{a}^\dagger = \widehat{1/a} \quad (a \neq 0), \quad z_\lambda^\dagger = z_\lambda \quad (\lambda \in L).$$

This gives the multiplicative inverse in the inverse-monoid sense: $xx^\dagger x = x$, $x^\dagger xx^\dagger = x^\dagger$, and $(x^\dagger)^\dagger = x$. For nonzero amplitude these equations hold in its ordinary multiplicative group; for a support element they are the idempotence identities $z_\lambda^2 = z_\lambda$. Uniqueness follows as well. A nonzero-amplitude x forces the amplitude and top support of its inverse. If $x = z_\lambda$, the second inverse identity forces the inverse to have zero amplitude, say z_μ ; the two identities then give $\lambda \leq \mu$ and $\mu \leq \lambda$. In particular

$$e^\dagger = e, \quad ee^\dagger = e, \quad \widehat{aa}^\dagger = \widehat{1} \quad (a \neq 0).$$

This acknowledges supported zero's genuine inverse in its local multiplicative group. Replacing the middle right-hand side by $\widehat{1}$ would instead perform the unital localization $S \rightarrow L$. Both operations and their connecting maps are proved in the semilattice chapter.

RD3. Exact transport on the distinct-root locus. For a finite set of labelled real roots define $D_{jk} = x_j - x_k$ and suppose the displayed differences are nonzero. The map $a \mapsto \widehat{a}$ identifies $\mathbb{R}$ with the top additive group of S , whose identity is e . It preserves addition, multiplication, the unit and additive inverses relative to e . For every nonzero difference it also preserves the displayed reciprocal. Consequently every rational identity whose denominators are products of these differences has exactly the same lifted identity. This follows by induction on additions, multiplications and reciprocal operations in its expression. The projection $p : S \rightarrow \mathbb{R}$ is the inverse on that group. Differentiation along a top-fibre curve $\widehat{f(t)}$ means $\widehat{f'(t)}$, so the identities commute with differentiation too. Order on this fibre is the transported real order, $\widehat{a} \leq \widehat{b}$ exactly when $a \leq b$; no order is asserted between distinct lower support fibres.

For principal-value sums one first uses the exact finite identities, then takes the original amplitude limit. This defines the topology and the limit in the top fibre by the bijection p . It neither adds nor removes any summand. In particular, retaining e in the coefficient semiring changes none of the source's identities on their stated distinct-root domain.

RD4. The precise monotonicity identity, including its environment. Rodgers–Tao work under the contradiction assumption $\Lambda < 0$; their Theorem `dynam` states, for $\Lambda < t \leq 0$,

$$x'_k = 2 \sum'_{j \neq k} \frac{1}{x_k - x_j}.$$

The zeros in this range are real and simple, by the cited Csordas–Smith–Varga result. Their finite-set energy is $E^K = \sum_{k,k' \in K, k \neq k'} (x_k - x_{k'})^{-2}$, and Lemma `ident(iii)` states exactly

$$(E^K)' = \sum_{\substack{j \notin K \\ k, k' \in K, k \neq k'}} \frac{4}{(x_k - x_{k'})^2 (x_k - x_j) (x_{k'} - x_j)}$$

$$-2 \sum_{\substack{k, k' \in K \\ k \neq k'}} \left(\frac{2}{(x_k - x_{k'})^2} - \sum_{\substack{k'' \in K \\ k'' \neq k, k'}} \frac{1}{(x_{k''} - x_k)(x_{k''} - x_{k'})} \right)^2.$$

The external sum is present. A finite subsystem's energy is not asserted to decrease without controlling it. The preceding transport proof applies to every term, square, sign and cutoff here. In section 8, Proposition `tlar-iter` controls those external terms across time intervals at most $1/(100 \log^2 T)$, with an interval shrunk by $\log^3 T$ at each end. Supported-zero inversion changes none of those terms, because no denominator there is supported zero.

This reading also makes the logical direction exact. Their contradiction excludes $\Lambda < 0$, leaving both $\Lambda = 0$ and $\Lambda > 0$. The latter is entirely compatible with their theorem and is the desired RH-disproof direction. A failure of some new extension of energy monotonicity, by itself, does not establish $\Lambda > 0$; that inference requires a calculation on the original arithmetic family at its time zero, rather than on a separately extended function of colliding roots.

RD5. The full collision defect of the reciprocal substitution. The failure of cancellation at a collision is a computable object. For the map from labelled roots to coefficients,

$$\sigma(x_1, x_2) = (a, b) = (x_1 + x_2, x_1 x_2), \quad P(Z) = Z^2 - aZ + b,$$

backward heat gives the exact coefficient vector $(a', b') = (0, -2)$. Away from $x_1 = x_2$, the root velocities $x'_1 = 2/(x_1 - x_2)$, $x'_2 = 2/(x_2 - x_1)$ give precisely that vector after $D\sigma$. At (r, r) , however,

$$D\sigma_{(r,r)}(v_1, v_2) = (v_1 + v_2, r(v_1 + v_2)).$$

The cokernel is a line, explicitly received by $\ell_r(\dot{a}, \dot{b}) = \dot{b} - r\dot{a}$. Its value on the actual heat vector is -2 . This is a nonzero obstruction to any finite differentiable labelled-root lift at the collision. The semilattice generalized reciprocal sets both colliding velocities to the supported value e ; their amplitude vector is $(0, 0)$, whose image under $D\sigma$ is $(0, 0)$, leaving the same defect $(0, -2)$. Thus the full coefficient heat vector survives, while a proposed finite root vector fails by this explicitly calculated cokernel class.

The replacement object supplied by this obstruction is the ramified root space $\{(h, Z) : Z^2 = 2h\}$, where $h = Z^2/2$. Its tangent map to time vanishes at $Z = 0$, and the vector field lifting ∂_h on $Z \neq 0$ is $Z^{-1}\partial_Z$. It has a simple pole in this coordinate. Both the ramification and the pole are calculated exactly; neither is silently removed by assigning a finite reciprocal at zero.

RD6. A real loss of monotonicity after totalizing energy. For the same actual polynomial heat solution $Q_h(Z) = Z^2 - 2h$ and $h > 0$, its two roots have gap $d(h) = 2\sqrt{2h}$. Therefore its ordered-pair energy is

$$E(h) = \frac{2}{d(h)^2} = \frac{1}{4h}, \quad E'(h) = -\frac{1}{4h^2} < 0.$$

At $h = 0$ the gap is the supported zero e . If one evaluates the same expression by the generalized inverse in RD2, the assigned energy is $2e^2 = e$, of amplitude zero. Consequently the totalized function $E^\dagger(0) = 0$, $E^\dagger(h) = 1/(4h)$ for $h > 0$ is not nonincreasing on $[0, \infty)$: for every $h > 0$ it exceeds its value at zero. This proves a precise version of the user's proposed monotonicity failure.

It also determines its logical consequence. Every zero of Q_h is real for every $h \geq 0$, including the double zero at zero. Hence this failure of the extended energy's monotonicity is compatible with real-rootedness at time zero. It cannot serve as the implication from failure of that extended energy law to failure of the real-zero condition. The original energy has $\lim_{h \downarrow 0} E(h) = +\infty$; the generalized-inverse value is a different, discontinuous extension, and the two are related by the explicit reciprocal and amplitude maps.

The same distinction is visible without orders or inequalities. In the ramification coordinate $q := Z$ one has $h = q^2/2$ and $E = 1/(2q^2)$. Its Laurent germ has valuation -2 and leading coefficient $1/2$. The valued germ field $\mathbb{C}((q))$, with its valuation and residue of q^2E , retains both data; evaluation of regular functions at $q = 0$ does not extend to it as a nonzero ring homomorphism. The split special-fibre receiver of the localized chart is L as proved in (6). In the original coordinate $r = iZ/2$ the same identities are $h = -2r^2$ and $E = -1/(8r^2)$, as computed in (4). That receiver retains support; the Laurent valuation retains the pole order as additional data. This identifies exactly which boundary information is needed if an energy argument is to cross the collision.

RD7. The differential equation retained at supported time zero. At a simple zero of the actual entire function, differentiating $H_t(x(t)) = 0$ gives $x' = H_{ZZ}/H_Z$, by $\partial_t H = -H_{ZZ}$. At a double zero (t_0, x_0) where $H_{ZZ} \neq 0$, this equation in undivided form is

$$H_Z x' = H_{ZZ}.$$

Its value there has left amplitude zero for every finite velocity, and nonzero right amplitude. In the top split fibre it is the unequal pair e and $\widehat{H_{ZZ}}$. Making e a unit by localization sends both to the top support and thus solves only the support image of the equation. Applying the amplitude map before localization retains the displayed nonzero differential defect. These are exact maps between the two equations; no arithmetic sign or nonreal zero is obtained by their identification.

RD8. An actual arithmetic sign at the origin. The moments in (7) evaluate a sign without a finite-family replacement. For every real t put $M_j(t) = \int_0^\infty u^{2j} e^{tu^2} \Phi(u) du > 0$. The complete original source gives

$$H_t(0) = M_0(t) > 0, \quad H'_t(0) = 0, \quad H''_t(0) = -M_1(t) < 0,$$

and hence

$$(H'_t(0))^2 - H_t(0)H''_t(0) = M_0(t)M_1(t) > 0.$$

All derivatives and signs follow from the domination and positivity proved with (7). Thus neither spatial zero $Z = 0$ nor this curvature test at that spatial point supplies a negative witness at time $t = 0$. The spatial point and the time parameter are retained separately. No inference about the sign at other spatial points is made.

RD9. The heat shift, its time-zero fibre, and the full support lattice. This calculation tests the inverse-monoid operation while retaining the distinguished time-zero fibre. Write $c_h \in \mathbb{R}$ for the prescribed heat shift c , to distinguish it from the third target coordinate c of the original polynomial F . For every value of this retained parameter define

$$Q_t^{[c_h]}(Z) = Z^2 - 2(t - c_h) = Z^2 - 2t + 2c_h, \quad \partial_t Q_t^{[c_h]} = -\partial_Z^2 Q_t^{[c_h]} = -2. \quad (11)$$

In this test family t marks its time-zero fibre. Its use as a coordinate does not assert equality with the time of the particular arithmetic function H_t .

First keep c_h indeterminate. There is an exact isomorphism of $\mathbb{R}[c_h]$ -algebras

$$\theta : \mathbb{R}[c_h, h, Z] \longrightarrow \mathbb{R}[c_h, t, Z], \quad \theta(h) = t - c_h, \quad \theta(Z) = Z, \quad \theta^{-1}(t) = h + c_h. \quad (12)$$

The generator formulas compose to the identity and hence prove the isomorphism. They send $Z^2 - 2h$ to $Q_t^{[c_h]}$ and induce an isomorphism of the quotient root algebras. The two distinguished evaluations satisfy

$$\text{ev}_{t=0} \circ \theta = \text{ev}_{h=-c_h}, \quad \text{ev}_{t=c_h} \circ \theta = \text{ev}_{h=0}. \quad (13)$$

Thus the isomorphism retains the marked fibre by specifying its image; it does not send time zero to the collision unless $c_h = 0$. Parameter specialization to any real c_h commutes with these maps.

Here are the heat maps in their exact receiving category. Let $A = \mathbb{R}[c_h, t, Z]$. For a real increment δ set

$$\mathcal{H}_\delta f = \sum_{j \geq 0} \frac{(-\delta)^j}{j!} \partial_Z^j f, \quad \mathcal{H}_\delta Q_t^{[c_h]} = Q_{t+\delta}^{[c_h]}. \quad (14)$$

The sum is finite on each polynomial. This is an additive $\mathbb{R}[c_h, t]$ -linear map. Multiplying the two finite operator sums and collecting ∂_Z^{2n} gives the coefficient $\sum_{j=0}^n (-\delta)^j (-\epsilon)^{n-j} / (j!(n-j)!) = (-\delta - \epsilon)^n / n!$. Consequently $\mathcal{H}_\delta \mathcal{H}_\epsilon = \mathcal{H}_{\delta+\epsilon}$, and $\mathcal{H}_{-\delta}$ is its inverse under composition. The second equality in (14) follows because $\partial_Z^2 Q = 2$ and all higher even derivatives vanish. In particular $Q_t^{[c_h]}$ is exactly the formal heat image $\exp(-t\partial_Z^2)(Z^2 + 2c_h)$.

For every original bounded distributive lattice L retain the carrier

$$G_L(A) = \{(0, \lambda) : \lambda \in L\} \cup \{(f, 1_L) : f \in A\}, \quad e = (0, 1_L), \quad \tau = (0, 0_L).$$

Its addition and multiplication are respectively $(f, \lambda) + (g, \mu) = (f + g, \lambda \vee \mu)$ and $(f, \lambda)(g, \mu) = (fg, \lambda \wedge \mu)$. The lifted heat map is

$$\widetilde{\mathcal{H}}_\delta(f, \lambda) = (\mathcal{H}_\delta f, \lambda), \quad p \widetilde{\mathcal{H}}_\delta = \mathcal{H}_\delta p, \quad \chi_L \widetilde{\mathcal{H}}_\delta = \chi_L. \quad (15)$$

It is well defined because any input with support below 1_L has amplitude zero, which the additive map sends to zero. The displayed addition formula proves additivity, and scalar linearity proves $G_L(\mathbb{R}[c_h, t])$ -linearity. It fixes every z_λ , including e , and its inverse is $\widetilde{\mathcal{H}}_{-\delta}$. These are automorphisms of additive semimodules. They are not asserted to be semiring maps: for $\delta \neq 0$, $\mathcal{H}_\delta(Z)^2 = Z^2$ whereas $\mathcal{H}_\delta(Z^2) = Z^2 - 2\delta$.

In contrast, the coordinate isomorphism θ , parameter specialization, and both evaluations in (13) are ring homomorphisms. Each lifts to the unital semiring homomorphism $(f, \lambda) \mapsto (\varphi(f), \lambda)$, retaining every support label. This formula proves all corresponding squares commute. The joint map (p, χ_L) is injective, so these lifted maps retain exactly the amplitude data of the stated ring maps together with the whole original support lattice.

Now fix a real c_h . The root algebra and its time-zero fibre are

$$B_{c_h} = \mathbb{R}[t, Z] / (Z^2 - 2(t - c_h)), \quad B_{c_h} / (t) = \mathbb{R}[Z] / (Z^2 + 2c_h). \quad (16)$$

The lifted quotient $G_L(B_{c_h}) \rightarrow G_L(B_{c_h} / (t))$ identifies precisely pairs with the same support label and amplitude difference in tB_{c_h} . Indeed this follows separately from the two coordinates of its formula, and proves the asserted kernel congruence in both directions. It does not erase a support label or replace a supported zero by τ .

The real algebras in (16) can be identified completely. If $c_h < 0$, write $a = \sqrt{-2c_h} > 0$. Evaluation at a and $-a$ gives an isomorphism with $\mathbb{R} \times \mathbb{R}$: for any desired values u, v , the degree-one polynomial $(u + v)/2 + (u - v)Z/(2a)$ has those values, and the kernel is $(Z^2 - a^2)$ by division. If $c_h = 0$ the algebra is $\mathbb{R}[\eta]/(\eta^2)$, with $Z \mapsto \eta$. If $c_h > 0$, evaluation $Z \mapsto i\sqrt{2c_h}$ gives an isomorphism with $\mathbb{C}$ as a real algebra: the images of $1, Z$ form a real basis, and the relation is satisfied. These isomorphisms lift under G_L as before. They calculate the entire time-zero fibre, including its nilpotent when it is singular.

At the collision time $t = c_h$ the fibre is always $\mathbb{R}[\eta]/(\eta^2)$. This agrees with the exact ramification coordinate $q = Z$, $t = c_h + q^2/2$. The point-value receiver $G_L(\mathbb{C})$ is the same inverse monoid for every c_h , with the inverse of RD2 and $e^\dagger = e$. The collision algebra also retains its nonzero nilpotent η ; that element does not acquire a generalized inverse in the algebra. Indeed $\eta b \eta = 0 \neq \eta$ for every b , so the first inverse identity fails. After applying G_L the same amplitude calculation proves this failure for $\widehat{\eta}$. Thus the point-value inverse operation and the full singular fibre are related by their actual evaluation map $\eta \mapsto 0$; the multiplicity has not been discarded.

For real t the roots themselves are

$$\begin{cases} \pm i\sqrt{2(c_h - t)}, & t < c_h, \\ 0 \text{ with multiplicity two}, & t = c_h, \\ \pm \sqrt{2(t - c_h)}, & t > c_h. \end{cases} \quad \{t : Q_t^{[c_h]} \text{ has only real roots}\} = [c_h, \infty). \quad (17)$$

This proves that the threshold is exactly c_h and proves its behaviour at equality. At the marked time zero the roots are nonreal for $c_h > 0$, a real double root for $c_h = 0$, and distinct real roots for $c_h < 0$. The evaluation map

$$G_L(\mathbb{C}[c_h, t, Z]) \longrightarrow G_L(\mathbb{C}), \quad (f, \lambda) \longmapsto (f(c_h, t, Z), \lambda)$$

detects each root by the equality $\widehat{Q_t^{[c_h]}}(Z) = e$. Composing with χ_L sends that value, and every supported nonzero value, to 1_L . This gives the exact map that forgets the root condition; the joint receiver (p, χ_L) retains it.

Finally, on $t > c_h$ the pair gap is $d = 2\sqrt{2(t - c_h)}$ and its ordered-pair energy is

$$E_{c_h}(t) = \frac{1}{4(t - c_h)}, \quad E'_{c_h}(t) = -\frac{1}{4(t - c_h)^2}, \quad E_{c_h}^\dagger(c_h) = 0. \quad (18)$$

The last term is the amplitude of the supported value $\widehat{2}(d^\dagger)^2 = e$ at $d = e$; it is not the limit of the first term. Therefore the same failure of nonincreasing behaviour on $[c_h, \infty)$ occurs for every shift. The root-to-coefficient map still has heat vector $(0, -2)$, and at the collision $(0, 0)$ its differential has image $\{(v, 0) : v \in \mathbb{R}\}$. The cokernel functional is $\ell_0(\dot{a}, \dot{b}) = \dot{b}$, so the unresolved finite-velocity class is still exactly -2 . Time translation changes none of these local identities.

The same full lattice L , the same point-value inverse monoid, and the same genuine local inverse $e^\dagger = e$ therefore occur with every real threshold c_h . In particular, both $c_h = 0$ and $c_h > 0$ occur, so these inverse and support data alone do not choose between zero and a positive threshold, even after restricting attention to nonnegative thresholds. The singularity at the marked time zero is retained explicitly by (16); it is present precisely when $c_h = 0$. This is a test of the proposed inference from the inverse operation. The initial amplitudes $Z^2 + 2c_h$ are not the arithmetic amplitudes H_0 , whose moments and singularity question remain unchanged. No conclusion about that arithmetic singularity or RH is obtained by replacing those amplitudes with this family.

10 The source statements and the question they decide

There is no implication from an arbitrary nonidentity holonomy operator to failure of the Riemann hypothesis in the source results used here. The exact spectral implication is instead

$$\left| e^{L(\rho-1/2)} \right| \neq 1, \quad L > 0, \quad \rho \text{ an actual nontrivial zeta zero} \iff \Re \rho \neq \tfrac{1}{2}. \quad (\text{HW1})$$

The requirement that this is the actual zero representation is part of the formula, not an omitted identification. We construct its Mellin observation, full jets, primitive section, and Weil pairing below. We also construct the maps of the supported-zero boundary and the prime-orbit covers into the same arithmetic distribution.

The relevant original statements have the following exact scope. Connes–Consani, *Spectral Triples and Zeta Cycles* [1], Section 6.2, Theorem **spectralreal** (author-source lines 1378–1436), identify the spectrum on

$$\mathcal{H}(L) = (\Sigma_{eL} \mathcal{E}(\mathcal{S}_0^{\text{ev}}))^\perp \subset L^2(\mathbb{R}_+^\times / e^{L\mathbb{Z}}), \quad \mathcal{E}\phi(x) = x^{1/2} \sum_{n \geq 1} \phi(nx), \quad (\text{HW2})$$

with critical-zero ordinates satisfying $e^{i\gamma L} = 1$. Here $\mathcal{S}_0^{\text{ev}}$ imposes both $\phi(0) = 0$ and $\int_{\mathbb{R}} \phi = 0$. Their *Hochschild homology, trace map and zeta-cycles* [2], final theorem and Remark **remtauto** (lines 641–703), retain critical multiplicities in the scaling-site sheaf quotient and explicitly allow nontrivial nilpotent Jordan terms at multiple critical zeros. Their all-zero realization in

Proposition `laplacian`, evaluation map `evalmaprho`, and Corollary `rhequiv` (lines 423–476) retains all multiplicities and has scalar spectral parameter

$$\Delta(\rho) = (\rho - \tfrac{1}{2})^2 - \tfrac{1}{4}. \quad (\text{HW3})$$

The critical-cycle Hilbert space and this all-zero realization are different specified receivers, joined here by their explicit Mellin jet maps, not by discarding the noncritical coordinates.

Connes–Consani’s *Knots, primes and class field theory* [4], Proposition `mappingtorus`(iii) and Theorem `mappingtorus1`, explicitly prove nontrivial prime-orbit monodromy; their Theorem `main`(iii) identifies it with arithmetic Frobenius. Those are unconditional source results, not assertions that these monodromies refute RH. Finally, the Weil criterion concerns the full explicit formula on every admissible test. We use its original constants from [1], equations `bombieriexplicit1bis`, `bombieriexplicit2bis`, and `bombtest`, and its precise positivity statement from [3], Appendix `apppositivity`, Proposition `mainprop`. Weil’s contribution and Yoshida’s compact-test refinement are credited by that source. The bounded archimedean theorem `mainthmintro` in [3] requires support $[2^{-1/2}, 2^{1/2}]$ and vanishing Fourier values at $i/2$ and 0; it is not a universal holonomy criterion.

11 The actual arithmetic map, including every local unit

Fix

$$\begin{aligned} D &= -x\partial_x, \quad V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi \text{ even}, \phi(0) = \int_{\mathbb{R}} \phi = 0\}, \\ \mathcal{B} &= \{F \in C^\infty(\mathbb{R}_+^\times) : \sup_{x>0} |x^b D^k F(x)| < \infty \ (b \in \mathbb{Z}, k \geq 0)\}, \\ \Theta\phi(x) &= 2 \sum_{n \geq 1} \phi(nx), \quad Q = \mathcal{B}/\Theta V, \quad \mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}. \end{aligned} \quad (\text{HW4})$$

The Fourier transform on $\mathbb{R}$ has kernel $e^{-2\pi i x \xi}$. Poisson summation gives $\Theta\phi(x) = x^{-1} \widehat{\Theta\phi}(1/x)$ on V , so $\Theta V \subset \mathcal{B}$, with every Euler derivative included. Absolute summation for $\Re s > 1$, followed by continuation, gives

$$\mathcal{M}\Theta\phi(s) = 2\zeta(s)M_+\phi(s), \quad M_+\phi(s) = \int_0^\infty \phi(x) x^s \frac{dx}{x}, \quad \mathcal{M}DF(s) = s\mathcal{M}F(s). \quad (\text{HW5})$$

Evenness and $\phi(0) = 0$ give holomorphy of $M_+\phi$ for $\Re s > -2$; its value at 1 is zero. Thus HW5 annihilates the full multiplicity of every nontrivial zero, including all derivatives below that multiplicity.

Set $a(s) = s(s-1)$ and retain the entire completed function

$$g(s) = 2\xi(s) = a(s)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad g(1-s) = g(s), \quad g(\bar{s}) = \overline{g(s)}, \quad g(0) = g(1) = 1. \quad (\text{HW6})$$

Let Z be any finite set of its actual zeros, stable under $\iota\rho = 1 - \bar{\rho}$, with their full original orders m_ρ . For an arbitrary retained $c_h \neq 0$, define

$$\begin{aligned} h(s) &= c_h \prod_{\rho \in Z} (s - \rho)^{m_\rho}, \quad A_h = \mathbb{C}[s]/(h) = \prod_{\rho \in Z} \mathbb{C}[t_\rho]/(t_\rho^{m_\rho}), \quad t_\rho = s - \rho, \\ g(\rho + t) &= u_\rho(t)t^{m_\rho}, \quad u_\rho(0) = 2\lambda_\rho \neq 0, \quad U_h = g/h, \quad u_h = [U_h] \in A_h^\times. \end{aligned} \quad (\text{HW7})$$

No factor u_ρ , c_h , or higher Taylor coefficient is replaced. The original spectral observation is

$$\mathcal{J}_h : Q \longrightarrow A_h, \quad [F] \longmapsto j_h \mathcal{M}F, \quad (j_h H)_{\rho, k} = \frac{H^{(k)}(\rho)}{k!}, \quad 0 \leq k < m_\rho. \quad (\text{HW8})$$

HW5 proves it is well defined.

Lemma 11.1 (Exact primitive full-jet section). *The map HW8 is onto. The following formulas construct a section using the actual completed zeta function and all its local units:*

$$\begin{aligned} \phi_*(x) &= (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}, & M_+ \phi_*(s) &= \frac{1}{2}a(s)\pi^{-s/2}\Gamma(s/2), \\ R_u &= \text{rem}_h([a]^{-1}u_h^{-1}u), & S_\rho H(x) &= x^{-\rho} \int_x^\infty H(t)t^{\rho-1} dt, & S_h &= c_h^{-1} \prod_{\rho \in Z} S_\rho^{m_\rho}, \quad (\text{HW9}) \\ \text{Prim}_h(u) &= a(D)S_h\Theta(R_u(D)\phi_*), & \mathcal{M}\text{Prim}_h(u) &= aU_h R_u. \end{aligned}$$

The transform has full jet u on Z , vanishes to the full actual zero order off Z , and vanishes at $s = 0, 1$.

Proof. The Gaussian substitution $y = \pi x^2$ and the gamma recurrence give the displayed Mellin identity, including its factor $1/2$. Gaussian moments give $\phi_*(0) = 0$ and $\int \phi_* = 0$. Hence $\mathcal{M}\Theta\phi_* = g$, and $\mathcal{M}\Theta(R_u(D)\phi_*) = gR_u$. The inverse $[a]^{-1}$ exists because $0, 1 \notin Z$; u_h^{-1} is a full truncated-power-series inverse. The remainder representative has degree less than $\deg h$.

For $H \in \mathcal{B}$ with $\mathcal{M}H(\rho) = 0$, the defining integral for $S_\rho H$ can also be written $-x^{-\rho} \int_0^x H(t)t^{\rho-1} dt$. At infinity use the first formula and any sufficiently large negative power bound for H ; at zero use the second and any sufficiently large positive power bound. They prove arbitrary power bounds at both ends. Differentiation yields $(D - \rho)S_\rho H = H$, and then every Euler derivative has the same bounds. Integration by parts gives $\mathcal{M}S_\rho H = (\mathcal{M}H)/(s - \rho)$. The apparent singularity is removable by the zero assumption. Iterate this argument: division at a selected zero lowers its vanishing order by exactly one and preserves the orders at other selected zeros. Thus S_h is defined on gR_u , in any order. Mellin injectivity, obtained by Fourier injectivity on any vertical line, proves independence of that order. The result is $U_h R_u$, and application of $a(D)$ proves HW9. In A_h , $[a]u_h[R_u] = u$. At unselected zeros $h \neq 0$, so the factor $U_h = g/h$ retains the entire original zero order. Finally the factor a gives the two endpoint vanishings. $\square$

For $L \in \mathbb{R}$, retain the original unshifted dilation and its exact critical shift:

$$U_{eL}F(x) = F(e^{-L}x), \quad \mathcal{U}_L = e^{-L/2}U_{eL}, \quad T_L = M_{e^{L(s-1/2)}} : A_h \longrightarrow A_h. \quad (\text{HW10})$$

Substitution in the Mellin integral proves $\mathcal{M}\mathcal{U}_L F(s) = e^{L(s-1/2)}\mathcal{M}F(s)$. The same dilation preserves V and commutes with Θ , so it acts on Q , and

$$\mathcal{J}_h \mathcal{U}_L = T_L \mathcal{J}_h, \quad T_{\log p} = p^{-1/2} \Pi_p, \quad \Pi_p = M_{p^s}. \quad (\text{HW11})$$

Thus the relevant holonomy has an explicit arithmetic observation and every original prime factor remains present. The linear section HW9 need not commute on representatives: its exact discrepancy is

$$\mathcal{U}_L \text{Prim}_h(u) - \text{Prim}_h(T_L u) \in \ker(j_h \mathcal{M}). \quad (\text{HW12})$$

Its Mellin transform is exactly $aU_h(e^{L(s-1/2)}R_u - R_{T_L u})$. This records the defect, including its vanishing full jets. We assert no stronger identification with ΘV here.

12 Phase, modulus, and nilpotent transport

In the original local basis $1, t_\rho, \dots, t_\rho^{m_\rho-1}$ let $N_\rho = M_{t_\rho}$. All coefficients of T_L are

$$T_L|_\rho = e^{L(\rho-1/2)} \sum_{j=0}^{m_\rho-1} \frac{L^j N_\rho^j}{j!}, \quad (T_L u)_{\rho,k} = e^{L(\rho-1/2)} \sum_{j=0}^k \frac{L^{k-j}}{(k-j)!} u_{\rho,j}. \quad (\text{HW13})$$

Writing $\rho = \frac{1}{2} + \delta + i\gamma$, its scalar character is

$$e^{\delta L} e^{i\gamma L}, \quad \left| e^{L(\rho-1/2)} \right| = e^{\delta L}. \quad (\text{HW14})$$

For $L > 0$ its modulus is one exactly when $\delta = 0$; it is the identity scalar exactly when also $\gamma L \in 2\pi\mathbb{Z}$. The full $T_L|_\rho$ is the identity exactly when these two conditions hold and $m_\rho = 1$. Indeed

$$e^{LN_\rho} - 1 = N_\rho \left(L + \frac{L^2 N_\rho}{2!} + \dots \right) \quad (\text{HW15})$$

has a unit second factor for $L \neq 0$. Thus its kernel and image are precisely $\ker N_\rho$ and $\text{im } N_\rho$. In particular a multiple critical zero can have nontrivial unipotent transport while satisfying HW1 with modulus one.

Proposition 12.1 (The metric obstruction and its exact quotient). *For a fixed $L > 0$, the full packet admits a positive definite Hermitian form invariant under T_L exactly when every $\rho \in Z$ is critical and every $m_\rho = 1$. The quotient*

$$A_h \longrightarrow A_h/\mathfrak{n}_h = \mathbb{C}^Z, \quad \mathfrak{n}_h = \prod_{\rho \in Z} (t_\rho), \quad (\text{HW16})$$

always retains the scalar characters and removes precisely the nilpotent layers. Its operator has an invariant positive definite Hermitian form exactly when every selected zero is critical.

Proof. An invariant positive definite form gives the same norm to $T_L^n v$ for every integer n . For an eigenvector of eigenvalue q , this forces $|q| = 1$. If $m_\rho > 1$, apply T_L^n to 1 in that local factor: its coefficient of $t_\rho^{m_\rho-1}$ is $q^n (nL)^{m_\rho-1} / (m_\rho - 1)!$. For $|q| = 1$ this is unbounded, while every coefficient functional is bounded in a finite-dimensional positive norm. This contradicts invariance. Conversely, when all $m_\rho = 1$ and all scalar moduli are one, the coordinate counting metric is invariant. On the quotient the same proof omits the nilpotent calculation and proves the stated equivalence. $\square$

For clarity, flat circle holonomy and periodic descent can be written without an identification by name. On the logarithmic line a scalar solution $v(t) = e^{\alpha t} v(0)$ has

$$v(t + L) = e^{\alpha L} v(t). \quad (\text{HW17})$$

It defines a section of the flat line with that multiplier. The constant fibre Hermitian metric descends through this gluing exactly when $|e^{\alpha L}| = 1$. Ordinary periodic functions require the stronger condition $e^{\alpha L} = 1$. Reversing pullback orientation replaces the multiplier by its inverse, leaving both criteria unchanged. With $\alpha = i\gamma$, any L for which $\gamma L \notin 2\pi\mathbb{Z}$ gives nonidentity unitary holonomy. At $L = 0$, $T_0 = 1$ for every scalar and every jet; a loop around a parameter called zero is not this zero-time operator.

Here is the direct arithmetic content of HW2. For the Fourier character $x^{-i\gamma}$ on the circle, inverse-pullback scaling by e^b has eigenvalue $e^{i\gamma b}$. Descent requires $e^{i\gamma L} = 1$. Unfolding its Hermitian pairing with $\Sigma_{eL} \mathcal{E}\phi$ gives

$$\int_0^\infty x^{i\gamma} \mathcal{E}\phi(x) \frac{dx}{x} = \zeta\left(\frac{1}{2} + i\gamma\right) M_+ \phi\left(\frac{1}{2} + i\gamma\right). \quad (\text{HW18})$$

First prove this identity where the sum is absolutely convergent; Poisson and the two zero moments continue it to the displayed line. Periodization converges there by rapid decay at both ends, so unfolding is legitimate. With the retained ϕ_* of HW9, the second factor is nonzero on that line: $a(\frac{1}{2} + i\gamma) \neq 0$, the gamma function has no zeros, and its argument has positive real part. Thus orthogonality for every ϕ is equivalent to the critical zero condition. Fourier series on the compact circle give all its characters, proving the exact source criterion with no inference that nonidentity phase is forbidden at other lengths.

13 The original residue and the Jacobian Weil trace

Define $u^\dagger(s) = \overline{u(1 - \bar{s})}$ on the reflected packet. Retain

$$R_h(u, v) = \sum_{\rho \in Z} \text{Res}_\rho \frac{u^\dagger(s)v(s)}{g(s)} ds, \quad J_g = M_{g'(s)}. \quad (\text{HW19})$$

The coefficient matrix, with row $t_{\iota\rho}^j$ and column t_ρ^k , is exactly

$$(-1)^j [t^{m_\rho-1-j-k}] u_\rho(t)^{-1}. \quad (\text{HW20})$$

Its antidiagonal entries are $(-1)^j/(2\lambda_\rho) \neq 0$; hence the raw pairing is perfect on all original jets. Reflection sends a residue to minus its complex conjugate, so $R_h(u, v) = -\overline{R_h(v, u)}$. There is no assertion that this skew Hermitian residue is a positive definite Hilbert metric.

The logarithmic derivative gives the exact trace:

$$T_h(u, v) = R_h(u, J_g v) = \sum_{\rho \in Z} m_\rho \overline{u(\iota\rho)} v(\rho), \quad (\text{HW21})$$

$$\ker J_g = \mathfrak{n}_h, \quad \text{rad } T_h = \mathfrak{n}_h.$$

Indeed $g'/g = m_\rho/t + u'_\rho/u_\rho$, and locally $g'v = m_\rho u_\rho(0)v(\rho)t^{m_\rho-1}$ modulo t^{m_ρ} . These identities prove every assertion, including the full radical. The multiplier in HW10 obeys

$$(e^{L(s-1/2)})^\dagger = e^{-L(s-1/2)}, \quad R_h(T_L u, T_L v) = R_h(u, v), \quad T_h(T_L u, T_L v) = T_h(u, v). \quad (\text{HW22})$$

This is true even for offcritical zeros and nilpotent jets. Before the shift the two pairings are e^L -similitudes, in particular p -similitudes for the original Π_p .

At a fixed reflected zero the value block is (m_ρ) . At a nonfixed reflected pair it is

$$W_{\rho, \iota\rho} = m_\rho \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad T_L = \begin{pmatrix} q & 0 \\ 0 & 1/\bar{q} \end{pmatrix} \text{ on values, } \quad q = e^{L(\rho-1/2)}. \quad (\text{HW23})$$

Both signs are present: values $(1, 1)$ have norm $2m_\rho$ and values $(1, -1)$ have norm $-2m_\rho$. Consequently T_h is positive semidefinite exactly when every selected zero is critical, irrespective of multiplicity. For a critical multiple zero, HW22 and the degenerate positive form coexist with HW13's nontrivial unipotent factor. The perfect residue still records all the lost derivative data.

14 The whole explicit formula and an attained negative test

For $F, H \in \mathcal{B}$, set

$$\mathcal{W}(F, H) = \sum_{\rho \text{ actual}} m_\rho \overline{\mathcal{M}F(1 - \bar{\rho})} \mathcal{M}H(\rho). \quad (\text{HW24})$$

This sum and its continuity require no hypothesis on the positions inside the original critical strip. Here is a complete convergence argument. Let $H_* = \Theta\phi_*$. Its Gaussian sum satisfies $|H_*(x)| \leq Cx^4 e^{-\pi x^2}$ for $x \geq 1$, since $\sum_{n \geq 1} n^4 e^{-\pi(n^2-1)} < \infty$. The Fourier identity for the seed and Poisson give $H_*(x) = x^{-1} H_*(1/x)$. Thus

$$g(s) = \int_1^\infty H_*(x)(x^s + x^{1-s}) \frac{dx}{x}, \quad \log \max_{|s| \leq R} |g(s)| \leq C(R+1) \log(R+2). \quad (\text{HW25})$$

For the bound substitute $y = \pi x^2$, bound the resulting gamma integral by a constant times $n!$ with $n = \lceil (R+5)/2 \rceil$, and use $n! \leq n^n$. To count zeros, take $2R < r < 3R$ with a zero-free circle, factor the finitely many zeros in that disc out of g , and apply the mean-value identity

to the logarithmic modulus of the remaining nonvanishing holomorphic factor. The average of $\log |re^{i\theta} - \rho|$ is $\log r$, by its convergent logarithmic power series. Since $g(0) = 1$, this is Jensen's circle identity [5], and it yields

$$\sum_{|\rho| < r} m_\rho \log(r/|\rho|) = \frac{1}{2\pi} \int_0^{2\pi} \log |g(re^{i\theta})| d\theta. \quad (\text{HW26})$$

Every zero with $|\rho| \leq R$ contributes at least $\log 2$. Hence their total multiplicity is $O((R+1)\log(R+2))$. Repeated integration by parts in $\log x$, using HW4, bounds $\mathcal{M}F(\sigma + it)$ by every inverse power of $1 + |t|$, uniformly for $0 \leq \sigma \leq 1$, continuously in finitely many seminorms of F . Dyadic summation proves absolute convergence and continuous sesquilinearity of HW24.

Here are all signs in the explicit formula. Write

$$b_F(x) = x^{1/2}F(x), \quad b_H(x) = x^{1/2}H(x), \quad K = b_F^* * b_H, \quad b^*(x) = \overline{b(x^{-1})}, \quad (\text{HW27})$$

where convolution uses dx/x . Define

$$\begin{aligned} P(K) &= \sum_p \sum_{k \geq 1} (\log p) p^{-k/2} (K(p^k) + K(p^{-k})), \\ W_{\mathbb{R}}(K) &= (\log 4\pi + \gamma_E) K(1) \\ &\quad + \int_1^\infty (K(x) + K(x^{-1}) - 2x^{-1/2}K(1)) \frac{x^{1/2}}{x - x^{-1}} \frac{dx}{x}, \\ P_e(F, H) &= \overline{\mathcal{M}F(1)} \mathcal{M}H(0) + \overline{\mathcal{M}F(0)} \mathcal{M}H(1). \end{aligned} \quad (\text{HW28})$$

The full Guinand–Weil explicit formula in the Connes–Consani conventions cited above is

$$\mathcal{W}(F, H) = P_e(F, H) - W_{\mathbb{R}}(K) - P(K). \quad (\text{HW29})$$

For compact smooth tests this is the cited human source result: the substitution $F = x^{-1/2}b_F$ identifies its transform at every actual zero, with reflection $1 - \bar{\rho}$. In particular the endpoint product at $s = 0$ is $\overline{\mathcal{M}F(1)} \mathcal{M}H(0)$, and the product at $s = 1$ is its opposite endpoint pairing; these are not squared absolute endpoint norms.

All terms extend continuously to $\mathcal{B}$. In logarithmic coordinates $b_F(e^t)$ and all derivatives decay faster than $e^{-A|t|}$ for every fixed A ; their convolution has the same property, by splitting $|u| + |t - u| \geq |t|$. The prime sum is then bounded by convergent sums of the form $\sum_{n \geq 2} (\log n) n^{-A}$, because prime powers are integers and each has at most $\log n / \log 2$ presentations even without unique factorization. Near $x = 1$, the numerator in $W_{\mathbb{R}}$ is $O(\log x)$, and the denominator is of first order, so that integral is continuous in the needed local derivatives. At infinity the same rapid decay applies. Cutting off in $\log x$ now approximates any element of $\mathcal{B}$ in every seminorm; the continuous zero sum already proved therefore extends HW29.

Theorem 14.1 (Exact global bridge). *For all $u, v \in A_h$,*

$$\mathcal{W}(\text{Prim}_h(u), \text{Prim}_h(v)) = T_h(u, v) = R_h(u, J_g v). \quad (\text{HW30})$$

Every actual offcritical zero gives a compact smooth primitive test on which the complete Weil form is strictly negative. Conversely, a strictly negative value of that complete form on an admissible test implies at least one actual offcritical zero. Nonidentity unitary phase, higher critical jets, and the two boundary weights do not satisfy that implication by themselves.

Proof. HW9 fixes all selected jets and makes the full transform vanish to each unselected actual zero order. Therefore HW24 reduces exactly to HW21; no unselected term is dropped without its computed zero value.

For an actual nonfixed reflection pair of multiplicity m , choose Z to consist of that pair, and choose u with values $(1, -1)$ and all positive-order Taylor coefficients zero. The section $F = \text{Prim}_h(u)$ is primitive and $\mathcal{W}(F, F) = -2m$. To obtain a compact primitive test, choose $\chi \in C_c^\infty(\mathbb{R})$ equal to one near zero and put $F_n(x) = \chi(\log x/n)F(x)$. The product rule and the arbitrary power bounds in HW4 prove $F_n \rightarrow F$ in $\mathcal{B}$. Pick a nonnegative nonzero $\psi \in C_c^\infty(\mathbb{R}_+^\times)$ and $r > 1$. If $A = \mathcal{M}\psi(0) > 0$, $B = \mathcal{M}\psi(1) > 0$, the two compact functions $\psi, U_r\psi$ have endpoint matrix $\begin{pmatrix} A & A \\ B & rB \end{pmatrix}$ of determinant $AB(r-1) > 0$. Let v_0, v_1 be its inverse linear combinations giving endpoint pairs $(1, 0), (0, 1)$. Then

$$\tilde{F}_n = F_n - \mathcal{M}F_n(0)v_0 - \mathcal{M}F_n(1)v_1 \quad (\text{HW31})$$

is compact, smooth, and primitive, and converges to F in $\mathcal{B}$. By continuity its full Weil value tends to $-2m$, so is strictly negative for all sufficiently large n . This gives the test required by the criterion using the exact arithmetic map and the original local units. For the converse, if all actual zeros are critical, every summand in HW24 on the diagonal is $m_\rho |\mathcal{M}F(\rho)|^2 \geq 0$. Absolute convergence proves nonnegativity, excluding a negative test. The final assertion follows from HW13–23 and the boundary calculation below. $\square$

15 Supported zero, the endpoint representation, and inversion

The full supported theta sum and its punctured part are

$$\Theta_e\phi(x) = \sum_{n \in \mathbb{Z}} \phi(nx), \quad P\phi(x) = \sum_{n \neq 0} \phi(nx), \quad a_\phi = \phi(0), \quad b_\phi = \int_{\mathbb{R}} \phi. \quad (\text{HW32})$$

The term $n = 0$ occurs exactly once and equals a_ϕ . Poisson summation, after removing that term on both sides, gives

$$P\phi(x) = x^{-1}P\hat{\phi}(1/x) + b_\phi/x - a_\phi. \quad (\text{HW33})$$

Splitting the Mellin integral at one proves

$$\begin{aligned} \mathcal{M}P\phi(s) &= \int_1^\infty P\phi(x)x^s \frac{dx}{x} + \int_1^\infty P\hat{\phi}(x)x^{1-s} \frac{dx}{x} \\ &\quad + \frac{b_\phi}{s-1} - \frac{a_\phi}{s}. \end{aligned} \quad (\text{HW34})$$

The two integrals are entire by rapid decay. Thus supported evaluation and its Fourier dual give the original residues $-a_\phi$ at 0 and b_ϕ at 1, with their signs.

For the full bounded distributive lattice Λ , keep

$$G_\Lambda(R) = \{(r, 1_\Lambda) : r \in R\} \cup \{z_\lambda = (0, \lambda) : \lambda \neq 1_\Lambda\}, \quad \tau = z_{0_\Lambda}, \quad e = z_{1_\Lambda}, \quad (\text{HW35})$$

with componentwise addition $(r+s, \lambda \vee \mu)$ and multiplication $(rs, \lambda \wedge \mu)$. In the global fixed-sector carrier use the original space

$$\mathcal{F}_\Lambda = \mathcal{S}^{\text{ev}}(\mathbb{R}) \oplus \bigoplus_{\lambda \neq 1_\Lambda} \mathbb{C}_\lambda. \quad (\text{HW36})$$

This is the fixed-sector space, not every stalk of the full semilocal coefficient sheaf. Its exact endpoint sequence is

$$0 \longrightarrow V \longrightarrow \mathcal{F}_\Lambda \xrightarrow{\beta_\Lambda} B_\Lambda = \mathbb{C}(0)_e \oplus \mathbb{C}(1)_{e^\vee} \oplus \bigoplus_{\lambda \neq 1_\Lambda} \mathbb{C}(0)_\lambda \longrightarrow 0, \quad (\text{HW37})$$

where $\beta_\Lambda(\phi, c) = (a_\phi, b_\phi, c)$; the inclusion of V has all lower-label coordinates zero. It is onto: choose an even Schwartz function nonzero at zero, and an even Schwartz function vanishing

there with nonzero integral; their endpoint matrix is triangular and invertible, and the remaining coordinates are independent. Dilation $\phi(x) \mapsto \phi(e^{-L}x)$, fixing each c_λ , has endpoint matrix $\text{diag}(1, e^L, 1, \dots, 1)$. The same critical shift as HW10 therefore gives

$$T_L^{\text{bd}} = \text{diag}(e^{-L/2}, e^{L/2}, e^{-L/2}, \dots, e^{-L/2}). \quad (\text{HW38})$$

The top endpoint pair preserves $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Its offunit scalar moduli occur unconditionally for $L > 0$; they belong to the residues in HW34, not to the zeros in HW7. This is an actual endpoint map, so the arithmetic distinction is expressed by an exact sequence rather than an absence of any relation.

The two zero notions also have distinct operator meanings. Precomposition by multiplication by supported zero on $\mathcal{S}(\mathbb{R}) + \mathbb{C}1$ is the idempotent

$$E\phi = \phi(0)1, \quad E^2 = E, \quad \widehat{E}(\psi + c\delta_0) = \left(\int_{\mathbb{R}} \psi + c \right) \delta_0 \quad (\text{HW39})$$

on its Fourier image $\mathcal{S}(\mathbb{R}) \oplus \mathbb{C}\delta_0$. The second identity follows from $\widehat{1} = \delta_0$. Neither idempotent is an invertible transport on the whole displayed space. If an idempotent endomorphism is made invertible, it becomes the identity, and its universal linear receiver is its image: the quotient

$$X \longrightarrow X / \ker E \cong \text{im } E, \quad [x] \longmapsto Ex. \quad (\text{HW40})$$

Indeed any intertwining map into a space where $E = 1$ satisfies $f(x) = f(Ex)$, factors uniquely through that map, and $\text{im } E$ has $E = 1$. For HW39 this is the constant line, not a newly obtained nontrivial-zero eigenspace.

The same calculation on the entire lattice gives

$$G_\Lambda(R)[e^{-1}] \cong \Lambda, \quad (r, \lambda) \longmapsto \lambda. \quad (\text{HW41})$$

Proof: an invertible idempotent is one; multiplication by e sends (r, λ) to $(0, \lambda)$, so any map inverting e factors through this support projection. The lattice with unit 1_Λ realizes that factorization, proving its universal property. Every label is retained, including τ and the supported top zero.

For the actual packet define its supported semimodule

$$\mathbf{A}_\Lambda = \{(u, 1_\Lambda) : u \in A_h\} \cup \{(0, \lambda) : \lambda \neq 1_\Lambda\}. \quad (\text{HW42})$$

Addition uses support join, and the action of $(c, \lambda) \in G_\Lambda(\mathbb{C})$ uses scalar multiplication and support meet. This is closed because a nontop element has zero amplitude. The lifted arithmetic action and its precise localization are

$$\widetilde{T}_L(u, \lambda) = (T_L u, \lambda), \quad e\mathbf{A}_\Lambda = \{(0, \lambda) : \lambda \in \Lambda\} \cong \Lambda, \quad \chi\widetilde{T}_L = \chi. \quad (\text{HW43})$$

The map is additive and scalar-linear by HW10, including all lower-label zero cases; its inverse is $\widetilde{T}_{-L}$. The localization identifies amplitudes with equal support: its fibre at top is the full A_h , and every lower fibre is a singleton. It therefore has identity action on the localized labels for every T_L , whether critical, offcritical, or nonsemisimple. For an ordinary additive-group coefficient module, e already acts as zero because $e + e = e$ and additive cancellation forces $ev = 0$. Inverting it forces that coefficient module to zero. These are two exact receivers, with different categories and explicitly given maps.

$$\begin{array}{ccccc}
\mathcal{B}/\Theta V & \xrightarrow{\mathcal{J}_h} & A_h & \longrightarrow & A_h/\mathfrak{n}_h \\
\mathcal{U}_L \downarrow & & \downarrow T_L & & \downarrow \text{diag}(e^{L(\rho-1/2)}) \\
\mathcal{B}/\Theta V & \xrightarrow{\mathcal{J}_h} & A_h & \longrightarrow & A_h/\mathfrak{n}_h
\end{array}$$

Figure 2: Actual arithmetic scaling and its reduced quotient. HW8–16 prove the arrows and both commuting squares. The full-unit primitive section is HW9; the Jacobian trace descends along the last arrow by HW21. Independently, HW43 retains every lattice label and records the localization which erases amplitudes. These are exact diagrams, not identifications of every support stalk with one vector space.

16 Actual prime holonomy and its trivial-character observation

We now retain the distinct arithmetic prime-circle holonomy of [4]. Let

$$K = \widehat{\mathbb{Z}}^\times, \quad \chi : K \rightarrow G, \quad X^\chi = (Y_{\mathbb{Q}} \times G)/K, \quad (z, g) \sim (zw, \chi(w)g), \quad (\text{HW44})$$

where G is finite abelian. At prime p , put

$$I_p = \chi(\mathbb{Z}_p^\times), \quad \tilde{p}_p = 1, \quad \tilde{p}_\ell = p \ (\ell \neq p), \quad g_p = \chi(\tilde{p}), \quad \ell_p = \log p. \quad (\text{HW45})$$

The source’s mapping-torus theorem gives, in the natural balanced fibre coordinate, the oriented model

$$((G/I_p) \times \mathbb{R})/((g, t) \sim (g_p^{-1}g, t + \ell_p)). \quad (\text{HW46})$$

Indeed before the finite quotient the relation is $(h, t) \sim (ph, t + \ell_p)$; reducing the compact coordinate gives $g_0 = \chi(h)^{-1}g$, which changes to $g_p^{-1}g_0$. Positive first return, expressed at $t = 0$, is multiplication by g_p . Fibre inversion turns this into the source’s displayed generator convention and reverses deck characters, explaining the two orientations. At unramified primes $g_p = \text{Frob}_p$ by the source’s Theorem `main(iii)`.

For a finite-dimensional unitary representation $r : G \rightarrow U(V_0)$, put

$$P_{I_p} = \frac{1}{|I_p|} \sum_{i \in I_p} r(i), \quad V_p = V_0^{I_p}, \quad T_p = r(g_p)|_{V_p}. \quad (\text{HW47})$$

Finite summation proves P_{I_p} is a self-adjoint projection onto exactly the invariant subspace. Thus T_p is unitary; its eigenvalues are roots of unity. They can be nonidentity without having nonunit modulus. Sections with inverse-pullback flow have the exact model

$$v(t + \ell_p) = T_p^{-1}v(t), \quad U_p(b)v(t) = v(t - b), \quad U_p(\ell_p) = T_p. \quad (\text{HW48})$$

For $\nu \in C_c^\infty(\mathbb{R})$, Fourier expansion on each eigenline of T_p yields

$$\text{Tr} \int_{\mathbb{R}} \nu(b) U_p(b) db = \ell_p \sum_{k \in \mathbb{Z}} \text{Tr}(T_p^k) \nu(k\ell_p). \quad (\text{HW49})$$

To verify the sign, for $T_p = e^{i\theta}$ the basis is $\ell_p^{-1/2} e^{i(2\pi n - \theta)t/\ell_p}$. Integration gives $\widehat{\nu}((2\pi n - \theta)/\ell_p)$ with the minus Fourier convention. The periodic Dirac series gives $\sum_n e^{-2\pi i n b/\ell_p} = \ell_p \sum_k \delta(b - k\ell_p)$; after multiplying by $\nu(b)e^{i\theta b/\ell_p}$ it is HW49. Rapid Fourier decay proves trace-class convergence for this single circle. This does not assert trace class for the direct sum of all prime circles.

For a logarithmic test $q \in C_c^\infty(\mathbb{R})$ with transform $\mathcal{L}q(s) = \int q(t)e^{-(s-1/2)t} dt$, choose an even smooth κ which vanishes near zero and equals one when $|t| \geq \frac{1}{2} \log 2$. Apply HW49 to

$\nu_q(t) = \kappa(t)e^{-|t|/2}q(-t)$. In $\mathbb{C}[G]$, before any character projection, the resulting prime distribution is

$$\mathcal{P}_G(q) = \sum_{p,k \geq 1} (\log p) p^{-k/2} \{E_{I_p} g_p^k q(-k \log p) + E_{I_p} g_p^{-k} q(k \log p)\}, \quad E_I = \frac{1}{|I|} \sum_{i \in I} i. \quad (\text{HW50})$$

Every term and orientation follows from HW49, and the sum is finite on this compact test. Taking $\text{Tr } r$ gives the representation-valued prime trace including inertia. The cutoff has no effect at a nonzero prime iterate.

The augmentation $\epsilon : \mathbb{C}[G] \rightarrow \mathbb{C}$ has exact section $c \mapsto cE_G$, and

$$\mathcal{P}_G(q) = P_{\text{fin}}(q)E_G + (1 - E_G)\mathcal{P}_G(q), \quad \epsilon \mathcal{P}_G(q) = P_{\text{fin}}(q), \quad (\text{HW51})$$

where

$$P_{\text{fin}}(q) = \sum_{p,k \geq 1} (\log p) p^{-k/2} (q(-k \log p) + q(k \log p)). \quad (\text{HW52})$$

Proof: $\epsilon(E_I g) = 1$ and $gE_G = E_G$; the kernel of augmentation is spanned by $g - 1$, and multiplication by $1 - E_G$ projects onto it. This proves exactly how nontrivial prime holonomy reaches the original zeta distribution: its trivial character has the original HW28 prime factors, and the other character data remain in an explicitly retained augmentation ideal. More generally, for $\Re s > 1$,

$$L_r(s) = \prod_p \det(1 - p^{-s} T_p | V_p)^{-1}, \quad -\frac{L'_r}{L_r}(s) = \sum_{p,k \geq 1} (\log p) \text{Tr}(T_p^k) p^{-ks}. \quad (\text{HW53})$$

Expand $-\log(1 - z) = \sum_{k \geq 1} z^k/k$ on every eigenline; $|\text{Tr}(T_p^k)| \leq \dim V_0$ proves absolute convergence and termwise differentiation there. The trivial representation gives precisely $\zeta(s)$. No replacement of a nontrivial character by a zeta zero is made.

At the global zero every element of K stabilizes the carrier point, so HW44 gives a singleton fibre G/G . Its coefficient space is

$$W = V_0^G, \quad P_G = \frac{1}{|G|} \sum_{g \in G} r(g). \quad (\text{HW54})$$

The same holds separately at every retained fixed support label. Evaluation at zero and additive integration land in W : equivariance proves the first, and invariance of the additive Haar measure under compact units proves the second. Tensoring HW37 with W retains their weights and every label. Thus the fixed-point coefficient is $\dim W$, not $\dim V_0$; a nontrivial character has zero invariant boundary coefficient.

17 The branch loop at a double point and the attained endpoint map

For the double-root family in the heat coordinate retain

$$A_h^{\text{dbl}} = \mathbb{C}[Z]/(Z^2 - 2h). \quad (\text{HW55})$$

For $h \neq 0$, evaluation at the ordered pair $(\sqrt{2h}, -\sqrt{2h})$ identifies the algebra with $\mathbb{C}^2$. Analytic continuation around $h = 0$ interchanges the square roots: write $h = re^{i\theta}$ and continue $\sqrt{2r}e^{i\theta/2}$ from $\theta = 0$ to 2π . Consequently the loop matrix is

$$M_{\text{loop}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad M_{\text{loop}}^* M_{\text{loop}} = 1, \quad M_{\text{loop}}^2 = 1. \quad (\text{HW56})$$

It is nonidentity and unitary. At $h = 0$, the fibre is $\mathbb{C}[Z]/(Z^2)$, retaining its nonzero nilpotent Z . These are the loop of the punctured family and the nonreduced central fibre, respectively. The regular bilinear multiplication trace, in basis $(1, Z)$, is $\text{diag}(2, 4h)$; this follows by multiplying $(a + bZ)(c + dZ)$ and taking twice its constant coefficient. For real h , its real symmetric signature changes across zero while the punctured complex loop HW56 stays the same permutation. If the involution $Z \mapsto -Z$ is inserted in the first argument, that displayed diagonal is instead $\text{diag}(2, -4h)$. Both choices and their signs are explicit; neither is silently the arithmetic HW21.

The existing supported-endpoint calculation supplies an attained map, not only a formal resemblance. In the eight coordinates $(0_+, 0_-, 1_+, 1_-, -1_+, -1_-, \infty_+, \infty_-)$, let Q_J exchange $1_+ \leftrightarrow -1_-, 1_- \leftrightarrow -1_+$, and $\infty_+ \leftrightarrow \infty_-$, fixing $0_\pm$. Define

$$j(A, B) = (0, 0, A/\sqrt{2}, A/\sqrt{2}, B/\sqrt{2}, B/\sqrt{2}, 0, 0). \quad (\text{HW57})$$

Direct substitution gives the exact pullback

$$j(A, B)^* Q_J j(C, D) = \bar{A}D + \bar{B}C. \quad (\text{HW58})$$

Thus j is an isometric embedding of the hyperbolic endpoint pair of HW28. No zero value or derivative is assigned by this map: its inputs are the two actual endpoint functionals.

For completeness those inputs are attained by a precise theta test, retaining the original Fourier convention. For $q \in C_c^\infty(\mathbb{R})$, put

$$\phi_q(x) = \int_{\mathbb{R}} e^{t/2} q(t) e^{-\pi e^{2t} x^2} dt. \quad (\text{HW59})$$

This is even Schwartz, by compactness of the t -support. The Gaussian integral gives $\phi_q(0) = \mathcal{L}q(0)$ and $\int_{\mathbb{R}} \phi_q = \mathcal{L}q(1)$. The transform of a Gaussian gives $\widehat{\phi_q} = \phi_{q(-\cdot)}$. The endpoint map $q \mapsto (\mathcal{L}q(0), \mathcal{L}q(1))$ is onto: for any nonnegative compact bump of nonzero mass, that bump and its nonzero translate have endpoint column vectors differing by the factors $e^{\pm b/2}$, whose two-by-two determinant is nonzero when $b \neq 0$. Hence HW57–59 realize every endpoint direction. For a primitive test both coordinates vanish, and its image under j is zero. The whole Weil form still has the two remaining distributions in HW29.

This final diagram records the exact boundary relationship:

$$\begin{array}{ccc} C_c^\infty(\mathbb{R}) & \xrightarrow{q \mapsto \phi_q} & \mathcal{S}^{\text{ev}}(\mathbb{R}) \\ & \searrow (\mathcal{L}q(0), \mathcal{L}q(1)) & \downarrow (\text{ev}_0, f) \\ & & \mathbb{C}^2 \xrightarrow{j} \mathbb{C}^8. \end{array} \quad (\text{HW60})$$

HW34 identifies its pole residues; HW37 is its full fixed-label endpoint extension; HW9 supplies the different full-zero-jet section; HW29 joins all of them to the original explicit formula. The nontrivial loop HW56 and the endpoint embedding HW60 therefore have proved receivers, while only an actual zero character with nonunit modulus, or an attained negative value of HW29, has the RH consequence in HW1 and Theorem 14.1.

18 The retained meromorphic transport residue and its jet receiver

The original eight-state calculation also contains a different collision clock $t = h - 1/8$. It must not be replaced by the double-root clock of HW55. Retain its full local family from EHM27–46 [6]:

$$\begin{aligned} \beta(t) &= \frac{\sqrt{9 + 64t} - 3}{2}, \quad \beta(\beta + 3) = 16t, \quad \sqrt{9} = 3, \\ \mathcal{L} &= \mathbb{C}\{t\}[\epsilon, T]/(\epsilon^2 - \beta\epsilon, T^2 + 16t - (6 + 3\beta)\epsilon), \quad \mathcal{L}_0 = \mathbb{C}[T]/(T^4), \quad \epsilon = T^2/6. \end{aligned} \quad (\text{HW61})$$

The two successive monic relations prove that $1, \epsilon, T, \epsilon T$ is a free basis before specialization. Over the punctured disc the map flipping the sign only at the root $\epsilon = \beta$ is exactly

$$S_b(\epsilon) = \epsilon, \quad S_b(T) = (1 - 2\epsilon/\beta)T, \quad S_b(\epsilon T) = -\epsilon T. \quad (\text{HW62})$$

The multiplier takes values $1, -1$ on the two roots, and its square is one by $\epsilon^2 = \beta\epsilon$. It preserves the T^2 relation and is its own inverse. Since $t/\beta = (\beta + 3)/16$, its operator residue in the stated free basis is

$$\delta = tS_b|_{t=0} = -\frac{T^3}{16} \frac{d}{dT}, \quad \delta(T) = -T^3/16, \quad \ker \delta = \langle 1, T^2, T^3 \rangle, \quad \text{im } \delta = \mathbb{C}T^3, \quad \delta^2 = 0. \quad (\text{HW63})$$

Indeed its value on T is $-\frac{3}{8}\epsilon T = -T^3/16$, and its other three free-basis values are zero. The derivation sends T^4 to $-T^6/4$, so descends to the quotient. This proves its Leibniz rule and all four displayed assertions. Its nonzero coefficient rules out a regular extension of S_b in this free family. The obstruction determines the actual nonzero derivation δ , rather than a vanished transport.

It also determines the unipotent algebra action $V_b = \exp(b\delta) = 1 + b\delta$ on $\mathcal{L}_0$. The identity $\delta(f)\delta(g) = 0$, since its image is (T^3) , and the Leibniz rule prove that V_b is multiplicative; V_{-b} is its inverse. This constructed action is not misidentified as a regular specialization of the meromorphic S_b . Its regular trace pairing for $T^\# = -T$ is $4\overline{f(0)}g(0)$, because every nonconstant multiplication operator is nilpotent. Thus $Q(\delta f, g) = Q(f, \delta g) = 0$, while $\delta \neq 0$.

There is a precise linear comparison with every actual jet factor, including its scalar spectral character. For the actual local factor at ρ with $m = m_\rho \geq 2$, and any fixed $L \neq 0$, set

$$\begin{aligned} \Psi_{\rho,L} : \mathcal{L}_0 &\longrightarrow (t_\rho^{m-2}) \subset A_\rho, & \Psi_{\rho,L}(f_0 + f_1T + f_2T^2 + f_3T^3) &= f_1t_\rho^{m-2} - 16Lf_3t_\rho^{m-1}, \\ \ker \Psi_{\rho,L} &= \langle 1, T^2 \rangle, & \Psi_{\rho,L}\delta &= LN_\rho\Psi_{\rho,L}, \quad \Psi_{\rho,L}V_b = e^{-bL(\rho-1/2)}T_{bL}\Psi_{\rho,L}. \end{aligned} \quad (\text{HW64})$$

It is onto the indicated two-dimensional filtered piece, since both displayed coefficients are nonzero. Apply both sides of the middle identity to $1, T, T^2, T^3$: only T has a nonzero value, equal to Lt_ρ^{m-1} . Exponentiating, with $N_\rho^2 = 0$ on this piece, gives the final equality, retaining the scalar phase and modulus explicitly. For a simple actual zero define $\Psi_{\rho,L} = 0$; the same intertwining identity holds with zero image, and the nonzero nilpotent residue is still retained in its source. For $m = 2$, the value of $\Psi_{\rho,L}f$ is f_1 ; for $m > 2$, it is zero. Therefore HW21 evaluates its Weil pairing exactly through those values, while HW20 retains its entire raw-residue pairing with complementary jets. This gives an explicit receiver for the nilpotent defect without assigning it an unproved offcritical arithmetic character.

19 The singular endpoint in the original arithmetic heat family

EP1. The endpoint question. The requested analogy concerns a singularity at arithmetic time zero. Such a calculation can be made directly at that endpoint; it need not begin with a numerical bound on the de Bruijn–Newman constant. We retain the entire original H_t of EH1. The following statements describe every actual real zero of H_0 , with its original multiplicity and all higher coefficients. They neither assert that a multiple zero exists nor replace H_t by a freely chosen polynomial. The backward heat equation and the positivity criterion used here are those of Rodgers–Tao and Connes–Consani, cited in EH1 and HW24–31.

EP2. The full local coefficient equation. Fix an actual real zero x_0 of H_0 , let m be its order, and write

$$H_0(x_0 + \xi) = \sum_{n \geq m} a_n \xi^n, \quad a_n = \frac{\partial_Z^n H_0(x_0)}{n!}, \quad a_m \neq 0. \quad (\text{EP1})$$

The coefficients are real. Joint analyticity, established from the original theta source in EH9, and $\partial_t H = -\partial_z^2 H$ give

$$[t^j \xi^\ell] H_t(x_0 + \xi) = (-1)^j \frac{(\ell + 2j)!}{j! \ell!} a_{\ell+2j}. \quad (\text{EP2})$$

Indeed repeated differentiation gives $\partial_t^j \partial_z^\ell H = (-1)^j \partial_z^{\ell+2j} H$; evaluation at zero and division by $j! \ell!$ proves the formula. All terms with $\ell + 2j < m$ vanish. No higher coefficient has been discarded by this equality.

Define the explicit monic polynomial

$$P_n(z) = \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(-1)^j n!}{j! (n-2j)!} z^{n-2j}. \quad (\text{EP3})$$

In the classical Hermite convention it is $H_n^{\text{Hermite}}(z/2)$; the explicit coordinate conversion avoids any change of heat scale. The source formula is DLMF18.5.13, Chapter18, by T.H. Koornwinder, R. Wong, R. Koekoek and R.F. Swarttouw. All identities and zero properties needed below are proved here. Substitute $t = u^2$, $\xi = uz$ into the convergent local Taylor series:

$$H_{u^2}(x_0 + uz) = \sum_{n \geq m} a_n u^n P_n(z). \quad (\text{EP4})$$

For z in any fixed compact set and u sufficiently small, this is an equality of convergent analytic functions. To justify regrouping, choose a closed polydisc strictly inside a domain of analyticity of $H_t(x_0 + \xi)$. For bounded z and small u , (u^2, uz) lies in that polydisc, where its double Taylor series converges absolutely. Regrouping by $n = 2j + \ell$ is consequently legitimate.

EP3. Every leading root is real and simple. The generating function of EP3 is

$$\sum_{n \geq 0} P_n(z) \frac{w^n}{n!} = e^{zw - w^2}. \quad (\text{EP5})$$

Multiplying the two exponential series proves it coefficient by coefficient. It gives $P'_n = nP_{n-1}$, $P_{n+1} = zP_n - 2nP_{n-1}$, and $2P''_n - zP'_n + nP_n = 0$. With $w_0(z) = e^{-z^2/4}$, Taylor expansion of $w_0(z - 2w)/w_0(z)$ also proves $P_n w_0 = (-2)^n w_0^{(n)}$. Repeated integration by parts therefore gives

$$\int_{\mathbb{R}} P_n(z) R(z) e^{-z^2/4} dz = 0 \quad (\deg R < n), \quad (\text{EP6})$$

since every boundary term is a polynomial times a decaying Gaussian. If the real polynomial P_n had fewer than n real sign-changing zeros, let R be the product of their distinct linear factors. Then $\deg R < n$, while $P_n R$ has constant nonzero sign except at its finitely many zeros: all remaining real roots have even multiplicity and conjugate nonreal factors have constant positive sign on the real line. Its weighted integral is nonzero, contrary to EP6. Thus P_n has n real simple roots. Denote them by $r_1 < \dots < r_n$. This proof includes the exact weight and scale.

EP4. Actual arithmetic root branches through the endpoint. Set $c = a_{m+1}/a_m$ and $d = a_{m+2}/a_m$, retaining these actual arithmetic numbers. Dividing EP4 by $a_m u^m$ gives an analytic function of (u, z) with value $P_m(z)$ at $u = 0$. At each r_j , its z derivative is nonzero. The analytic implicit function theorem gives a unique root $z_j(u)$ with $z_j(0) = r_j$. For real u , conjugation and uniqueness make $z_j(u)$ real. To verify that these are the complete nearby zeros, choose a small disc about x_0 with no other zero of H_0 . Uniform convergence of H_t on its boundary implies $|H_t - H_0| < |H_0|$ there for small t . The straight-line homotopy has no boundary zero, so its argument-principle integral is constant and equals m . The m distinct implicit branches for small positive u exhaust that count.

Their first three terms, including the arithmetic correction, are

$$x_j(t) = x_0 + r_j\sqrt{t} + 2ct + r_j(2d - c^2)t^{3/2} + O(t^2), \quad t \downarrow 0. \quad (\text{EP7})$$

Here and below the remainders refer to this fixed actual root and its full analytic germ. For proof, substitute $z_j = r_j + A_ju + B_ju^2 + O(u^3)$ in $P_m + cuP_{m+1} + du^2P_{m+2} + O(u^3) = 0$. At a root of P_m , EP5 gives

$$P_{m+1} = -2P'_m, \quad P'_{m+1} = 0, \quad P''_m = \frac{1}{2}r_jP'_m, \quad P_{m+2} = -2r_jP'_m.$$

The successive equations give $A_j = 2c$ and $B_j = r_j(2d - c^2)$, proving EP7. In particular the order- t translation is common to every member of the colliding cluster.

For negative time put $t = -u^2$, $\xi = uz$. The leading polynomial is $i^m P_m(z/i)$, by EP2, so its roots are ir_j . For $m \geq 2$ at least two r_j are nonzero. The same implicit argument gives nonreal roots of the actual H_{-u^2} near x_0 for all sufficiently small positive u . At time zero the cluster itself consists of the real root x_0 ; at small positive time all its members are real. Thus a singular real cluster at time zero has an explicitly calculated one-sided heat transition. Its existence alone yields no nonreal zero of H_0 . This is a calculation on the actual family, not on EP3 as an independent replacement family.

EP5. The entire finite-cluster energy, including its finite part. Use the ordered-pair convention of Rodgers–Tao and keep all m roots in the cluster:

$$E_{x_0}(t) = \sum_{i \neq j} \frac{1}{(x_i(t) - x_j(t))^2}.$$

Its endpoint expansion is

$$E_{x_0}(t) = \frac{m(m-1)}{8t} + \frac{m(m-1)}{4}(c^2 - 2d) + O(t). \quad (\text{EP8})$$

This is the cluster contribution; the external interactions in RD4 remain part of the full Rodgers–Tao identity. Here is a complete derivation of its leading constant. For $P_m(z) = \prod_j (z - r_j)$, logarithmic differentiation of the product with its i th factor removed gives

$$\sum_{j \neq i} \frac{1}{r_i - r_j} = \frac{r_i}{4}, \quad \sum_{j \neq i} \frac{1}{(r_i - r_j)^2} = \frac{m-1}{6} - \frac{r_i^2}{48}. \quad (\text{EP9})$$

For the second identity, the differentiated differential equation in EP5 gives $P'''_m(r_i)/P'_m(r_i) = r_i^2/4 - (m-1)/2$. On the other hand this ratio is three times the square of the first sum minus three times the second sum. This proves EP9. The coefficient of z^{m-2} in EP3 is $-m(m-1)$ and that of z^{m-1} is zero, so $\sum_i r_i^2 = 2m(m-1)$. Summing EP9 gives $\sum_{i \neq j} (r_i - r_j)^{-2} = m(m-1)/8$. EP7 makes every gap $u(r_i - r_j)(1 + (2d - c^2)u^2) + O(u^4)$; substitution proves EP8 initially with an $O(u)$ remainder. The symmetric sum of all ordered pairs is unchanged by $u \mapsto -u$, because these two parameters describe precisely the same zero multiset of H_{u^2} . Its convergent Laurent expansion is therefore even. The remainder improves to $O(u^2) = O(t)$, as stated. For $m = 1$ the sum is empty and both displayed coefficients vanish.

For $m \geq 2$, at $t = 0$ every internal gap is supported e , so the inverse-monoid assignment sends each inverse square and their nonempty finite sum to e . For $m = 1$ the sum is empty: it is τ in the ambient semiring, or e if explicitly formed in the top additive group. For $m \geq 2$ the amplitude assigned this way is zero, whereas the Laurent germ EP8 has the positive residue $m(m-1)/8$. Both data are retained: the assignment and the Laurent pole are related by evaluation on the punctured chart, whose split special fibre is the full L . They are not identified by taking a finite limit. This calculates the exact amount of collision energy lost by that finite assignment.

EP6. The local nilpotent fibre and its Weil value. At the same actual endpoint put $\rho = 1/2 + ix_0/2$. The local algebra is

$$A_\rho = \mathbb{C}[v]/(v^m), \quad v = s - \rho = i\xi/2. \quad (\text{EP10})$$

Reflection on the point set sends ξ to $\bar{\xi}$; the induced conjugate-linear involution on its coordinate algebra fixes the generator ξ and sends v to $-v$. The regular multiplication trace is $\text{Tr}(M_f) = mf(0)$: in the ordered basis $1, v, \dots, v^{m-1}$, its matrix is triangular with diagonal $f(0)$. Consequently the exact local Weil contribution is

$$T_\rho(f, g) = m\overline{f(0)}g(0), \quad \text{rad } T_\rho = (v). \quad (\text{EP11})$$

This agrees with the actual complete-formula pullback HW21, HW24–31, including its multiplicity and both endpoint removals. It is nonnegative at this real-zero collision, and retains a nilpotent radical rather than imposing strict positivity on it. The shifted arithmetic holonomy is $e^{Lix_0/2} \exp(LM_v)$, retaining its entire nilpotent factor. It preserves EP11, since the scalar has modulus one and the nilpotent factor has constant value one.

The split lift $G_L(A_\rho)$ also retains the full nilpotent element $\hat{v}$, with $\hat{v}^m = e$. For $m > 1$ it cannot have a reflexive multiplicative inverse: projecting $\hat{v}b\hat{v} = \hat{v}$ would give $v^2p(b) = v$. Iteration gives $v = v^m p(b)^{m-1} = 0$, contradicting $v \neq 0$. Supported e itself still satisfies $e^\dagger = e$. The exact unital localization at either e or $\hat{v}$ is L : inverting $\hat{v}$ inverts its power e , and after localizing at e every top-supported amplitude becomes the top unit. This proves both universal maps and preserves all labels.

EP7. The original meromorphic holonomy residue through the pairing. HW61–64 retain the different eight-state collision clock and the original nonzero derivation $\delta = -T^3\partial_T/16$ on $\mathbb{C}[T]/(T^4)$. For an actual factor of order $m \geq 2$, their exact linear map is

$$\Psi_{\rho,L}(f) = f_1 v^{m-2} - 16L f_3 v^{m-1}, \quad \Psi\delta = LM_v\Psi, \quad L \neq 0. \quad (\text{EP12})$$

For $f = T$, the image of its nonzero residue is $\Psi\delta T = Lv^{m-1}$. It is nonzero and lies in the radical of the actual Weil trace. In particular, for every packet vector g ,

$$T_h(\Psi\delta T, g) = 0, \quad R_\rho(\Psi\delta T, 1) = \frac{(-1)^{m-1}L}{u_\rho(0)} \neq 0 \quad \text{at the fixed reflected root, } L \in \mathbb{R} \setminus \{0\}. \quad (\text{EP13})$$

The second equality follows by extracting v^{-1} from $(-1)^{m-1}Lv^{m-1}/(u_\rho(v)v^m)$; it preserves the original unit $u_\rho(0) = 2\lambda_\rho$. For the first, every value of $\Psi\delta T$ is zero, and HW21 applies. Thus the holonomy defect remains detected by the perfect residue pairing, while its Weil pairing is evaluated as zero. It has not been deleted. The exact intervening map is multiplication by g' in HW19–21.

EP8. An endpoint nonreal zero and its exact persistence map. A different endpoint datum has a different consequence. For every actual nonreal zero z_0 of H_0 , choose a closed disc around it disjoint from the real axis, with zero-free boundary. Compactness gives $\min_{\partial D} |H_0| > 0$, and joint analyticity gives $\max_{\partial D} |H_t - H_0| < \min_{\partial D} |H_0|$ for all sufficiently small real $|t|$. The same homotopy/argument-principle proof as in EP4 retains the positive zero count in that disc. Thus an actual nonreal zero at time zero persists to positive times. This explains how an endpoint disproof would subsequently yield $\Lambda > 0$, without assuming that a positive-time bound has to be the starting calculation.

The endpoint defect actually evaluated in EP8 and EP13 is instead a pole of a reciprocal or an infinitesimal transport in a retained real-zero fibre. Its exact local Weil value is EP11–13. These calculations do not exhibit an actual nonreal zero or a negative complete Weil test. They do determine the pole, its finite part, the full nilpotent and support fibres, and the exact receiving

maps, which are the endpoint data required to test the proposed analogy without erasing its singularity.

EP9. The arithmetic unit receives the finite collision energy. The local unit in HW7 is determined by the same original coefficients:

$$u_\rho(v) = 16(-2i)^m \sum_{j \geq 0} a_{m+j} (-2i)^j v^j. \quad (\text{EP14})$$

Indeed $g(s) = 2\xi(s) = 16H_0(Z)$ and the local variable is $\xi_{\text{local}} = Z - x_0 = -2iv$. Dividing this exact equality by v^m gives EP14, including its factor 16. In particular

$$u_\rho(0) = 16(-2i)^m a_m, \quad \frac{u'_\rho}{u_\rho}(0) = -2ic, \quad (\log u_\rho)''(0) = 4(c^2 - 2d). \quad (\text{EP15})$$

Differentiating EP14 proves these identities; the logarithmic second derivative is independent of the chosen local logarithm. Hence

$$\text{FP}_{t=0} E_{x_0}(t) = \frac{m(m-1)}{16} (\log u_\rho)''(0). \quad (\text{EP16})$$

The raw-residue coefficient in EP13 is correspondingly $(-1)^{m-1} L / [16(-2i)^m a_m]$. In that equation Ψ is extended by zero to all other packet factors before taking the global T_h . For a check retaining the differential, set $F(\xi) = f(i\xi/2)$ and $G(\xi) = g(i\xi/2)$ for the two local jet arguments. Then

$$R_\rho(f, g) = \frac{i}{32} \text{Res}_{\xi=0} \frac{\overline{F(\xi)} G(\xi)}{H_0(x_0 + \xi)} d\xi. \quad (\text{EP17})$$

This follows from $ds = i d\xi/2$ and $2\xi(s) = 16H_0(Z)$; the use of ξ as a local variable does not change the named completed function. Inserting the actual Jacobian multiplier $g'(s) = -32iH'_0(Z)$ cancels the prefactor and gives the residue $m\overline{F(0)}G(0)$, exactly EP11. Both the energy's finite part and the two pairings are thus calculated through the same retained arithmetic unit.

EP10. The original Rodgers–Tao positive potential at supported zero. Their original section7, immediately before equation vlog, uses, on $a > 0$, the potential

$$V(a) = a^{-2} + 2a - 3 = \frac{(a-1)^2(2a+1)}{a^2} \geq 0. \quad (\text{EP18})$$

The identity follows by expansion of its numerator. Their energy uses $a = |x_k - x_j|/|\xi_k - \xi_j|$, with distinct actual roots and fixed distinct classical locations ξ_k, ξ_j ; its stated domain therefore has $a > 0$. Here we compute exactly what happens at $a = 0$.

In the original $G_L(\mathbb{R})$ let a carry top support, let $a^\dagger$ be its inverse-monoid reciprocal, and let all differences below be formed in the top additive group with identity e . Write $\eta = aa^\dagger$, an idempotent. The two expressions in EP18 have the respective extensions

$$V_{\text{expanded}} = (a^\dagger)^2 + 2a - 3, \quad V_{\text{factored}} = (a-1)^2(2a+1)(a^\dagger)^2. \quad (\text{EP19})$$

Distributing without cancelling a gives the exact defect

$$V_{\text{expanded}} - V_{\text{factored}} = (2a-3)(1-\eta). \quad (\text{EP20})$$

For proof, expand the second numerator as $2a^3 - 3a^2 + 1$, multiply by $(a^\dagger)^2$, and use $a^2(a^\dagger)^2 = \eta$ and $a^3(a^\dagger)^2 = a\eta$. No division at zero is used. On $a > 0$, $\eta = 1$ and the defect is the top additive zero e . At $a = e$, $\eta = e$ and the amplitudes are

$$p(V_{\text{expanded}}(e)) = -3, \quad p(V_{\text{factored}}(e)) = 0, \quad p(V_{\text{expanded}} - V_{\text{factored}}) = -3. \quad (\text{EP21})$$

This is a negative value of the expanded, totalized potential. Its exact origin is the idempotent complement $1 - \eta$, which vanishes on the unit locus and has amplitude 1 at the collision. Both presentations are retained. Their difference is precisely the supported-zero defect EP20. Universal unital localization at e sends both values to the same top of L and preserves every other support label, whereas the top-fibre amplitude records the difference.

For the original weighted pair the assigned expanded value is $-3/|\xi_k - \xi_j|^2$. For a nonempty colliding cluster, each such summand is negative, although the actual punctured potential has a positive pole. This extension fails the potential's original nonnegativity property at zero. It does not alter the Rodgers–Tao theorem, which uses only distinct-root values, and it is not yet a negative value of the arithmetic Weil form. The exact Weil value of the retained critical collision and its transported nilpotent is EP11–17. The two explicit receivers record where the negative assigned potential and the zero Weil contribution occur.

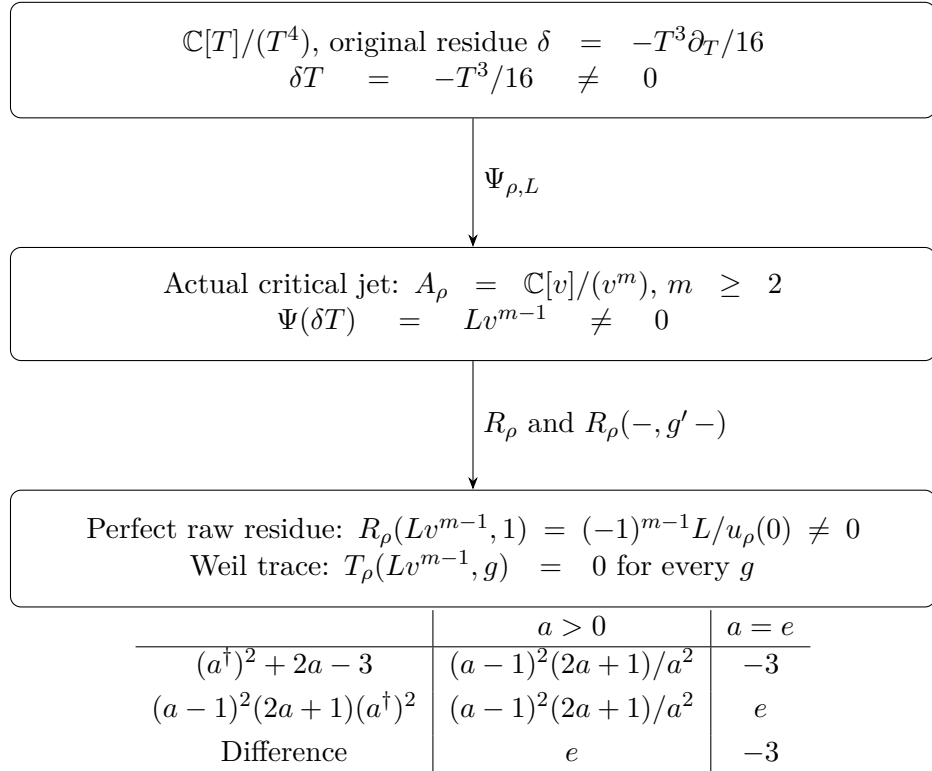


Figure 3: The endpoint defect has two explicitly computed receivers. The upper diagram retains the actual eight-state residue, its full jet image and its nonzero raw-residue value (HW61–64 and EP12–17). For a fixed reflected zero its Weil image lies in the nilpotent radical. The table computes the two extensions of Rodgers–Tao’s exact positive potential (EP18–21); entries retain top support and their differences are in the top additive group. Neither calculation identifies a negative assigned potential with a negative Weil test.

20 The complete conductor of an original arithmetic heat collision

HC1. Original objects and the purpose of the comparison. Use precisely the Rodgers–Tao family and arithmetic coordinate of EH1 and EP1–17. Fix any actual real zero x_0 of H_0 , of order $m \geq 1$. Put $\xi = Z - x_0$, $\mathcal{O} = \mathbb{C}\{t\}$ and $\mathcal{K} = \text{Frac } \mathcal{O}$. The full coefficients a_n , $c = a_{m+1}/a_m$ and $d = a_{m+2}/a_m$ remain those of EP1. The existence of a multiple zero is not asserted. Every calculation below applies to every actual zero, including $m = 1$.

EP2–7, proved from the original heat equation, give all the local branches after the exact base change $t = u^2$:

$$\theta_j(u) = x_j(u^2) - x_0 = r_j u + 2cu^2 + r_j(2d - c^2)u^3 + O(u^4), \quad \theta_j(-u) = \theta_{m+1-j}(u). \quad (\text{HC1})$$

The equality of branches follows by uniqueness in the analytic implicit function theorem: the same entire function is evaluated at $u^2 = (-u)^2$, and the leading coefficient changes from r_j to $-r_j = r_{m+1-j}$. Each θ_j here is the complete analytic germ, not the displayed truncation.

Let $r = \lfloor m/2 \rfloor$ and $\varepsilon = m - 2r \in \{0, 1\}$. Index the positive Hermite roots by $b_1 < \dots < b_r$ and let θ_a be the corresponding branches. Set

$$\begin{aligned} E_a(t) &= \frac{\theta_a(u) + \theta_a(-u)}{2}, \\ R_a(t) &= \frac{\theta_a(u) - \theta_a(-u)}{2u}, \\ Q_a(t, \xi) &= (\xi - E_a(t))^2 - tR_a(t)^2. \end{aligned} \quad (\text{HC2})$$

Evenness makes E_a, R_a elements of $\mathcal{O}$; $R_a(0) = b_a \neq 0$ and $E_a(t) = 2ct + O(t^2)$. When $\varepsilon = 1$, the central branch is even in u ; write it $\theta_0(t)$ and set $Q_0 = \xi - \theta_0(t)$. It has $\theta_0(t) = 2ct + O(t^2)$.

HC2. The full finite algebra in the original coordinate. Define

$$W(t, \xi) = \prod_{a=1}^r Q_a(t, \xi) Q_0(t, \xi)^\varepsilon, \quad A = \mathcal{O}[\xi]/(W). \quad (\text{HC3})$$

The convention omits Q_0 when $\varepsilon = 0$. This monic polynomial has degree m , has all the actual nearby zeros, and $W(0, \xi) = \xi^m$. Thus $1, \xi, \dots, \xi^{m-1}$ is an $\mathcal{O}$ -basis and $A/tA = \mathbb{C}[\xi]/(\xi^m)$.

For completeness the factorization retains the full analytic unit:

$$H_t(x_0 + \xi) = U(t, \xi)W(t, \xi), \quad U(0, 0) = a_m \neq 0. \quad (\text{HC4})$$

Here is a direct construction. Take a small circle around $\xi = 0$ containing no other zero of H_0 , and shrink the t disc so its boundary has no zero of H_t . EP4's argument-principle calculation counts exactly m zeros inside. On that boundary H_t/W is jointly analytic. Its Cauchy integral in the circle defines a jointly analytic function in the interior. For nonzero sufficiently small t the roots are simple by HC1, so the removable-singularity theorem identifies this Cauchy integral with H_t/W everywhere. The equality extends to $t = 0$ by continuity. At $(0, 0)$ its value is a_m ; on a smaller neighbourhood it is a unit. This proves HC4 without discarding any coefficient of H .

HC3. An exact inclusion into the separated branch algebra. For each paired component introduce its own parameter u_a with $u_a^2 = t$. Define

$$B = \prod_{a=1}^r \mathbb{C}\{u_a\} \times \mathcal{O}^\varepsilon, \quad \iota : A \longrightarrow B, \quad \xi \longmapsto (E_a(t) + u_a R_a(t))_{a=1}^r \times \theta_0(t). \quad (\text{HC5})$$

The last component is omitted if $\varepsilon = 0$. This is a unital $\mathcal{O}$ -algebra map in the original coordinate. Each Q_a is irreducible over $\mathcal{K}$: its discriminant $4tR_a^2$ has odd t valuation and cannot be a square

in $\mathcal{K}$. The factors are distinct because their leading root slopes have different squares. The linear factor, if present, is different from every quadratic factor. Therefore evaluation into the product of the fraction fields is injective modulo W , which proves injectivity of ι .

The component map is onto. Indeed $u_a = (\xi - E_a(t))/R_a(t)$ in the quotient by Q_a , and every convergent series in u_a is uniquely $f(t) + u_a g(t)$. The central component is $\mathcal{O}$ itself. Consequently B is finite free of rank m over $\mathcal{O}$, and is also finite as an A -module. It is the integral closure of A in its product of fraction fields. To verify that assertion directly, each factor $\mathbb{C}\{u_a\}$ is a discrete valuation ring: every nonzero series is a power of u_a times a unit. A Laurent series of negative valuation cannot satisfy a monic equation with coefficients of nonnegative valuation, because its highest power would be the unique term of smallest valuation. This proves integral closedness, also for the central factor. An element integral over A is integral over $\mathcal{O}$ by finiteness and therefore has nonnegative valuation in every factor. It belongs to B . Conversely finiteness of B over A proves integrality by the determinant argument for multiplication on a finite generating module. These two inclusions prove the assertion.

The original algebra A is retained throughout; B is an auxiliary finite algebra with the displayed inclusion and a computed quotient.

HC4. Every elementary divisor of the original inclusion. Use the basis of B consisting first of its component identities (including the central one), then $u_1, \dots, u_r$. Let $J(t)$ be the matrix of HC5 from the ordered original basis $1, \xi, \dots, \xi^{m-1}$. Its entries are explicitly

$$\begin{aligned} J_{a,k}^{\text{even}}(t) &= \frac{\theta_a(u)^k + \theta_a(-u)^k}{2}, \\ J_{a,k}^{\text{odd}}(t) &= \frac{\theta_a(u)^k - \theta_a(-u)^k}{2u}, \\ J_{0,k}(t) &= \theta_0(t)^k \quad (\varepsilon = 1), \quad 0 \leq k < m. \end{aligned} \tag{HC6}$$

These formulas retain the full original branches. Their precise divisibility is sufficient to give the factorization

$$J(t) = C(t)D(t), \quad D(t) = \text{diag}_{0 \leq k < m} t^{\lfloor k/2 \rfloor}, \quad C(t) \in \text{GL}_m(\mathcal{O}). \tag{HC7}$$

Here C is defined entry by entry by HC6 divided by the stated column power of t ; it is not omitted from any metric or map.

For proof, $\theta_a(u)$ has order one, so the even part of its k th power is divisible by $u^{2\lfloor k/2 \rfloor}$ and its odd part divided by u by $u^{2\lfloor k/2 \rfloor}$. The central branch has order at least one in t . Thus all entries defining C are analytic. To check its determinant, order the even columns before the odd columns for this check only. For $k = 2\ell$ the even rows at $t = 0$ are $b_a^{2\ell}$, and the central row is 1 for $\ell = 0$ and 0 otherwise. The odd rows in these columns are $4\ell c b_a^{2\ell-1}$, with value zero when $\ell = 0$. For $k = 2\ell + 1$ the even and central rows vanish and the odd rows are $b_a^{2\ell+1}$. The diagonal blocks are consequently Vandermonde matrices on the distinct numbers b_a^2 , together with 0 in the even block when the central branch is present; the odd block has the extra nonzero row factors b_a . Both blocks are square and invertible. Thus $\det C(0) \neq 0$, proving HC7 in the original column order as well.

Multiplication by C^{-1} induces the exact module isomorphism

$$B/A \simeq \bigoplus_{k=0}^{m-1} \mathcal{O}/(t^{\lfloor k/2 \rfloor}), \quad \text{length}_{\mathcal{O}}(B/A) = \left\lfloor \frac{(m-1)^2}{4} \right\rfloor. \tag{HC8}$$

The summand with exponent zero is the zero module. The length is $r(r-1)$ for $m = 2r$ and r^2 for $m = 2r+1$, by summing the exponents. HC7, rather than a rank comparison alone, proves every elementary divisor.

HC5. The conductor with its full arithmetic factors. The conductor is the ideal $\mathfrak{c} = \{b \in B : bB \subseteq A\}$, regarded in B via HC5. For any component a let $W_a = W/Q_a$ and evaluate at its original branch; for the central component use $W_0 = W/Q_0$. Then

$$\begin{aligned}\mathfrak{c}_a &= W_a(t, \theta_a(u_a))\mathbb{C}\{u_a\} = u_a^{2(r-1)+\varepsilon}\mathbb{C}\{u_a\}, \\ \mathfrak{c}_0 &= W_0(t, \theta_0(t))\mathcal{O} = t^r\mathcal{O} \quad (\varepsilon = 1).\end{aligned}\tag{HC9}$$

The first equalities retain the exact unit factors as well as the orders. The conductor is the product of these component ideals.

To prove it, an element supported in just component a belongs to A precisely when it is the image of a polynomial divisible by all the other factors Q_b . The ring $\mathcal{O}[\xi]$ is a unique factorization domain (polynomial Gauss lemma over the discrete valuation ring); its distinct monic irreducible factors therefore have product W_a dividing that polynomial. Conversely every multiple of W_a vanishes in the other components. The projection $A \rightarrow B_a$ is onto by HC5, so the ideal of possible a -components is exactly $W_a(\theta_a)B_a$. Multiplying by the component idempotents of B proves both necessity and sufficiency for the conductor definition.

For $b \neq a$, HC1 gives

$$Q_b(t, \theta_a(u_a)) = (b_a^2 - b_b^2)u_a^2 + O(u_a^3), \quad Q_0(t, \theta_a(u_a)) = b_a u_a + O(u_a^2).\tag{HC10}$$

Every displayed leading coefficient is nonzero. On the central component $Q_b(t, \theta_0(t)) = -b_b^2 t + O(t^2)$. Multiplying gives exactly the orders in HC9. For $m = 1$ this gives conductor $\mathcal{O}$; for $m = 2$ it gives $B = A$. All components and all multiplicities are included.

HC6. The special-fibre map and the loss measured by its kernel. At $t = 0$ the inclusion has the explicit specialization

$$\bar{\iota} : \mathbb{C}[\xi]/(\xi^m) \longrightarrow \prod_{a=1}^r \mathbb{C}[u_a]/(u_a^2) \times \mathbb{C}^\varepsilon, \quad \xi \longmapsto (b_a u_a)_a \times 0.\tag{HC11}$$

For $m \geq 2$ its kernel is exactly (ξ^2) and its image has dimension two, spanned by the images of $1, \xi$. Its cokernel has dimension $m - 2$. For $m = 1$ it is an isomorphism. Indeed powers ξ^k with $k \geq 2$ map to zero, while the two displayed vectors are independent because some $b_a \neq 0$. This proves the assertions directly and records the precise failure of injectivity after special-fibre base change. No nilpotent is silently removed from the original A/tA .

The trace form also pulls back exactly. For real t the coefficient involution fixes ξ and conjugates coefficients. On B in the stated component basis its Hermitian trace matrix is

$$G_B(t) = \text{diag}(\underbrace{2, \dots, 2}_r, \underbrace{1}_\varepsilon, \underbrace{2t, \dots, 2t}_r), \quad G_A(t) = J(t)^* G_B(t) J(t).\tag{HC12}$$

The entry 1 is omitted when $\varepsilon = 0$. In a quadratic component multiplication by u_a has matrix $\begin{pmatrix} 0 & t \\ 1 & 0 \end{pmatrix}$; its trace is zero and the trace of u_a^2 is $2t$. This proves G_B . Over $\mathcal{K}$ the inclusion is an algebra isomorphism, so regular traces agree there; both sides of HC12 are analytic, giving the equality also at zero. In particular

$$G_A(0) = \text{diag}(m, 0, \dots, 0).\tag{HC13}$$

Thus the entire nilpotent radical of the original local Weil trace EP11 is retained. The appearance of several dual-number factors in the auxiliary special fibre does not replace that original trace.

HC7. The discriminant, including its first arithmetic correction. Let $\mathfrak{D}_A(t)$ denote the determinant of the regular bilinear trace Gram matrix in $1, \xi, \dots, \xi^{m-1}$. At real t it is also the determinant in HC12. Put $M = m(m-1)/2$ and $b = 2d - c^2$. Then

$$\boxed{\mathfrak{D}_A(t) = D_m t^M (1 + m(m-1)bt + O(t^2)), \quad D_m = 2^M \prod_{j=1}^m j^j.}\tag{HC14}$$

The empty-product conventions give $D_1 = 1$ and a constant discriminant for the one-dimensional algebra.

To prove the full coefficient, evaluate the original power basis at all the m branches. Its Vandermonde determinant squared is

$$\mathfrak{D}_A(t) = \prod_{i < j} (\theta_i(u) - \theta_j(u))^2. \quad (\text{HC15})$$

HC1 makes every gap $u(r_i - r_j)(1 + bu^2) + O(u^4)$. The common $2cu^2$ translation cancels. Consequently the product is $u^{2M} \prod_{i < j} (r_i - r_j)^2 (1 + 2Mbu^2 + O(u^3))$. It is invariant under $u \mapsto -u$ because that permutation preserves the entire multiset of branches. Its analytic expansion in t therefore improves the remainder to $O(t^2)$ after removing t^M .

For the constant, the monic Hermite recurrence proved in EP5 is $P'_n = nP_{n-1}$ and $P_n = zP_{n-1} - 2(n-1)P_{n-2}$. Define $R_n = \text{Res}(P_n, P_{n-1})$ by products over roots. Exchanging the two monic factors introduces sign $(-1)^{n(n-1)} = 1$. Evaluation at the roots of P_{n-1} gives $R_n = (-2(n-1))^{n-1} R_{n-1}$, with $R_1 = 1$. The derivative product formula for a discriminant then gives $\text{disc } P_n = (-1)^{n(n-1)/2} n^n R_n = 2^{n(n-1)/2} \prod_{j=1}^n j^j$. This proves HC14. The Hermite convention and human-source attribution remain precisely DLMF18.5.13, with its authors and explicit coordinate conversion given in EP3.

HC12 also gives $\mathfrak{D}_A = (\det J)^2 2^{2r} t^r$. Taking orders and using HC8 proves the exact identity

$$M = r + 2 \text{length}_{\mathcal{O}}(B/A). \quad (\text{HC16})$$

Thus the discriminant order separates into the branch ramification order and twice the explicitly computed conductor quotient length.

The same coefficient is measured by the original cluster energy:

$$\left. \frac{d}{dt} \log \frac{\mathfrak{D}_A(t)}{D_m t^M} \right|_{t=0} = m(m-1)(2d - c^2) = -4 \text{FP}_{t=0} E_{x_0}(t) = -\frac{m(m-1)}{4} (\log u_\rho)''(0). \quad (\text{HC17})$$

The quotient inside the logarithm is an analytic unit with value one, so this derivative requires no choice of logarithm. The two last equalities follow by substituting EP8 and EP15–16; they retain the original arithmetic local unit, not a substituted polynomial source.

HC8. The support lift retains every connecting map. For any of the original bounded distributive support lattices L , apply G_L to HC5. The map

$$G_L(A) \longrightarrow G_L(B), \quad (a, 1) \longmapsto (\iota(a), 1), \quad (0, \lambda) \longmapsto (0, \lambda) \quad (\text{HC18})$$

is injective: its amplitude map is injective and it fixes every support label. Addition and multiplication are preserved by the original definitions. Its arithmetic reflection is exactly ι . Under the unital localization at supported e both sides map to L , and the induced map on L is the identity. This follows directly from the universal localization proved in SH: every top-supported arithmetic value becomes the top unit, while every zero-support label remains. The full diagram commutes:

$$\begin{array}{ccc} G_L(A) & \hookrightarrow & G_L(B) \\ p_A \downarrow & & \downarrow p_B \\ A & \xrightarrow{\iota} & B \end{array} \quad (\text{HC19})$$

The support receivers commute in the separate diagram:

$$\begin{array}{ccc} G_L(A) & \hookrightarrow & G_L(B) \\ \chi_A \downarrow & & \downarrow \chi_B \\ L & \xlongequal{\quad} & L. \end{array}$$

The quotient HC8, conductor HC9 and trace HC12 are computed before this amplitude collapse. The support localization cannot reconstruct them from its value alone; their exact receiver is the retained arithmetic reflection, which does not extend through that localization. For example e acts as zero on the arithmetic A -module B/A , so its base change along the localization is zero: on a tensor $1 \otimes v$, the equality $1 = e$ gives $1 \otimes v = 1 \otimes ev = 0$.

HC9. The relative cotangent defect and its exact endpoint. The full time family has the two-term cotangent complex

$$L_{A/\mathcal{O}} = [A \xrightarrow{W_\xi} A d\xi], \quad \text{degrees } -1, 0. \quad (\text{HC20})$$

One construction starts with the free graded algebra over $\mathcal{O}[\xi]$ on one exterior generator η of degree -1 , with differential $d\eta = W$. Its underlying two-term complex is exact in degree -1 because multiplication by the monic W is injective; its degree-zero cokernel is A . Taking relative differentials and tensoring with A sends $d\eta$ to $W_\xi d\xi$, proving HC20.

Multiplication by W_ξ is injective on A : in the product of fraction fields of HC5 its value on each branch is the product of all nonzero differences from the other branches. It is invertible there, and A injects into that product. Thus

$$H^{-1}(L_{A/\mathcal{O}}) = 0, \quad \Omega = H^0(L_{A/\mathcal{O}}) = A/(W_\xi) d\xi. \quad (\text{HC21})$$

The module Ω is a finite torsion $\mathcal{O}$ -module with

$$\text{length}_{\mathcal{O}} \Omega = M = \frac{m(m-1)}{2}. \quad (\text{HC22})$$

For proof, the determinant of multiplication by W_ξ in A is the product of its branch values, hence $(-1)^M \mathfrak{D}_A(t)$. A square matrix over the discrete valuation ring with nonzero determinant has cokernel length equal to the determinant valuation: successive row and column elimination selects an entry of least valuation, clears its row and column by division, and iterates to a diagonal matrix of powers of t times units. Its cokernel has the corresponding sum of exponents, equal to the valuation of the determinant. HC14 now proves HC22.

Specializing the actual complex, rather than its cohomology alone, gives $[A_0 \xrightarrow{m\xi^{m-1}} A_0 d\xi]$, where $A_0 = \mathbb{C}[\xi]/\xi^m$. The free $\mathcal{O}$ -resolution in HC21 gives the exact base-change calculation

$$\begin{aligned} \text{Tor}_1^{\mathcal{O}}(\Omega, \mathbb{C}) &\simeq (\xi) \subset A_0, \\ \Omega \otimes_{\mathcal{O}} \mathbb{C} &\simeq A_0/(\xi^{m-1}) d\xi. \end{aligned} \quad (\text{HC23})$$

For $m = 1$ both sides vanish. For $m \geq 2$ both displayed spaces have dimension $m - 1$. The first map sends the cycle represented by $a\eta$ to its original coefficient $a \in (\xi)$. Consequently the endpoint radical of the original Weil trace EP11 is precisely this explicit derived base-change group. It occurs even though the relative family has zero H^{-1} . This extends the programme's collision Jacobian/cotangent comparison RD27–28a to the original arithmetic time family; the complete argument is given here, with its original coordinates.

HC10. A perfect observation of the torsion throughout the family. For $f \in A$ let $\lambda_A(f)$ be the coefficient of ξ^{m-1} in its unique representative of degree less than m . The bilinear form $\lambda_A(fg)$ is perfect over $\mathcal{O}$. In the power basis its entries with $i + j < m - 1$ are 0 and those with $i + j = m - 1$ are 1: products of degree less than m require no reduction. Reversing the columns therefore gives a triangular matrix with diagonal 1. Its determinant is $(-1)^M$, a unit. This proves perfection without deleting the terms above the anti-diagonal.

The exact trace identity is

$$\text{Tr}_{A/\mathcal{O}}(M_f) = \lambda_A(W_\xi f). \quad (\text{HC24})$$

On a nonzero fibre, Lagrange interpolation says $\lambda_A(g) = \sum_j g(\theta_j)/W_\xi(\theta_j)$: the j th interpolation basis polynomial has leading coefficient $1/W_\xi(\theta_j)$. Taking $g = W_\xi f$ gives HC24 there. Both sides

are analytic in t , so the identity holds at zero too. This proves the trace-to-dual factorization through the actual Jacobian multiplier W_ξ .

Omit the symbol $d\xi$ when identifying Ω with its coefficient module in the following explicitly defined pairing:

$$\mathcal{L} : \Omega \times \Omega \longrightarrow \mathcal{K}/\mathcal{O}, \quad ([a], [b]) \longmapsto \lambda_A \left(\frac{ab}{W_\xi} \right) \pmod{\mathcal{O}}. \quad (\text{HC25})$$

The inverse of W_ξ is taken in $A \otimes \mathcal{K}$, where it exists by HC21. Adding $W_\xi c$ to either representative changes the value by $\lambda_A(cb) \in \mathcal{O}$, proving well-definedness. If the value is zero for all $b \in A$, perfection of λ_A says that a/W_ξ belongs to A , and therefore $a \in W_\xi A$. This proves nondegeneracy. It is perfect: a cyclic module $\mathcal{O}/(t^q)$ has dual into $\mathcal{K}/\mathcal{O}$ equal to $t^{-q}\mathcal{O}/\mathcal{O}$, of the same length q . Applying the elementary divisor argument of HC22 to Ω shows that the injective duality map just proved has equal-length source and target, hence is an isomorphism.

At the endpoint the corresponding finite duality is

$$(\xi) \times A_0/(\xi^{m-1}) \longrightarrow \mathbb{C}, \quad (a, [b]) \longmapsto \lambda_{A_0}(ab). \quad (\text{HC26})$$

It is well defined because $\xi\xi^{m-1} = 0$. The bases $\xi, \dots, \xi^{m-1}$ and $1, \dots, \xi^{m-2}$ have a reversed identity pairing matrix, so it is perfect. Thus the trace-invisible endpoint directions have both an exact family torsion observation HC25 and an exact endpoint dual receiver HC26. These are bilinear residue pairings, with the trace relation HC24; they are not assigned positivity that their definitions do not give.

HC11. The full prime spectrum at the actual collision. The preceding calculation also determines the new prime fibres; it must not be read only at the level of complex vector spaces. Write $z_\lambda = (0, \lambda)$ and let $\mathfrak{e} = (e) \subset G_L(R)$ for any ring R . The multiplication $ez_\lambda = z_\lambda$ gives

$$\mathfrak{e} = \{z_\lambda : \lambda \in L\}, \quad \sqrt{\mathfrak{e}} = p^{-1}(\text{Nil } R). \quad (\text{HC27})$$

Indeed every element of the principal ideal has zero amplitude, every zero label is its displayed multiple, and a top-supported a has a power in this ideal exactly when a is nilpotent. For the original $A_0 = \mathbb{C}[\xi]/(\xi^m)$ this is

$$\sqrt{(e)} = Q_{(\xi)} = p^{-1}((\xi)), \quad V(e) = \{Q_{(\xi)}\}. \quad (\text{HC28})$$

When $m \geq 2$, (e) itself is not prime in this collision algebra: the nonzero top-supported elements $\widehat{\xi}$ and $\widehat{\xi^{m-1}}$ multiply to e , and neither belongs to (e) . For $m = 1$ the amplitude ring is a field and (e) is prime. This states precisely how the supported-zero prime in the integral coefficient ring changes under the actual nonreduced specialization. It retains the supported ideal, its radical and its prime receiver.

Here is the complete classification, including every lower label. A semiring prime omitting e contains no top-supported amplitude: if it contained $\widehat{a}$, multiplying by e would put e in that prime. Its zero labels form a proper lattice prime ideal J , and the corresponding semiring prime is $P_J = \{z_\lambda : \lambda \in J\}$. Closure under addition is closure under joins, and multiplicative primality is primality for meets. Conversely these conditions verify directly that every P_J is a semiring prime. A prime containing e contains every zero label; its top amplitudes form a ring prime $\mathfrak{p}$, since addition, multiplication and additive inversion (multiplication by -1) are preserved. Its exact form is $Q_{\mathfrak{p}} = p^{-1}(\mathfrak{p})$. Conversely each such inverse image is prime. Every P_J is contained in every $Q_{\mathfrak{p}}$, because its elements have zero amplitude. This reproves the complete support/arithmetical specialization order.

For the original special-fibre map HC11 the resulting spectral map is

$$\begin{array}{ccc} \text{Spec}_{\text{lat}} L \blacktriangleright \{q_1, \dots, q_r, q_0 \text{ if } \varepsilon = 1\} & \longrightarrow & \text{Spec}_{\text{lat}} L \blacktriangleright \{q\}, \\ P_J & \longmapsto & P_J, \\ q_a \text{ and } q_0 & \longmapsto & q. \end{array} \quad (\text{HC29})$$

Here $q = Q_{(\xi)}$, and q_a is the prime of the a th dual-number factor (with all other product components in the ideal); q_0 is the central field-factor prime. Evaluation of HC11 proves every inverse image in HC29. The coefficient object on the left is $G_L(B_0)$ with its one complete support lattice L , not the larger product of independent support carriers. Thus the exact number $r + \varepsilon$ of arithmetic branch points, their common image, the shared support primes and the conductor are all retained. Localizing e keeps the identity on the displayed lattice spectrum; it removes the arithmetic points through the same universal map as HC18–19. The branch and holonomy information is carried by the original inclusion and its arithmetic fibres before that map.

HC12. The discriminant is measured by the holonomy commutator. Use the original coefficient holonomy $M_A(t)$ and the complete scaling multiplier $S_L(t)$ of HH9 and HH29. For fixed real L and small positive t , the reflected trace is positive definite, so it defines the Hilbert–Schmidt norm on its endomorphisms. The exact expansion proved in HH33–38, combined with HC17, gives

$$\begin{aligned} \|[M_A(t), S_L(t)]\|_{\text{HS}, T_t}^2 &= 2L^2 m(m-1)t \\ &+ \left[4L^2 \partial_t \log \frac{\mathfrak{D}_A(t)}{D_m t^M} \Big|_{t=0} - \frac{L^4}{3} m(m-1)(2m-3) \right] t^2 + O(t^3). \end{aligned} \quad (\text{HC30})$$

No metric is replaced: HH18 identifies this exact norm with the Euclidean norm of value vectors at the same positive time, and HC12 gives its full original-coordinate matrix. HC17 evaluates the derivative by the original a_n . Thus the conductor discriminant, the arithmetic energy finite part and the measured noncommutativity of the two transports have an explicit common coefficient. This is a quantitative receiver for the original geometry; no sign of the complete global Weil form is inferred from this positive-time local norm.

21 Parameter holonomy, the reflected trace, and arithmetic scaling

HH1. Original arithmetic family and scope. We use precisely the convention of Brad Rodgers and Terence Tao, *The de Bruijn–Newman constant is non-negative*, arXiv:1801.05914v5, author source equations `hoz`, `sas`, `phidef`, `htdef` and the sentence following `htdef`:

$$\begin{aligned} H_t(Z) &= \int_0^\infty e^{tu^2} \Phi(u) \cos(Zu) du, & H_0(Z) &= \tfrac{1}{8} \xi_R(\tfrac{1}{2} + iZ/2), \\ \Phi(u) &= \sum_{n \geq 1} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) e^{-\pi n^2 e^{4u}}, & \partial_t H_t &= -\partial_Z^2 H_t. \end{aligned} \quad (\text{HH1})$$

Here $\xi_R(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)/2$. The subscript prevents confusion with the local coordinate ξ below. Fix an actual real zero x_0 of H_0 and retain its actual order $m \geq 1$ and every coefficient

$$\begin{aligned} H_0(x_0 + \xi) &= \sum_{n \geq m} a_n \xi^n, & a_n &= \partial_Z^n H_0(x_0)/n!, & a_m &\neq 0, \\ c &= a_{m+1}/a_m, & d &= a_{m+2}/a_m, & b &= 2d - c^2. \end{aligned} \quad (\text{HH2})$$

This describes each actual real zero without asserting that an actual multiple zero exists. At $t = 0$ the constructions below are original completed-zeta jets and their reflected Weil trace contribution. At $t \neq 0$ they are the explicitly defined trace and multiplier on the actual deformed H_t cluster. An explicit formula for the original zeta function is not asserted for these deformed fibres.

For completeness the analyticity used here follows directly from HH1. On compact subsets of complex (t, Z) every differentiated integrand is bounded by a constant times $e^{Cu^2 + Cu}(1 +$

$u)^N e^{-c_1 e^{4u}}$ for $u \geq 1$, with $c_1 > 0$. To obtain this estimate, absorb polynomial factors of n into $e^{-\pi n^2 e^{4u}/2}$, and sum the remaining Gaussian tail uniformly. The series and its derivatives are uniformly convergent on bounded u intervals. The displayed bound is integrable and remains so after any finite number of t and Z derivatives. Differentiation under the integral thus proves joint holomorphy and the heat equation. For real t all local coefficients are real. These facts also prove the uniform convergence on a small fixed root-isolating circle that is used below.

HH2. All analytic branches, with their original coefficients. Define

$$P_n(z) = \sum_{j=0}^{\lfloor n/2 \rfloor} \frac{(-1)^j n!}{j!(n-2j)!} z^{n-2j}. \quad H_{u^2}(x_0 + uz) = \sum_{n \geq m} a_n u^n P_n(z). \quad (\text{HH3})$$

The second equality is convergent for bounded z and small u : the heat equation gives $[t^j \xi^\ell] H_t = (-1)^j (\ell + 2j)! a_{\ell+2j} / (j! \ell!)$, and the locally absolutely convergent double Taylor series can be regrouped by $n = 2j + \ell$. This P_n is the classical Hermite polynomial $H_n^{\text{Hermite}}(z/2)$; the explicit finite formula is also DLMF 18.5.13, Chapter 18 by T.H. Koornwinder, R. Wong, R. Koekoek and R.F. Swarttouw. All zero properties used here have the following proof. The generating identity $\sum P_n(z) w^n / n! = e^{zw - w^2}$ gives

$$P'_n = n P_{n-1}, \quad P_{n+1} = z P_n - 2n P_{n-1}, \quad 2P''_n - z P'_n + n P_n = 0, \quad P_n e^{-z^2/4} = (-2)^n \partial_z^n e^{-z^2/4}. \quad (\text{HH4})$$

Integration by parts makes P_n orthogonal to each polynomial of degree less than n for the positive weight $e^{-z^2/4}$. If it had fewer than n distinct real sign-changing roots, multiply it by the product of those linear factors. The product has constant nonzero sign away from finitely many points, contradicting this orthogonality. Hence P_m has exactly m real simple roots $r_1 < \dots < r_m$. Its parity gives $r_{\sigma j} = -r_j$, where $\sigma j = m + 1 - j$.

Apply the analytic implicit-function theorem to $H_{u^2}(x_0 + uz)/(a_m u^m)$ at $(0, r_j)$. It gives convergent germs $\theta_j(u) = u z_j(u)$, with

$$H_{u^2}(x_0 + \theta_j(u)) = 0, \quad \theta_j(u) = r_j u + 2c u^2 + r_j b u^3 + O(u^4), \\ \theta_j(-u) = \theta_{\sigma j}(u), \quad \theta_j(\bar{u}) = \overline{\theta_j(u)}. \quad (\text{HH5})$$

Here the coefficients beyond the displayed terms remain the unique convergent coefficients specified by HH3; no polynomial replacement of H is made. To compute the displayed terms, at a root of P_m , HH4 yields $P_{m+1} = -2P'_m$, $P'_{m+1} = 0$, $P''_m = r_j P'_m / 2$, $P_{m+2} = -2r_j P'_m$. Substitution of $z_j = r_j + A_j u + B_j u^2 + O(u^3)$ gives $A_j = 2c$ and $B_j = r_j b$. Complex conjugation and the change $u \mapsto -u$ give the last two identities by uniqueness of the corresponding implicit germ. Every difference $\theta_j - \theta_k$ equals $u(r_j - r_k + O(u))$, and is nonzero for small $u \neq 0$. An isolating circle about x_0 , followed by the argument principle and uniform convergence, retains exactly m zeros for small t . These m branches therefore exhaust the original local cluster.

HH3. The original algebra, its full unit, and its evaluation map. Put $\mathcal{O} = \mathbb{C}\{t\}$ and let $\mathcal{K}$ be its fraction field. Define

$$W(t, \xi) = \prod_{j=1}^m (\xi - \theta_j(u)), \quad t = u^2, \quad H_t(x_0 + \xi) = U(t, \xi) W(t, \xi), \quad \mathcal{A} = \mathcal{O}[\xi] / (W). \quad (\text{HH6})$$

The product is even in u by HH5, so its coefficients are convergent germs in t . Its coefficients are real, $W(0, \xi) = \xi^m$, and U is a holomorphic unit with $U(0, \xi) = \sum_{j \geq 0} a_{m+j} \xi^j$. One can obtain the latter assertion without discarding any unit: divide the jointly holomorphic function $H_{u^2}(x_0 + \xi)$ successively by $\xi - \theta_j(u)$. Holomorphic division by one such factor follows from the integral of its ξ derivative along a line segment. At each remaining root the quotient vanishes for $u \neq 0$, hence also at $u = 0$ by analyticity. All m divisions are therefore valid. The resulting

quotient is even in u , since this is so away from the zero set, and hence everywhere by analytic continuation. It descends to $U(t, \xi)$, with $U(0, 0) = a_m \neq 0$.

In the original basis $1, \xi, \dots, \xi^{m-1}$ set

$$V(u)_{jk} = \theta_j(u)^k \quad (1 \leq j \leq m, 0 \leq k < m). \quad V(-u) = PV(u), \quad (Py)_j = y_{\sigma j}. \quad (\text{HH7})$$

Its determinant is $\prod_{j < k} (\theta_k - \theta_j)$, nonzero off $u = 0$. For a coefficient column f , $V(u)f$ is its exact vector of values. This retains every coefficient of every actual branch and the entire original nonreduced central algebra $\mathcal{A}/(t) = \mathbb{C}[\xi]/(\xi^m)$.

HH4. The exact Gauss–Manin connection and the unit contribution. The degree-zero Gauss–Manin connection here means differentiation of functions along the finite etale root cover, on the punctured disc. Its exact formula in $\mathcal{A}_{\mathcal{K}}$ is

$$\begin{aligned} \nabla f &= \partial_t f + q \partial_{\xi} f, & q &= -\frac{W_t}{W_{\xi}} = \frac{H_{\xi\xi}}{H_{\xi}} = \frac{W_{\xi\xi}}{W_{\xi}} + 2\frac{U_{\xi}}{U} \pmod{W}, \\ \nabla f &= f' + \Gamma_{\mathcal{A}} f, & \Gamma_{\mathcal{A}}(t) &= V(u)^{-1} \frac{1}{2u} \partial_u V(u). \end{aligned} \quad (\text{HH8})$$

Every inverse in this equation is taken in the etale algebra over $\mathcal{K}$, where W_{ξ} is invertible; reduction modulo W is performed after the indicated operations. Indeed differentiating $W(t, \theta_j) = 0$ gives $\theta'_j = -W_t/W_{\xi}$. Differentiating the actual equation $H = 0$ and using HH1 gives $\theta'_j = H_{\xi\xi}/H_{\xi}$. Differentiating $H = UW$ twice in ξ proves the last expression at every simple root, and hence in the etale algebra. The expression $\partial_t + q\partial_{\xi}$ annihilates W modulo W , so it is independent of polynomial representative. In the original power basis the k th column is the coefficient column of $k\xi^{k-1}q$ modulo W , including a zero column for $k = 0$. The chain rule applied to Vf proves the matrix formula. Its invariance under $u \mapsto -u$ follows from HH7, so it is meromorphic in t . In particular the term $2U_{\xi}/U$ is part of the actual connection and cannot be suppressed.

HH5. Complete loop holonomy. For a path in the punctured t disc, choose its continuous square-root lift with endpoints u_0, u_1 . Horizontality is precisely $(V(u)f)' = 0$. Thus the exact coefficient transport and positive once-around holonomy at a base point $t_0 = u_0^2$ are

$$C_{1 \leftarrow 0} = V(u_1)^{-1}V(u_0), \quad M_{\mathcal{A}}(t_0) = V(u_0)^{-1}PV(u_0), \quad M_{\mathcal{A}}^2 = I. \quad (\text{HH9})$$

These formulas are independent of the initial choice of square root. A loop of winding number n has holonomy $M_{\mathcal{A}}^n$, because the square-root endpoint is $(-1)^n u_0$. There are $r = \lfloor m/2 \rfloor$ transpositions and $\epsilon = m - 2r \in \{0, 1\}$ fixed branches. The multiplicities of the holonomy eigenvalues $+1, -1$ are therefore $r + \epsilon, r$. In particular a multiple real zero gives nonidentity parameter holonomy, although its central reflected trace is nonnegative, as will be calculated below. This holonomy is a meromorphic automorphism of $\mathcal{A}_{\mathcal{K}}$; regularity on the original integral lattice $\mathcal{A}$ has not been assumed.

HH6. A logarithmic lattice and the exact integral-lattice defect. Over the u cover let

$$e_j = \prod_{k \neq j} \frac{\xi - \theta_k(u)}{\theta_j(u) - \theta_k(u)}. \quad (\text{HH10})$$

Their evaluations are the coordinate unit vectors, so they are horizontal idempotents. For each positive leading root $r_a > 0$ choose its pair $(a, +), (a, -)$ with leading roots $r_a, -r_a$ and set

$$\mathbf{e}_a = e_{a,+} + e_{a,-}, \quad \mathbf{f}_a = u(e_{a,+} - e_{a,-}), \quad \mathcal{L} = \bigoplus_{a=1}^r (\mathcal{O}\mathbf{e}_a \oplus \mathcal{O}\mathbf{f}_a) \oplus \mathcal{O}^{\epsilon}\mathbf{e}_0. \quad (\text{HH11})$$

Here $\mathbf{e}_0$ is the central idempotent if m is odd; the last summand is absent otherwise. All these elements are invariant under $u \mapsto -u$ and so belong to $\mathcal{A}_{\mathcal{K}}$. Products from different summands

vanish, $\mathbf{e}_a^2 = \mathbf{e}_a$, $\mathbf{e}_a \mathbf{f}_a = \mathbf{f}_a$, $\mathbf{f}_a^2 = t \mathbf{e}_a$ and $1 = \sum_a \mathbf{e}_a + \epsilon \mathbf{e}_0$. They are a basis over $\mathcal{K}$ and define an $\mathcal{O}$ algebra. They also give the explicit matrices

$$\nabla|_a = \partial_t + \begin{pmatrix} 0 & 0 \\ 0 & 1/(2t) \end{pmatrix}, \quad S_{\mathcal{L}}|_a = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}; \quad \nabla \mathbf{e}_0 = 0, \quad S_{\mathcal{L}} \mathbf{e}_0 = \mathbf{e}_0. \quad (\text{HH12})$$

Indeed $\nabla e_j = 0$ and $\partial_t u = u/(2t)$. This proves regular singularity by exhibiting a logarithmic lattice, with no claim that the original power basis has a simple-pole matrix.

For clarity the exact original-coordinate inclusion $J : \mathcal{A} \hookrightarrow \mathcal{L}$ is

$$(Jf)_{a,\text{even}} = \frac{f(\theta_{a,+}(u)) + f(\theta_{a,-}(u))}{2}, \quad (Jf)_{a,\text{odd}} = \frac{f(\theta_{a,+}(u)) - f(\theta_{a,-}(u))}{2u}, \\ (Jf)_0 = f(\theta_0(u)). \quad (\text{HH13})$$

The two displayed numerators have the required parity, so these are holomorphic germs in t . For $f = \xi^k$ they give the exact matrix columns. The map is injective because it is injective off zero and $\mathcal{A}$ is free over $\mathcal{O}$. Its determinant is nonzero, so its cokernel has finite length. In particular

$$M_{\mathcal{A}} = J^{-1} S_{\mathcal{L}} J, \quad \mathcal{D}_M = \frac{\mathcal{A} + M_{\mathcal{A}} \mathcal{A}}{\mathcal{A}} = \frac{J \mathcal{O}^m + S_{\mathcal{L}} J \mathcal{O}^m}{J \mathcal{O}^m} \subset \mathcal{L}/\mathcal{A}. \quad (\text{HH14})$$

This is an exact finite-length object, with the exact sequence $0 \rightarrow \mathcal{A} \rightarrow \mathcal{A} + M_{\mathcal{A}} \mathcal{A} \rightarrow \mathcal{D}_M \rightarrow 0$. The middle module is the smallest $M_{\mathcal{A}}$ -stable lattice containing $\mathcal{A}$: applying the involution interchanges its two summands. Its rank is m , and it is free since a finite torsion-free module over the discrete valuation ring $\mathcal{O}$ is free. No assertion that this sum is an algebra is needed. $\mathcal{D}_M = 0$ is exactly the assertion that the loop preserves the original lattice. Equations HH13–14 compute this defect from the full actual coefficients even when it is not zero.

The scalar even and odd branch germs

$$b_a(t) = \frac{\theta_{a,+}(u) + \theta_{a,-}(u)}{2}, \quad R_a(t) = \frac{\theta_{a,+}(u) - \theta_{a,-}(u)}{2u} \quad (\text{HH15})$$

satisfy $b_a(t) = 2ct + O(t^2)$, $R_a(t) = r_a(1 + bt) + O(t^2)$ and $R_a(0) = r_a \neq 0$. They retain all higher coefficients, and the original coordinate is exactly

$$\xi = \sum_a (b_a(t) \mathbf{e}_a + R_a(t) \mathbf{f}_a) + \epsilon \theta_0(t) \mathbf{e}_0. \quad (\text{HH16})$$

The scalar θ_0 is even in u and is real for real t . This provides the concrete relation between the auxiliary lattice and the full original algebra without identifying their special fibres.

HH7. The reflected trace and its three exact inertias. For real t define the conjugate-linear involution $\#$ by conjugating coefficients and fixing ξ . Put

$$T_t(f, g) = \text{Tr}_{\mathcal{A}_t}(M_{f\#g}), \quad K_{\mathcal{A}}(t)_{k\ell} = \text{Tr}_{\mathcal{A}_t}(M_{\xi^{k+\ell}}) = \sum_{j=1}^m \theta_j(u)^{k+\ell}. \quad (\text{HH17})$$

The last expression extends analytically to $t = 0$. For nonzero t it follows by diagonalizing the multiplication operators via V ; at zero direct multiplication on $\mathbb{C}[\xi]/\xi^m$ gives $\text{Tr}(M_f) = mf(0)$. Thus $T_t(f, g) = f^* K_{\mathcal{A}}(t) g$ for real t , and the matrix is real symmetric. For $t > 0$ choose $u = \sqrt{t}$; every root is real. For $t < 0$ choose $u = i\sqrt{-t}$. HH5 then gives $\overline{\theta_j(u)} = \theta_{\sigma_j}(u)$; each noncentral pair is nonreal for sufficiently small negative t . Consequently

$$K_{\mathcal{A}}(t) = V(u)^T V(u) = \begin{cases} V(u)^* V(u), & t > 0, \\ V(u)^* P V(u), & t < 0, \end{cases} \quad K_{\mathcal{A}}(0) = \text{diag}(m, 0, \dots, 0). \quad (\text{HH18})$$

The first equality is a holomorphic bilinear identity for complex t ; the other two are the Hermitian identities for the indicated real fibres. Congruence by the invertible V proves the full inertia, written as (positive, negative, zero):

$$\text{inertia } T_t = \begin{cases} (m, 0, 0), & t > 0, \\ (1, 0, m-1), & t = 0, \\ (r + \epsilon, r, 0), & t < 0. \end{cases} \quad \text{rad } T_0 = (\xi). \quad (\text{HH19})$$

On $\mathcal{L}$ the trace Gram matrix is exactly

$$K_{\mathcal{L}}(t) = \bigoplus_{a=1}^r \begin{pmatrix} 2 & 0 \\ 0 & 2t \end{pmatrix} \oplus (1)^\epsilon, \quad K_{\mathcal{A}}(t) = J(t)^T K_{\mathcal{L}}(t) J(t). \quad (\text{HH20})$$

For proof, $\#$ fixes $\mathbf{e}_a, \mathbf{f}_a$ for both real signs of t ; at negative t , conjugation of u and exchange of the idempotents cancel. The algebra relations give the traces $2, 0, 2t$. The congruence follows first off zero by HH13 and then at zero analytically. At $t = 0$ the map on fibres induced by J need not be injective. Accordingly HH20 does not replace the retained algebra $\mathbb{C}[\xi]/\xi^m$ by r dual-number factors.

HH8. Flatness, loop isometry, and the half-plane metric defect. Differentiating the holomorphic identity $K = V^T V$ proves

$$K' = \Gamma_{\mathcal{A}}^T K + K \Gamma_{\mathcal{A}}, \quad C_{1 \leftarrow 0}^T K(t_1) C_{1 \leftarrow 0} = K(t_0). \quad (\text{HH21})$$

For real t the connection matrix is real, so the first equality also states compatibility with the Hermitian trace on each real interval not containing zero. At a fixed nonzero real t , HH9 and HH18 give $M_{\mathcal{A}}^* K M_{\mathcal{A}} = K$, because P is an isometry both for I and for P . Thus the loop is unitary for the positive trace at $t > 0$, and preserves the indefinite trace at $t < 0$.

There is also an exact defect when moving between the two real structures. Let $t_+ > 0, t_- < 0$, choose the upper-half-plane path whose square-root lift goes from $u_+ = \sqrt{t_+}$ to $u_- = i\sqrt{-t_-}$, and let $C = V(u_-)^{-1} V(u_+)$. Then

$$K(t_+) - C^* K(t_-) C = V(u_+)^* (I - P) V(u_+) = 2V(u_+)^* \Pi_- V(u_+), \quad \Pi_- = (I - P)/2. \quad (\text{HH22})$$

This is a positive semidefinite matrix of rank r . Indeed the first real fibre has value Gram matrix I , while the second has value Gram matrix P ; substitution gives the equation, and Π_- is the orthogonal projection on the antisymmetric pair directions. Its image and kernel are therefore exact: the defect vanishes precisely when the transported value vector has equal values within each pair. HH21 still holds with transpose. There is no inconsistency: HH22 measures the different endpoint conjugations in the Hermitian forms.

HH9. The full-unit residue and the original Jacobian receiver. Introduce the entire deformed function and its exact arithmetic coordinate

$$g_t(s) = 16H_t(-2i(s - \tfrac{1}{2})), \quad s = \rho + i\xi/2, \quad \rho = \tfrac{1}{2} + ix_0/2, \quad g_0(s) = 2\xi_{\mathbb{R}}(s). \quad (\text{HH23})$$

For real t , reflection is $s \mapsto 1 - \bar{s}$, or $\xi \mapsto \bar{\xi}$ on points. On coordinate algebras $\xi^\# = \xi$, whereas $v^\# = -v$ for $v = s - \rho = i\xi/2$. The raw residue pairing and Jacobian multiplier are

$$R_t(f, g) = \frac{i}{32} \sum_{\text{cluster}} \text{Res}_\xi \frac{f^\#(\xi) g(\xi)}{U(t, \xi) W(t, \xi)} d\xi, \quad \mathcal{J}_t = M_{g'_t(s)} = -32i M_{H_\xi(t, x_0 + \xi)},$$

$$R_t(f, \mathcal{J}_t g) = T_t(f, g). \quad (\text{HH24})$$

The factor $i/32$ is exactly $ds/16 = (i/32)d\xi$. The pairing is well-defined modulo W , since changing a numerator by a multiple of W adds a holomorphic differential. It is perfect on

every fibre. At a simple root its weight $i/(32H_\xi)$ is nonzero. At $t = 0$ the matrix has entries $(i/32)[\xi^{m-1-k-\ell}]U(0, \xi)^{-1}$, zero when $k + \ell > m - 1$ and equal to $i/(32a_m)$ on the antidiagonal. Its determinant is therefore nonzero. This is a perfect sesquilinear residue pairing; with these real-coordinate conventions it is skew-Hermitian, not the Hermitian trace itself.

The last equality in HH24 follows by cancellation of the constants and the identity $H_\xi/H = W_\xi/W + U_\xi/U$. The unit term has no residue. At each root of multiplicity n , the residue of $f^\# g W_\xi/W$ is $n(f^\# g)$ evaluated at that root, exactly the regular multiplication trace. This proves the receiver at every fibre, including the original nonreduced central fibre. In matrix form, for $t \neq 0$ and $C_t = I$ or P as in HH18,

$$R_t = \frac{i}{32} V^* C_t \operatorname{diag}(H_\xi(t, x_0 + \theta_j)^{-1}) V, \quad \mathcal{J}_t = V^{-1} \operatorname{diag}(-32iH_\xi(t, x_0 + \theta_j)) V. \quad (\text{HH25})$$

Their product is precisely K_A , with all units retained.

The compatibility with the connection has its own exact unit-sensitive formula. In the punctured algebra set

$$\kappa = -\frac{H_{\xi\xi\xi}}{H_\xi} + \left(\frac{H_{\xi\xi}}{H_\xi}\right)^2 \pmod{W}. \quad (\text{HH26})$$

For horizontal f, g on a real interval, $\frac{d}{dt} R_t(f, g) = -R_t(f, M_\kappa g)$. Indeed the values of f, g are constant along each branch, while $(\partial_t + q\partial_\xi) \log H_\xi = -H_{\xi\xi\xi}/H_\xi + H_{\xi\xi}^2/H_\xi^2$ by HH1 and HH8. Differentiating the weights in HH25 proves the identity. The Jacobian satisfies $[\nabla, \mathcal{J}_t] = M_\kappa \mathcal{J}_t$ as a multiplier; its covariant derivative cancels this residue variation and recovers the trace compatibility HH21.

HH10. Exact specialization to the arithmetic jet and its Weil value. At $t = 0$ the coordinate map

$$\begin{aligned} \mathbb{C}[v]/(v^m) &\longrightarrow \mathcal{A}_0, & v &\longmapsto i\xi/2, & g_0(\rho + v) &= v^m u_\rho(v), \\ u_\rho(v) &= 16(-2i)^m \sum_{j \geq 0} a_{m+j} (-2i)^j v^j \end{aligned} \quad (\text{HH27})$$

is an isomorphism of the reflected algebras, including the full analytic unit in the displayed local equation. Consequently

$$T_0(f, g) = m \overline{f(0)} g(0), \quad (\log u_\rho)''(0) = 4(c^2 - 2d). \quad (\text{HH28})$$

The equality with the original local Weil contribution means exactly the multiplicity- m summand at the fixed reflected zero $\rho = 1 - \bar{\rho}$ in the spectral side $\sum_\rho m_\rho \overline{F(1 - \bar{\rho})} G(\rho)$. The complete explicit formula retains its other zeros, its prime terms, its Archimedean term and its endpoint terms; none is replaced by the finite cluster. The original convention and criterion are Alain Connes and Caterina Consani, *Weil positivity and trace formula, the archimedean place*, arXiv:2006.13771v1, author source introduction and Appendix mainprop; the full original-map derivation is also given in HW19–31 of this collection. The local equality here is proved independently by HH24 and HH27. It asserts no positivity of the remaining spectral packet.

HH11. Arithmetic scaling and its full deformed multiplier. For real L define, without any change of original coordinate,

$$\begin{aligned} S_L(t) &= M_{\exp(iL(x_0 + \xi)/2)}, & D_L(u) &= \operatorname{diag}(d_j), \\ d_j &= \exp(iL(x_0 + \theta_j(u))/2), & S_L &= V^{-1} D_L V. \end{aligned} \quad (\text{HH29})$$

The multiplier defines a holomorphic algebra section: it can be reduced modulo the monic W , or its entire convergent power series can be applied to the holomorphic multiplication matrix of ξ . At the arithmetic endpoint this is exactly

$$S_L(0) = e^{L(\rho-1/2)} \exp(LM_v) = e^{iLx_0/2} \sum_{k=0}^{m-1} \frac{L^k M_v^k}{k!}. \quad (\text{HH30})$$

It includes the original nilpotent factor on every jet layer. This is the shifted multiplicative arithmetic character convention used in Connes–Consani’s spectral realizations. Their paper *Hochschild homology, trace map and zeta cycles*, arXiv:2207.10419, author source Remark **remtauto**, explicitly retains the Jordan blocks at multiple critical zeros. HH30 derives the entire finite matrix here, so no simplicity assumption is imported.

For every real t , $(\exp(iL(x_0 + \xi)/2))^\# \exp(iL(x_0 + \xi)/2) = 1$. Thus

$$S_L(t)^* K(t) S_L(t) = K(t). \quad (\text{HH31})$$

This also preserves the raw residue by the same product identity. For $t > 0$ each d_j has modulus one. For $t < 0$ the paired values obey $d_{\sigma j} = 1/\bar{d}_j$ and $|d_j| = e^{-L \operatorname{Im} \theta_j/2}$. When $L \neq 0$, each noncentral pair then has reciprocal nonunit moduli. These are zeros of the actual deformed g_t , not zeros of g_0 at the arithmetic endpoint. They preserve an indefinite trace with inertia HH19. At $t = 0$ the nilpotent part in HH30 preserves the radical (v) and is invisible to the value trace, but remains detected by the perfect residue HH24.

In the auxiliary pair basis $(\mathbf{e}_a, \mathbf{f}_a)$ the exact same multiplier is

$$S_{L,a}(t) = e^{iL(x_0 + b_a(t))/2} \begin{pmatrix} \cos z_a & iu \sin z_a \\ i \sin z_a / u & \cos z_a \end{pmatrix}, \quad z_a = LR_a(t)u/2. \quad (\text{HH32})$$

The entries are analytic in t by parity, and the value at zero is $e^{iLx_0/2} \begin{pmatrix} 1 & 0 \\ iLR_a/2 & 1 \end{pmatrix}$. To prove HH32, write the multiplier as $A + B\mathbf{f}_a$ on this factor, using $\mathbf{f}_a^2 = t$, and take its exponential series. Multiplication by $A + B\mathbf{f}_a$ has matrix $\begin{pmatrix} A & tB \\ B & A \end{pmatrix}$. This proves every entry and its limit. The auxiliary central matrix does not replace the length- m original matrix HH30.

HH12. The exact relation, rather than an identification, of the two actions. The connection and the multiplier satisfy

$$[\nabla, S_L] = \frac{iL}{2} M_q S_L, \quad M_{\mathcal{A}} S_L M_{\mathcal{A}}^{-1} = V^{-1} \operatorname{diag}(d_{\sigma j}) V, \quad V[M_{\mathcal{A}}, S_L] V^{-1} = PD_L - D_L P. \quad (\text{HH33})$$

The first formula is the Leibniz rule and $\nabla \xi = q$; it includes the entire unit contribution in HH8. The other two follow by substituting HH9 and HH29. Thus these actions commute exactly when each exchanged pair has equal exponential values. Nonidentity parameter holonomy alone neither says that these values have nonunit modulus nor gives a negative value of the arithmetic endpoint trace.

HH13. A quantitative action defect containing the finite collision energy. For $t > 0$ the actual trace T_t is positive definite, so its Hilbert–Schmidt operator norm is defined. The map V is an isometry onto $\mathbb{C}^m$ with its usual inner product. Therefore HH33 gives the exact formula

$$\| [M_{\mathcal{A}}, S_L] \|_{\text{HS}, T_t}^2 = \sum_j |d_j - d_{\sigma j}|^2 = 8 \sum_{a=1}^r \sin^2 \left(\frac{L(\theta_{a,+}(u) - \theta_{a,-}(u))}{4} \right). \quad (\text{HH34})$$

There is one nonzero entry per row of $PD - DP$; a pair contributes twice the same absolute square. The identity $|e^{i\alpha} - e^{i\beta}|^2 = 4 \sin^2((\alpha - \beta)/2)$ gives exactly the factor 8 and the denominator 4 above.

For each fixed real L its full second-order expansion is

$$\begin{aligned} \| [M_{\mathcal{A}}, S_L] \|_{\text{HS}, T_t}^2 &= 2L^2 m(m-1)t \\ &\quad + m(m-1) \left(4L^2(2d - c^2) - \frac{L^4}{3}(2m-3) \right) t^2 + O(t^3). \end{aligned} \quad (\text{HH35})$$

Here the remainder is a convergent-germ remainder for the fixed actual root and fixed L . To prove every constant, HH3 has $[z^{m-2}]P_m = -m(m-1)$ and $[z^{m-4}]P_m = m(m-1)(m-2)(m-3)/2$,

with missing coefficients taken as zero. Newton's identities, or direct expansion of the product, give

$$\sum_{a=1}^r r_a^2 = m(m-1), \quad \sum_{a=1}^r r_a^4 = 2m(m-1)(2m-3). \quad (\text{HH36})$$

For example the full fourth power sum is $2m^2(m-1)^2 - 2m(m-1)(m-2)(m-3) = 4m(m-1)(2m-3)$; parity halves it. The identities also hold for $m = 1, 2, 3$ by the same product expansion. Exact parity in HH5 improves the gap expansion to $\theta_{a,+} - \theta_{a,-} = 2r_a u + 2r_a b u^3 + O(u^5)$. Insert this in $\sin^2 z = z^2 - z^4/3 + O(z^6)$ and multiply by 8. The contribution of pair a is $2L^2 r_a^2 t + (4L^2 r_a^2 b - L^4 r_a^4/6)t^2 + O(t^3)$. Summing by HH36 proves HH35. For $m \geq 2$ and fixed $L \neq 0$ this norm is strictly positive for all sufficiently small $t > 0$, despite both actions being isometries. For $m = 1$ or $L = 0$ it is identically zero.

The finite energy term can be calculated from the same retained coefficients, with the ordered-pair convention:

$$E_{x_0}(t) = \sum_{j \neq k} \frac{1}{(\theta_j(u) - \theta_k(u))^2} = \frac{m(m-1)}{8t} - \frac{m(m-1)b}{4} + O(t). \quad (\text{HH37})$$

For a complete verification, put $A_j = \sum_{k \neq j} (r_j - r_k)^{-1}$ and $B_j = \sum_{k \neq j} (r_j - r_k)^{-2}$. Logarithmic differentiation of the product for P_m , followed by HH4 and its derivative, gives $A_j = r_j/4$ and $3(A_j^2 - B_j) = P_m'''(r_j)/P_m'(r_j) = r_j^2/4 - (m-1)/2$. Thus $B_j = (m-1)/6 - r_j^2/48$. HH36 gives $\sum_j B_j = m(m-1)/8$. Each gap equals $u(r_j - r_k)(1 + bu^2) + O(u^4)$, so substitution gives the stated pole and finite part, initially with $O(u)$ remainder. The full ordered sum is even under $u \mapsto -u$, which permutes the roots. Its Laurent series is even, improving the remainder to $O(u^2) = O(t)$. For $m = 1$ the sum is empty and all coefficients are zero. This finite cluster statement makes no assertion about the external interactions in the full Rodgers–Tao dynamics.

Combining HH27–28 and HH37 gives the exact arithmetic-unit receiver and the following equivalent form of the t^2 coefficient in HH35:

$$\begin{aligned} \text{FP}_{t=0} E_{x_0} &= \frac{m(m-1)}{16} (\log u_\rho)''(0), \\ [t^2] \|[M_A, S_L]\|_{\text{HS}, T_t}^2 &= -16L^2 \text{FP}_{t=0} E_{x_0} - \frac{L^4}{3} m(m-1)(2m-3). \end{aligned} \quad (\text{HH38})$$

This is a numerical relation between parameter transport, the explicit exponential multiplier, and the original arithmetic unit. Its proof retains all higher branch coefficients through HH3, HH7–8, HH13 and HH34; those coefficients determine all subsequent terms as convergent analytic functions. It supplies neither a nonreal zero of H_0 nor a negative test for the complete original Weil form.

HH14. The full supported fixed locus of the actual endpoint action. Let Λ be the original bounded distributive support lattice, with every one of its labels retained; use Λ here to distinguish it from the real scaling parameter L . For the original arithmetic jet $A_0 = \mathbb{C}[\xi]/(\xi^m)$ define its supported module exactly as in the original lattice-split construction:

$$\hat{A}_0 = \{(0, \lambda) : \lambda \in \Lambda\} \cup \{(f, 1_\Lambda) : f \in A_0\}, \quad (f, \lambda) + (g, \mu) = (f + g, \lambda \vee \mu). \quad (\text{HH39})$$

The scalars $(a, \lambda) \in G_\Lambda(\mathbb{C})$ act by $(a, \lambda)(f, \mu) = (af, \lambda \wedge \mu)$. Nonzero amplitudes occur only at top support, so the operations are closed and satisfy the semimodule axioms by distributivity of the lattice and the original vector-space operations. In particular actual absence $(0, 0_\Lambda)$ and supported zero $(0, 1_\Lambda)$ remain distinct when the lattice is nontrivial. Lift HH30 by

$$\hat{S}_L(f, \lambda) = (S_L(0)f, \lambda). \quad (\text{HH40})$$

This is a semimodule automorphism: the matrix is complex linear and invertible and the support coordinate is unchanged. It is not asserted to be a unital algebra automorphism.

Set $\alpha = iL/2$, $N = M_\xi$, and $\lambda_0 = e^{iLx_0/2}$. If $\lambda_0 \neq 1$, the original full-jet inverse is the finite expression

$$(S_L(0) - I)^{-1} = \frac{1}{\lambda_0 - 1} \sum_{j=0}^{m-1} \left[-\frac{\lambda_0}{\lambda_0 - 1} (e^{\alpha N} - I) \right]^j. \quad (\text{HH41})$$

Indeed $N^m = 0$, so $(e^{\alpha N} - I)^m = 0$, and multiplication by $S_L - I = (\lambda_0 - 1)I + \lambda_0(e^{\alpha N} - I)$ telescopes to the identity. No jet coefficient is omitted by this terminating inverse. If $\lambda_0 = 1$ and $L \neq 0$, instead

$$S_L(0) - I = \alpha N h(\alpha N), \quad h(z) = \sum_{k=0}^{m-1} \frac{z^k}{(k+1)!}, \quad h(\alpha N)^{-1} = \sum_{j=0}^{m-1} (I - h(\alpha N))^j. \quad (\text{HH42})$$

The product $\alpha N h(\alpha N)$ is exactly $e^{\alpha N} - I$, because its last potentially extra term contains $N^m = 0$. The constant term of h is one; the same terminating geometric identity proves its displayed inverse. Thus the full fixed amplitude space and the full lifted fixed set are

$$F_L = \ker(S_L(0) - I) = \begin{cases} \{0\}, & e^{iLx_0/2} \neq 1, \\ \mathbb{C}\xi^{m-1}, & e^{iLx_0/2} = 1, \ L \neq 0, \\ A_0, & L = 0, \end{cases} \quad (\text{HH43})$$

$$\text{Fix}(\widehat{S}_L) = \{(0, \lambda) : \lambda \in \Lambda\} \cup \{(f, 1_\Lambda) : f \in F_L\}.$$

For the middle case, multiplication by the unit in HH42 does not change $\ker N$; writing the full polynomial of degree below m proves $\ker N = \mathbb{C}\xi^{m-1}$. This also covers $m = 1$, when the fixed amplitude is the entire one-dimensional algebra. If $m > 1$, the resonant operator itself is still nonidentity: its difference from I has rank $m - 1$ by HH42. The scalar phase alone therefore does not determine the fixed amplitudes at resonance, while every zero-support label is fixed in all cases. For $m > 1$ the nonzero resonant fixed direction lies in the Weil radical, but $R_0(\xi^{m-1}, 1) = i/(32a_m) \neq 0$ by HH24. This is its exact retained residue receiver.

Source scope. The exact author-source equations and human citations used in this derivation are given in HH1, HH3 and HH28–30. The retained original sources are Rodgers–Tao v5, Connes–Consani’s archimedean Weil paper and their Hochschild/zeta-cycles paper, and the DLMF Hermite formula with its named authors. The complete local input EP1–21 and full global receiver HW19–31 precede this derivation in the present reader. The entire lattice definition is stated in SH and HH39. The branch, connection, holonomy, trace and fixed-locus calculations have their proofs here. No paper-wide reading beyond the recorded source passages is claimed.

22 Prime holonomy on the original compact tests and full support

We now carry the endpoint fixed-locus calculation through the original compact tests. This uses the complete Weil form of HW24–31, with the two global moments, all prime powers and the Archimedean term retained. The underlying Connes–Consani source conventions are cited there and in HH28–30. The two-prime joint spectral receiver is also proved in the earlier programme source Two-prime torus receiver, TP1–6. The exact fixed-locus and compact-test maps needed here have complete proofs below.

Let

$$\mathcal{V} = \left\{ F \in C_c^\infty(\mathbb{R}) : \int F(t)e^{t/2} dt = \int F(t)e^{-t/2} dt = 0 \right\}, \quad \mathcal{L}F(s) = \int F(t)e^{(s-1/2)t} dt. \quad (\text{PFH1})$$

For real b define $\tau_b F(t) = F(t - b)$. Direct change of variables gives

$$\mathcal{L}(\tau_b F)(s) = e^{b(s-1/2)} \mathcal{L}F(s), \quad \tau_b \mathcal{V} = \mathcal{V}, \quad Q(\tau_b F, \tau_b G) = Q(F, G). \quad (\text{PFH2})$$

For the second equality the two zero moments are multiplied by $e^{b/2}$ and $e^{-b/2}$, respectively. For the third equality, every correlation $\int F(t) \overline{G(t-a)} dt$ is unchanged by translating both arguments. The complete finite-prime terms and every term of the Archimedean distribution in HW29 are expressions in precisely these correlations; the endpoint terms still vanish by PFH1. This proves invariance of the full analytic form, not merely one prime contribution. Equivalently, on its spectral side the factors at a reflected pair cancel exactly:

$$\overline{e^{b((1-\bar{\rho})-1/2)}} e^{b(\rho-1/2)} = 1.$$

This identity does not require $\Re \rho = 1/2$. It therefore asserts an isometry of the actual form, whose positivity is still the arithmetic question, rather than assuming a positive inner product.

The same transport acts on the original even Schwartz source through the explicit map

$$(\mathcal{E}F)(x) = |x|^{-1/2} F(\log |x|) \quad (x \neq 0), \quad (\mathcal{E}F)(0) = 0. \quad (\text{PFH3})$$

The support of F is compact, so $\mathcal{E}F$ vanishes in a neighbourhood of zero and outside a compact set; it is an even smooth Schwartz function. Its full integral is twice the positive moment in PFH1, hence zero. Change of variables proves

$$\mathcal{E} \tau_b F(x) = e^{-b/2} (\mathcal{E}F)(e^{-b}x), \quad \mathcal{M}(\Theta \mathcal{E}F)(s) = 2\zeta(s) \mathcal{L}F(s), \quad \Re s > 1, \quad (\text{PFH4})$$

where $\mathcal{M}\phi(s) = \int_0^\infty \phi(x) x^{s-1} dx$ and $\Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx)$. Absolute convergence in $\Re s > 1$ justifies the second equality term by term. The factor two and the half density are retained. Dilation commutes with the full theta sum by the same termwise identity.

For a prime p , set $b = -\log p$. The actual source operator is $\phi(x) \mapsto p^{1/2} \phi(px)$ and its Mellin factor is $p^{1/2-s}$. At an actual zero ρ its scalar value is $p^{1/2-\rho}$. On the entire length- m_ρ jet the operator is

$$T_p|_{A_\rho} = p^{1/2-\rho} \sum_{k=0}^{m_\rho-1} \frac{(-\log p)^k M_v^k}{k!}, \quad A_\rho = \mathbb{C}[v]/(v^{m_\rho}). \quad (\text{PFH5})$$

This is exactly HH30 with $L = -\log p$ at a critical zero, and the same Taylor calculation at any other actual zero. It is the opposite orientation to positive L ; no phase sign is suppressed.

There is now an exact conclusion for the full supported common fixed locus. For any nonempty finite packet of actual completed-zeta zeros let $E = \prod_\rho A_\rho$, with its entire zero-support lattice L attached as in HH39. Then

$$\boxed{\ker(T_2 - I) \cap \ker(T_3 - I) = 0, \quad \text{Fix}(\widehat{T}_2) \cap \text{Fix}(\widehat{T}_3) = \{(0, \lambda) : \lambda \in L\}.} \quad (\text{PFH6})$$

To prove it, if both scalar characters were one at a zero ρ , then $(\rho - 1/2) \log 2$ and $(\rho - 1/2) \log 3$ would belong to $2\pi i\mathbb{Z}$. A nonzero $\rho - 1/2$ would imply a rational relation between $\log 2$ and $\log 3$, hence an equality of nonzero integer powers of 2 and 3, impossible by unique factorization. Thus $\rho = 1/2$ would be required. But the original theta summands in HH1 are positive for $u \geq 0$ because $2\pi n^2 e^{4u} - 3 > 0$. Consequently $H_0(0) > 0$ and $\xi_R(1/2) \neq 0$. For each actual primary factor at least one of $T_2 - I, T_3 - I$ has a nonzero constant coefficient. Its full nilpotent inverse is the terminating expression HH41 with the corresponding scalar and parameter. The common fixed amplitude is therefore zero on every factor. Both operators fix every zero label, proving PFH6.

This fixed-locus result is unconditional and includes noncritical zeros if any exist. It does not erase their locations from the full action: the two scalar characters jointly distinguish ρ , by the

same irrationality calculation applied to $\rho - \sigma$. Moreover their nilpotent action retains every jet. Once the primary scalar $\lambda_p = p^{1/2-\rho}$ is specified, the exact finite logarithm recovers

$$M_v = -\frac{1}{\log p} \sum_{j=1}^{m_\rho-1} \frac{(-1)^{j+1}}{j} (\lambda_p^{-1} T_p - I)^j. \quad (\text{PFH7})$$

The exponential and logarithmic formal identities hold modulo v^{m_ρ} , so their series terminate and are inverse there. Thus the common fixed support is insensitive to the modulus, while the joint original action and its jet logarithm retain the exact arithmetic data. PFH2–5 give its maps back to the compact tests and the complete Weil form. This identifies what must be retained when using a supported fixed point in a purity or positivity calculation.

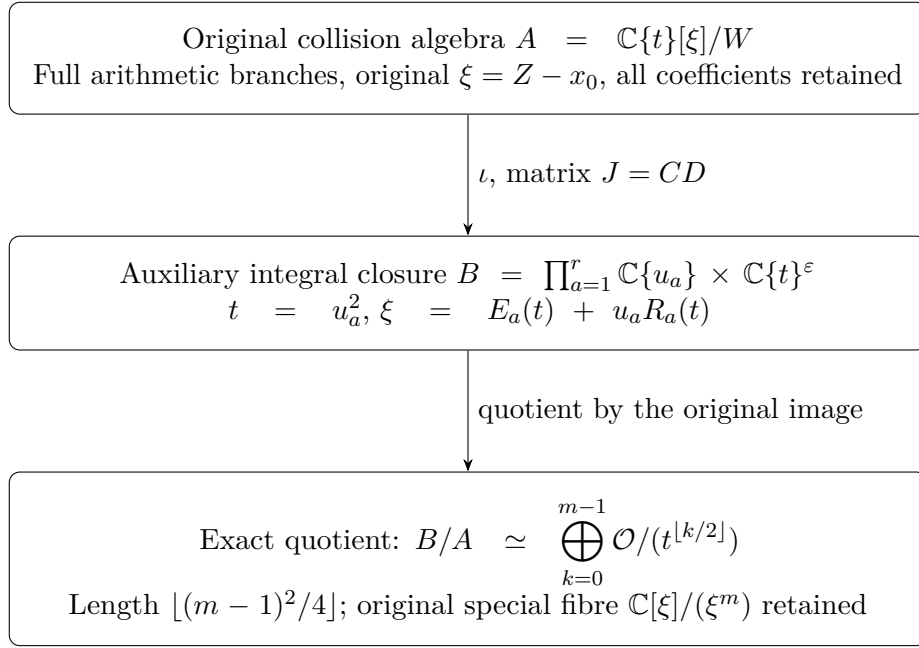
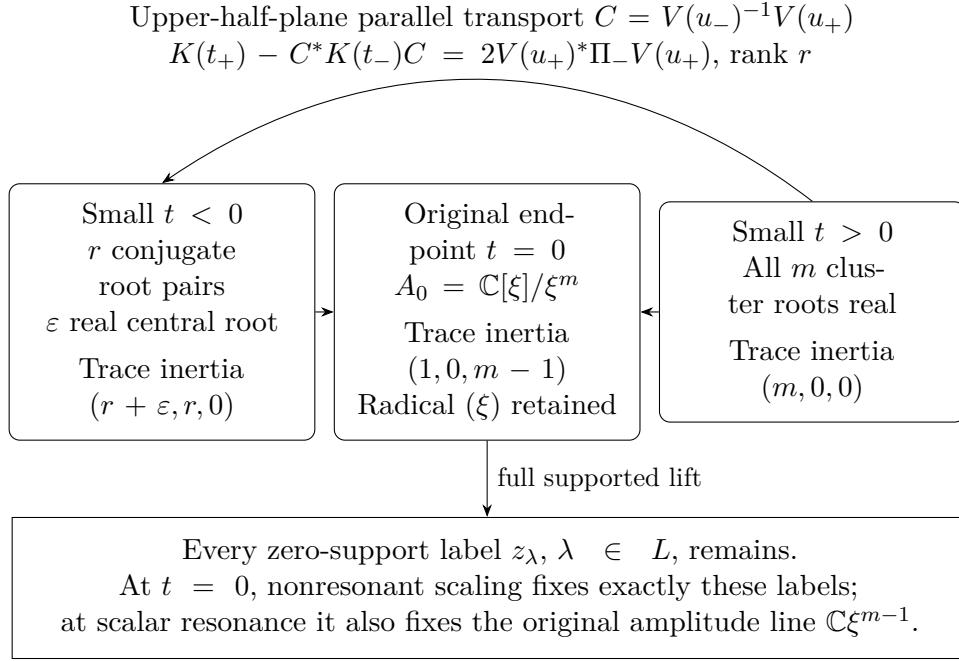


Figure 4: The full actual collision, not a replacement polynomial. HC1–7 define every arrow using the original H_t . HC8 evaluates the complete finite-module defect. HC9 retains every conductor unit, HC11 gives the possibly noninjective special-fibre map, and HC12–17 connect the discriminant to the original energy and completed-zeta unit. The source heat equation is Rodgers–Tao v5; the exact Hermite leading polynomial and its convention are cited and proved in EP2–6.



$$M_A = V^{-1}PV, \quad S_L = V^{-1} \text{diag}(e^{iL(x_0 + \theta_j)/2})V,$$

$$\|[M_A, S_L]\|_{\text{HS}, T_t}^2 = 8 \sum_{a=1}^r \sin^2 \left(\frac{L(\theta_{a,+} - \theta_{a,-})}{4} \right), \quad t > 0.$$

Figure 5: Exact geometry of the actual local heat family, with $r = \lfloor m/2 \rfloor$ and $\varepsilon = m - 2r$. The specialization arrows refer to analytic families and their limiting Gram matrices; they are not asserted to be isomorphisms of different fibres. HH17–22 prove the three signatures and the rank- r transport defect. HH29–38 give the complete two-action calculation and its original energy coefficient. HH39–43 give the fixed locus with every support label. Only the middle fibre is identified with the original completed-zeta packet contribution; the other fibres retain the actual H_t deformation. The human source conventions are Rodgers–Tao v5 and Connes–Consani’s explicit formula and multiplicity-sensitive spectral action, cited precisely in HH1 and HH28–30.

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